33. (a) (i) Explain the phenomenon of total internal reflection. State two conditions that must be satisfied for total internal reflection to occur. Derive the relation between the critical angle and the refractive index of a medium.

(ii) A ray of light is travelling from a denser medium of refractive index 1·5 to a rarer medium of refractive index 1·0. Calculate critical angle for total internal reflection. CBSE 26C 5

(i) When a ray of light enters from a denser medium to a rarer medium, it bends away from the normal. As the angle of incidence increases at the interface, angle of refraction also increases till the angle of incidence is $\frac{\pi}{2}$ (i.e the refracted ray grazes the surface of the interface). If the angle of incidence is further increased, the incident ray totally gets reflected back. This is called total internal reflection. | 2

Conditions

1. Light must travel from a denser medium to a rarer medium. | ½

2. At the interface, the angle of incidence must be greater than the critical angle. | ½

Relation

When light goes from a denser medium (refractive Index n_2) towards a rarer medium (refractive Index n_2), then for the value of critical angle, we have relation-

$n_2 \sin i_c = n_1 \sin 90^\circ$ | ½

$\sin i_c = \frac{n_1}{n_2}$

or, $\frac{n_2}{n_1} = \frac{1}{\sin i_c}$

or,

$n_{21} = \frac{1}{\sin i_c}$ | ½

(ii) Calculating critical angle

$n_2 = 1.5 = \frac{3}{2}$, $n_1 = 1$

$\sin i_c = \frac{1}{n_{21}} = \frac{2}{3}$ | ½

Therefore,

$i_c = \sin^{-1}\left(\frac{2}{3}\right)$ | ½

9. A ray of light is incident at an angle of 60° on face AB of prism ABC with $\angle A = 30°$. The ray emerging out of the prism makes an angle of 30° with the incident ray. The angle of emergence ($\angle e$) is equal to :

(A) 0°

(B) 30°

(C) 60°

(D) 90°

CBSE 26C

9. A ray of light is incident at an angle of 60° on face AB of prism ABC with $\angle A = 30°$. The ray emerging out of the prism makes an angle of 30° with the incident ray. The angle of emergence ($\angle e$) is equal to :

(A) 0°

(B) 30°

(C) 60°

(D) 90°

CBSE 26C

(A) 0^0

24. Derive an expression for refractive index of an equilateral prism in terms of the angle of minimum deviation and the angle of the prism. Plot a graph to show the variation of the angle of deviation as a function of angle of incidence. *3*

CBSE 26C

24. Derive an expression for refractive index of an equilateral prism in terms of the angle of minimum deviation and the angle of the prism. Plot a graph to show the variation of the angle of deviation as a function of angle of incidence. 3

CBSE 26C

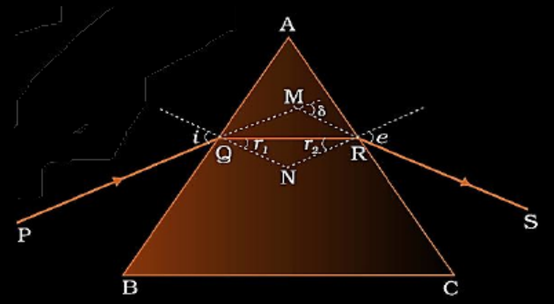
In quadrilateral AQNR,

$\angle A + \angle QNR = 180°$

From triangle QNR,

$r_1 + r_2 + \angle QNR = 180°$

Comparing the above equations,



½

$r_1 + r_2 = A$

The total deviation δ is the sum of deviations at the two faces:

$\delta = (i - r_1) + (e - r_2)$

$\delta = i + e - A$ ½

At minimum deviation,

$\delta = D_m$, $i = e$ *which implies* $r_1 = r_2$

$r = \frac{A}{2}$

$D_m = 2i - A$ or $i = \frac{A+D_m}{2}$ ½

The refractive index of the prism is

$n = \frac{\sin i}{\sin r}$ 1

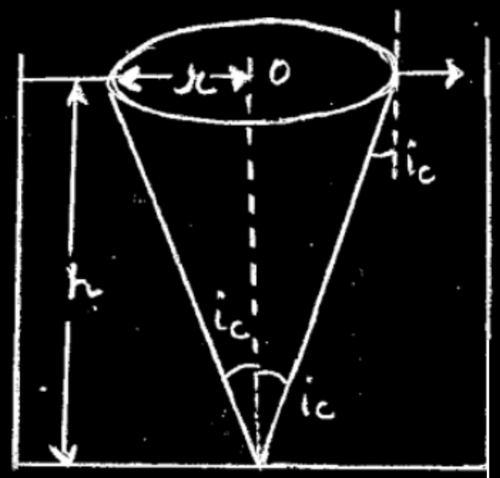
$n = \frac{\sin\left(\frac{A+D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$ ½



33. (a) A small bulb is placed at the bottom of a tank containing water to a depth of 1·00 m. Find the area of the surface of water through which light from the bulb can emerge from. The refractive index of water is (4/3). *3*

(b) An object is placed at a distance of 15 cm from a convex lens of focal length 10 cm. On the other side of the lens, a convex mirror is placed coaxially at a distance of 10 cm from the lens. If the image formed by this combination coincides with the object itself, draw the ray diagram and find the focal length of the convex mirror. *2*

CBSE 26C



1

$\sin i_c = \frac{1}{n}$

$\tan i_c = \frac{r}{h}$ ½

$r = h \tan i_c$ & $\tan i_c = \frac{1}{\sqrt{n^2-1}}$

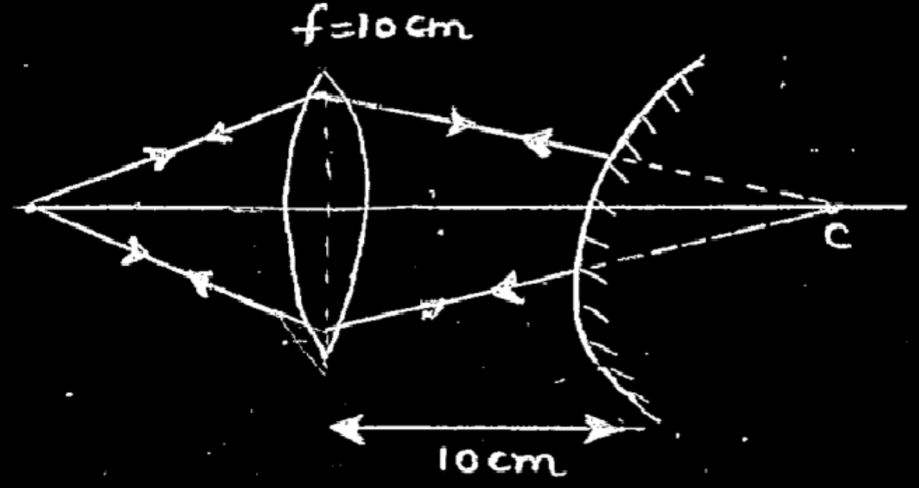
$A = \pi r^2 = \pi h^2 \tan^2 i_c$ ½

$A = \frac{\pi h^2}{n^2-1}$ ½

$A = \frac{\pi \times 1}{\frac{16}{9}-1}$ ½

$= \frac{9\pi}{7}$ m^2 $= \frac{9\times 3.14}{7}$ =4.04 m^2

(b)



1

$\frac{1}{v} - \frac{1}{u} = \frac{1}{f}$

$\frac{1}{v} = \frac{1}{10} - \frac{1}{15}$

$= \frac{1}{30}$

$v = 30$ cm ½

Radius of curvature of the mirror,

$R = 30 - 10 = 20$ cm

$f = \frac{R}{2}$ ½

$f = 10$ cm

24. (a) The distance of an object from a concave mirror is increased from 12 cm to 15 cm. It is observed that the new image becomes $\left(\frac{2}{5}\right)^{\text{th}}$ the size of the initial image. Calculate the focal length of the mirror.

(b) A light ray is incident on the horizontal upper face of a glass cube making an angle of 45° with the face. While emerging, the ray just grazes the vertical face of the cube. Find the refractive index of the glass. CBSE 26C 3

(a)

For $u = -12\ cm$

$$m_1 = \frac{h_1}{h_o} = \frac{f}{f+12} \quad \ldots(1)$$ ½

For $u = -15\ cm$

$$m_2 = \frac{h_2}{h_o} = \frac{f}{f+15} \quad \ldots(2)$$ ½

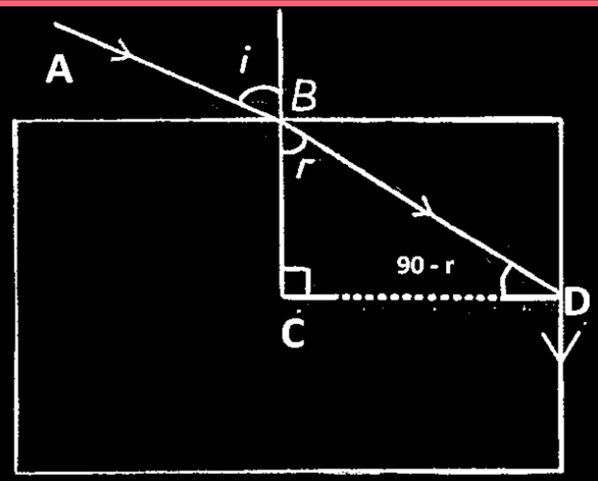
Dividing (1) by (2)

$$\frac{m_1}{m_2} = \frac{h_1}{h_2} = \frac{f+15}{f+12}$$

$$h_2 = h_1\left(\frac{2}{3}\right)$$

On solving,

$$f = -10\ cm$$ ½



At air-glass interface,

$$n = \frac{\sin i}{\sin r}$$

$$n = \frac{\sin 45^\circ}{\sin r}$$

$$\sin r = \frac{\sin 45^\circ}{n} = \frac{1}{\sqrt{2}\,n} \quad \ldots(1)$$ ½

At glass-air interface, light suffers Total Internal Reflection (TIR) ($i_C = 90°\text{-}r$).

$$n = \frac{1}{\sin i_C}$$

$$n = \frac{1}{\sin(90^\circ - r)} = \frac{1}{\cos r}$$

$$n = \frac{1}{cosr}$$

$$\cos r = \frac{1}{n} \quad \ldots(2)$$ ½

Squaring and adding (1) and (2)

$$1 = \frac{1}{2n^2} + \frac{1}{n^2}$$

On solving,

$$n = \sqrt{\frac{3}{2}}$$ ½

3. A ray of light passes from a medium of refractive index $\sqrt{2}$ into the medium of refractive index 1. For the ray to just graze along the surface of separation, the angle of incidence should be :

(A) $\frac{\pi}{3}$

(B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$

(D) $\frac{2\pi}{3}$

CBSE 26C

3. A ray of light passes from a medium of refractive index $\sqrt{2}$ into the medium of refractive index 1. For the ray to just graze along the surface of separation, the angle of incidence should be :

(A) $\frac{\pi}{3}$ (B) $\frac{\pi}{4}$

(C) $\frac{\pi}{6}$ (D) $\frac{2\pi}{3}$

CBSE 26C

(B) $\pi/4$

19. A tank is filled with a liquid to a height of 12·5 m. The apparent depth of a needle lying at the bottom of the tank is measured to be 9·0 m. Calculate the speed of light in the liquid. CBSE 2026 *2*

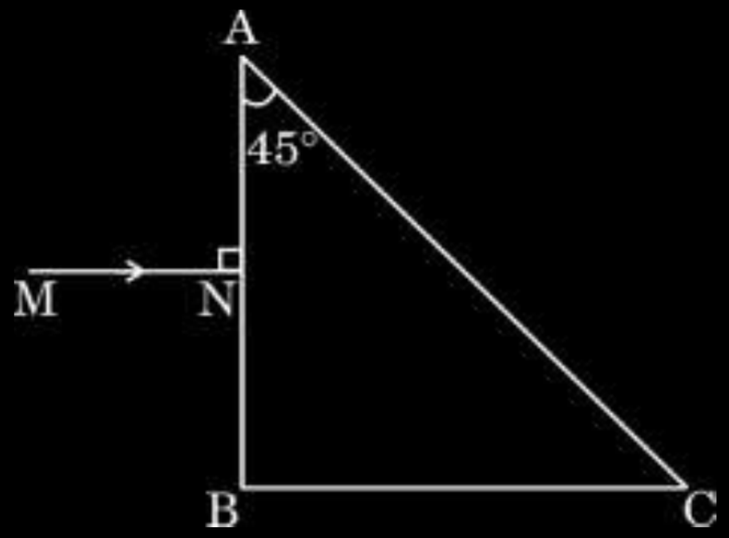
19. A tank is filled with a liquid to a height of 12·5 m. The apparent depth of a needle lying at the bottom of the tank is measured to be 9·0 m. Calculate the speed of light in the liquid. CBSE 2026 *2*

$$n = \frac{\text{speed of light in vacuum(c)}}{\text{speed of light in medium(v)}} \quad \ldots\ldots\ (i)$$ ½

$$n = \frac{\text{real depth}}{\text{apparent depth}}$$ ½

$$n = \frac{12.5}{9} \quad \ldots\ldots\ldots\

18. A ray of light MN is incident normally on the face corresponding with side AB of a prism with an isosceles right-angled triangular base ABC. Trace the path of the ray as it passes through the prism when the refractive index of the prism material is (i) $\sqrt{2}$, and (ii) $\sqrt{3}$. *2*



CBSE 2026

(i)

$$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}}$$

$$i_c = 45^0$$

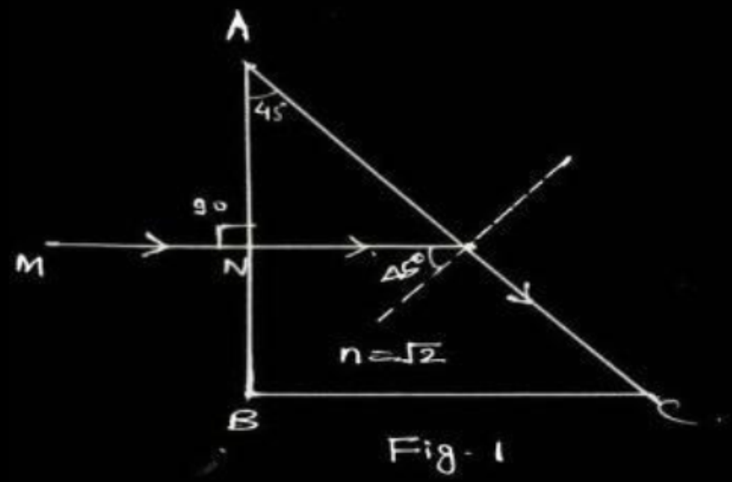


Fig - 1

1

Since $i = i_c$ so the emergent ray grazes alongside AC.

(ii)

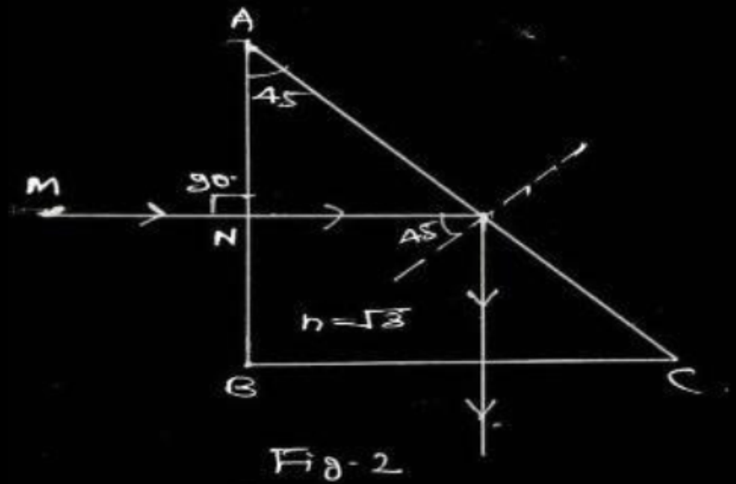


Fig - 2

1

$$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{3}}$$

$$i_c < 45^0$$

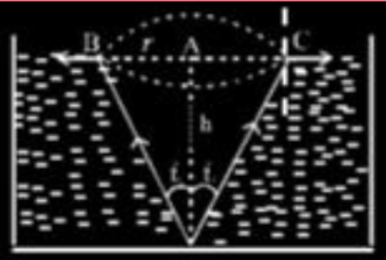
Since $i > i_c$ so, the ray suffers from Total Internal Reflection (TIR)

19. A small bulb is placed at the bottom of a tank, containing a transparent liquid of refractive index $\sqrt{2}$, to a depth of 1 m. Calculate the area of the surface of the liquid through which light from the bulb emerges. *2*

CBSE 2026

19. A small bulb is placed at the bottom of a tank, containing a transparent liquid of refractive index $\sqrt{2}$, to a depth of 1 m. Calculate the area of the surface of the liquid through which light from the bulb emerges. *2*

CBSE 2026



$r = h \times \tan i \quad [\text{as } i = i_c]$ ½

$\sin i_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}} \Rightarrow i_c = 45°$ ½

$r = h$

$\text{Area} = \pi r^2$ ½

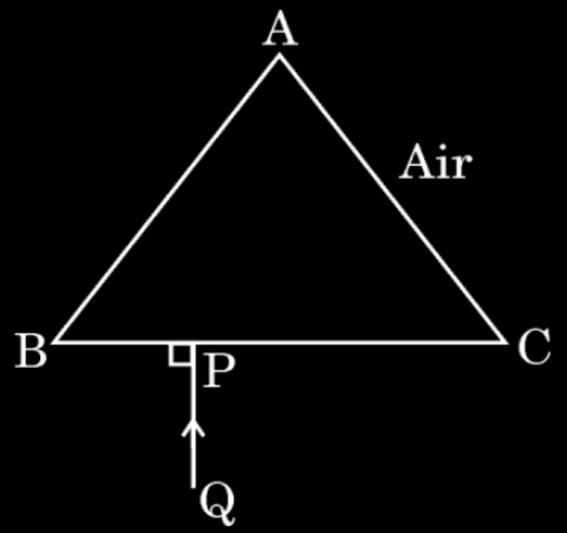
$= 3.14 \times (1)^2$

$\text{Area} = 3.14\ m^2$ ½

31. (a) (i) Define refractive index of a medium in terms of speed of light.

(ii) Derive the relation for the refractive index (μ) of a prism in terms of angle of minimum derivation (δ_m) and angle of prism (A).

(iii) A ray of light QP is incident normally on the face BC of a triangular prism ABC of refractive index 1·5 kept in air, as shown in the figure. Trace the path of the ray as it passes through the prism and give relevant explanation. *5*



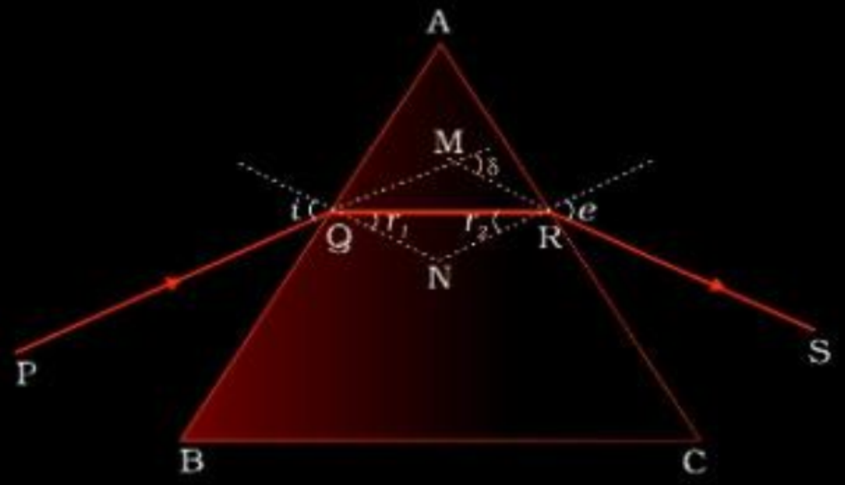
CBSE 2026

(i) Refractive index of a medium is the ratio of the speed of light in vacuum to the speed of light in the medium. — 1

Alternatively

$$n_{21} = \frac{c}{v}$$

(ii)



In the quadrilateral AQNR, two of the angles (at the vertices Q and R) are right angles. Therefore, the sum of the other angles of the quadrilateral is 180°.

$\angle A + \angle QNR = 180°$

From the triangle QNR,

$r_1 + r_2 + \angle QNR = 180°$

Comparing these two equations, we get

$r_1 + r_2 = A$ .....(i) — ½

The total deviation δ is the sum of the deviations at the two faces.

$\delta = i + e - A$ .....(ii) — ½

For $\delta = \delta_m$; i = e which implies $r_1 = r_2$

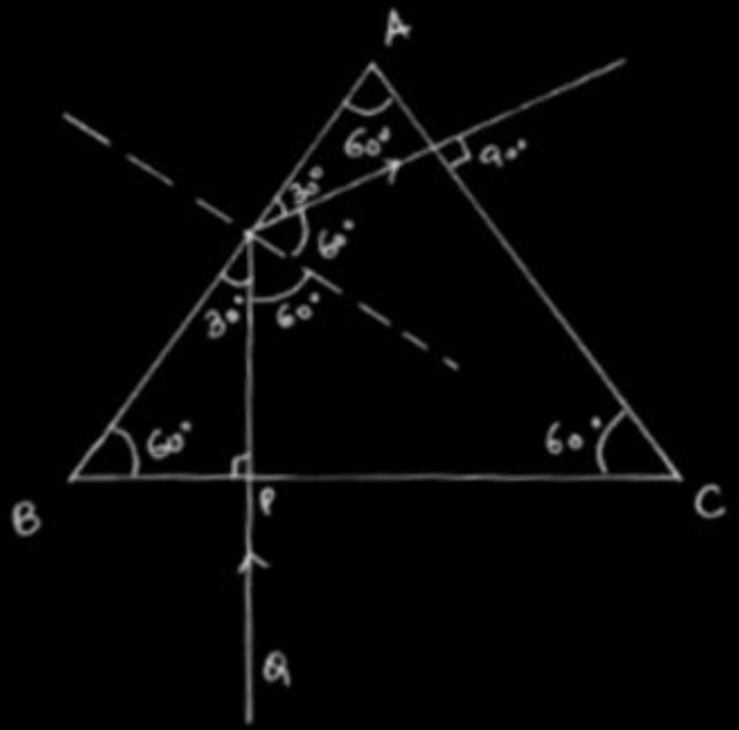
From equation (i) and (ii) — ½

$$r = \frac{A}{2}; i = \frac{A + \delta_m}{2}$$

Refractive index of the prism

$$n_{21} = \frac{n_2}{n_1} = \frac{\sin\frac{(A + \delta_m)}{2}}{\sin\frac{A}{2}}$$ — ½

(iii)

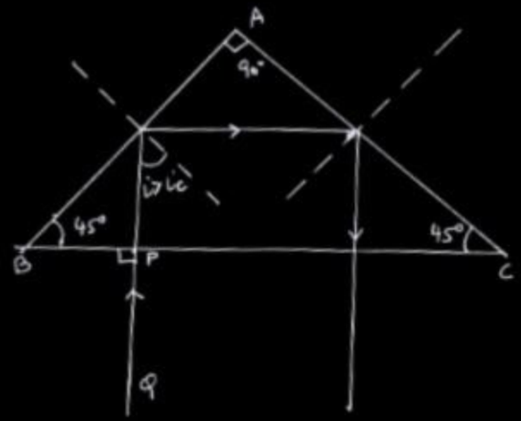


1

The ray QP strikes the face BC normally and hence goes undeviated. As $i > i_c$, it undergoes TIR. It strikes the face AC normally and comes out undeviated.

1

Alternatively



The ray QP strikes the surface BC normally and hence goes undeviated. As $i > i_c$, it undergoes TIR. As at face AC, $i > i_c$, it undergoes TIR. The ray comes out of the face BC undeviated.

(Award full credit of this part if the student takes the value of A other than 60° or 90° and traces the path correctly and according to the angle taken)

OR

7. In which of the following, total internal reflection does ***not*** occur ?

(A) Twinkling of stars

(B) Brilliance of diamonds

(C) Optical fibre

(D) Reflecting prism

CBSE 2026

7. In which of the following, total internal reflection does ***not*** occur ?

(A) Twinkling of stars

(B) Brilliance of diamonds

(C) Optical fibre

(D) Reflecting prism

CBSE 2026

(A) Twinkling of stars

26. A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface. *3*

CBSE 2026

26. A ray of light is travelling through a rectangular glass slab (refractive index $\frac{3}{2}$) and is incident on the horizontal glass-air surface at the critical angle for the two media. The slab is then brought in contact with water (refractive index $\frac{4}{3}$) such that a thin horizontal layer of water is formed on the surface of the slab. Find the angle at which the ray will emerge into air from the water-air surface. *3*

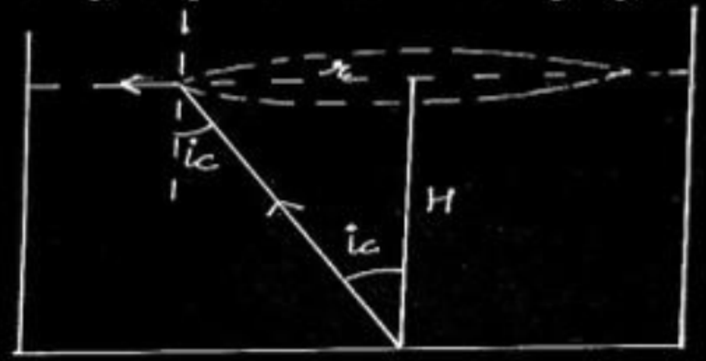
CBSE 2026

$\sin i_C = \frac{1}{n}$ $= \frac{2}{3}$	1
For glass – water surface $\frac{3}{2}\sin i_C = \frac{4}{3}\sin r$	½
$\frac{3}{2} \times \frac{2}{3} = \frac{4}{3}\sin r$ $\sin r = \frac{3}{4}$	
For water – air surface. $\frac{4}{3}\sin r = \sin e$	½
$\frac{4}{3} \times \frac{3}{4} = \sin e$ $\sin e = 1$	½
$e = 90^\circ$	½

23. Define the term 'critical angle'. A small bulb is placed at the bottom of a tank containing a liquid of refractive index μ at a depth H. It is observed that the light emerges from a circular area of radius r of the surface. Obtain the expression for r in terms of H and μ. CBSE 2026 *3*

23. Define the term 'critical angle'. A small bulb is placed at the bottom of a tank containing a liquid of refractive index μ at a depth H. It is observed that the light emerges from a circular area of radius r of the surface. Obtain the expression for r in terms of H and μ. CBSE 2026 *3*

Critical angle: The angle of incidence corresponding to the angle of refraction 90° for a given pair of media when light goes from denser to rarer medium. — 1



½

$$\text{Sin } i_c = \frac{1}{\mu} = \frac{r}{\sqrt{r^2 + H^2}}$$ — ½

$$\mu^2 r^2 = r^2 + H^2$$ — ½

$$(\mu^2 - 1)r^2 = H^2$$

$$r = \frac{H}{\sqrt{\mu^2 - 1}}$$ — ½

9. A ray of yellow light undergoes total internal reflection when it is incident at the interface of two media. This ray is successively replaced by the ray of blue, green and red lights. Which of the following statements is true if the angle of incidence is same in all cases ?

(A) Total internal reflection occurs for blue light only.

(B) Total internal reflection occurs for red light only.

(C) Total internal reflection occurs for both green and blue lights.

(D) None of them will undergo total internal reflection.

CBSE 2026

9. A ray of yellow light undergoes total internal reflection when it is incident at the interface of two media. This ray is successively replaced by the ray of blue, green and red lights. Which of the following statements is true if the angle of incidence is same in all cases ?

(A) Total internal reflection occurs for blue light only.

(B) Total internal reflection occurs for red light only.

(C) Total internal reflection occurs for both green and blue lights.

(D) None of them will undergo total internal reflection.

CBSE 2026

(C) Total internal reflection occurs for both green and blue lights.

23. A ray of light is incident at angle of 45° on one face of a prism with an equilateral triangular base. If it passes symmetrically through the prism, find the :

(a) angle of minimum deviation for the prism, and

(b) refractive index of the material of the prism. *3*

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23. A ray of light is incident at angle of 45° on one face of a prism with an equilateral triangular base. If it passes symmetrically through the prism, find the :

(a) angle of minimum deviation for the prism, and

(b) refractive index of the material of the prism. *3*

CBSE 2026

(a) As ray of light passes symmetrically through the prism $\angle r_1 = \angle r_2 = r \ , \ i = e$ $\delta_m = 2i - A$ $= 2 \times 45° - 60°$ $= 30°$	½ ½ ½
(b) $r = \frac{A}{2} = 30°$ $\mu = \sin \frac{(\frac{A+\delta_m}{2})}{sin\frac{A}{2}}$ $= \sin \frac{(\frac{60° + 30°}{2})}{sin30°}$ $= \frac{sin45°}{sin30°}$ $= \sqrt{2}$	½ ½ ½
(Alternatively) $\mu_a sini = \mu_g sinr$ $1 \times sin45° = \mu_g \times sin30°$ $\mu_g = \sqrt{2}$	

4. The process that allows light to travel through a tube which is not straight is – **1**

(A) reflection of light
(B) dispersion of light
(C) scattering of light
(D) total internal reflection

CBSE 2026

4. The process that allows light to travel through a tube which is not straight is – **1**

(A) reflection of light
(B) dispersion of light
(C) scattering of light
(D) total internal reflection

CBSE 2026

(D) total internal reflection of light

19. A ray of light in air is incident at angle ∠i on a face of an equilateral glass prism and is refracted through the prism. As ∠i is varied, it is observed that the ray undergoes minimum deviation, when the ∠i is three-fourth of the angle of the prism. Calculate the speed of light in the prism. CBSE 2026 **2**

19. A ray of light in air is incident at angle ∠i on a face of an equilateral glass prism and is refracted through the prism. As ∠i is varied, it is observed that the ray undergoes minimum deviation, when the ∠i is three-fourth of the angle of the prism. Calculate the speed of light in the prism. CBSE 2026 **2**

Angle of incidence $i = \frac{3}{4}A = 45^o$	1/2
At minimum deviation $r = \frac{A}{2} = 30^o$	1/2
$n = \frac{\sin i}{\sin r} = \frac{\sin 45^o}{\sin 30^o}$	
$n = \sqrt{2}$	1/2
$n = \frac{c}{v}$	
$v = \frac{3}{\sqrt{2}} \times 10^8$ m/s = 2.12×10^8m/s	1/2

19. A ray of light is incident on the face of a triangular prism of refracting angle 60° and it just suffers total internal reflection at the other face. Find the angle of incidence for the ray if the refractive index of the material of the prism is $\sqrt{2}$. CBSE 2026 **2**

19. A ray of light is incident on the face of a triangular prism of refracting angle 60° and it just suffers total internal reflection at the other face. Find the angle of incidence for the ray if the refractive index of the material of the prism is $\sqrt{2}$. CBSE 2026 **2**

$\sqrt{2}(\sin i_c) = 1$ — ½

$(\sin i_c) = \frac{1}{\sqrt{2}}$

$i_c = 45^0$

$r_1 + r_2 = A$ (Here $r_2 = i_c$)

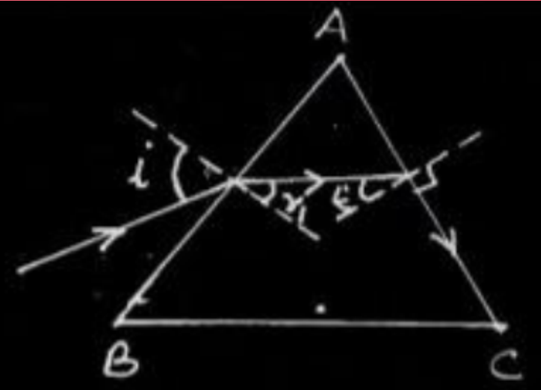
$r_1 = A - i_c$ — ½

$= 60^0 - 45^0$

$= 15^0$ — ½

$\text{Sin } i = \sqrt{2}\sin 15^0$

$i = \sin^{-1}[\sqrt{2}\sin 15^0] = \sin^{-1}(0.366)$ — 1/2



21. A ray of light is incident at an angle 2A on one of the faces of a glass prism of refracting angle A. After refraction from this face, the ray strikes the opposite face which is silvered. The reflected ray from this face retraces on its path.

Trace the ray diagram and find the relation between the refractive index of material of prism and the angle of prism. **2**

CBSE 2026

21. A ray of light is incident at an angle 2A on one of the faces of a glass prism of refracting angle A. After refraction from this face, the ray strikes the opposite face which is silvered. The reflected ray from this face retraces on its path.

Trace the ray diagram and find the relation between the refractive index of material of prism and the angle of prism. **2**

CBSE 2026



1

From Snell's Law

$n_1 . \sin i = n_2 . sinr$ ½

$A = r_1 + r_2$

$A = r_1 \quad \because r_2 = 0$

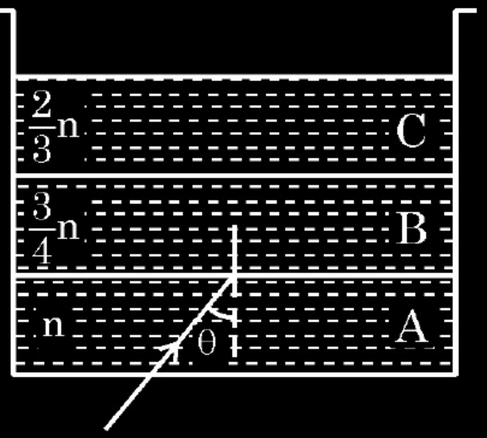
$1 \times sin2A = n \times sinA$

$2\ sinA\ cosA = n \times sinA$

$n = 2cosA$ ½

28. (a) State the two conditions under which total internal reflection occurs. **3**

(b)



A transparent container contains layers of three immiscible transparent liquids A, B and C of refractive indices n, $\frac{3}{4}$ n and $\frac{2}{3}$ n, respectively. A laser beam is incident at the interface between A and B at an angle θ as shown in figure. Prove that the beam does not enter region C at all for $\sin\theta \geq \frac{2}{3}$.

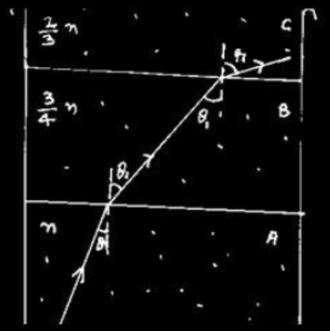
CBSE 2026

(a) (i) Light must travel from denser medium to rarer medium. ½

(ii) Angle of incidence must be greater than Critical angle. ½

$i > i_c$

(b)



For refraction at AB interface

$$\frac{\sin\theta}{\sin\theta_1} = \frac{3n/4}{n}$$

$$\sin\theta = \frac{3}{4}\sin\theta_1 \quad \text{---------- (1)} \qquad ½$$

For refraction at BC interface

$$\frac{\sin\theta_1}{\sin r} = \frac{2n/3}{3n/4}$$

$$\sin\theta_1 = \frac{8}{9}\sin r \quad \text{---------- (2)} \qquad ½$$

From equations (1) and (2)

$$\frac{3}{2}\sin\theta = \sin r \qquad ½$$

for $\sin\theta = \frac{2}{3}$

$\sin r = 1$ i.e $r = 90^0$ ½

Ray grazes the surface

Hence for $\sin\theta \geq \frac{2}{3}$ does not enter the region C at all

32. (a) A ray of light is incident on one face of a triangular glass prism of refractive index n and is refracted out from the opposite face. Deduce the expression for refractive index of glass prism in terms of the angle of minimum deviation and angle of the prism. Write the condition on n for refraction to take place on the opposite face. *5*

CBSE 2025 C

$$A + \delta = i + e \qquad \text{½}$$

and $A = r_1 + r_2$ ½

at angle of minimum deviation , $\delta = D_m$

$i = e$, $r_1 = r_2 = r$ ½

$A + D_m = 2i$

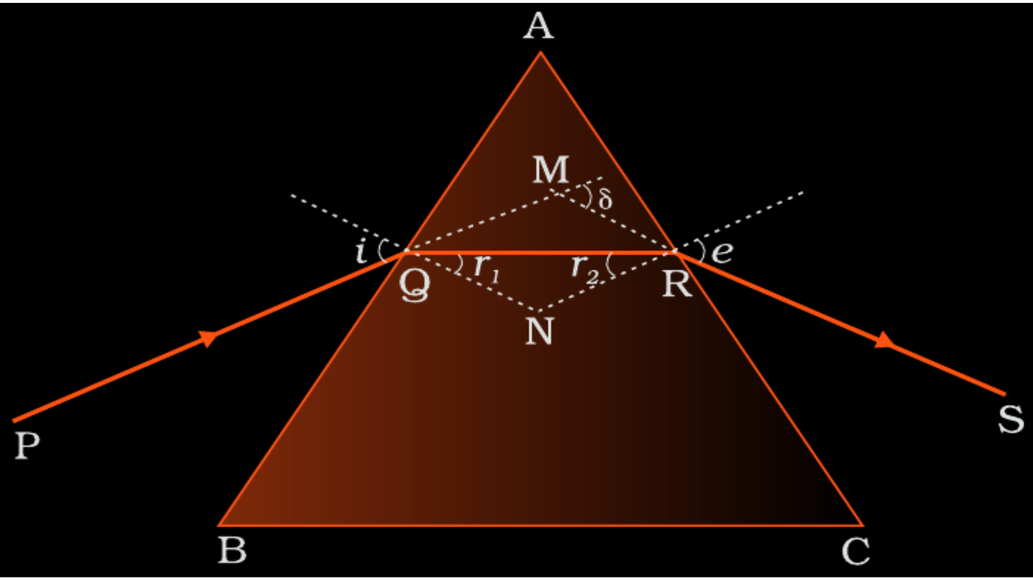
$$i = \frac{A + D_m}{2}$$

$$A = 2r \Rightarrow r = \frac{A}{2} \qquad \text{½}$$

From Snell's law

$$n = \frac{\sin i}{\sin r} \qquad \text{½}$$

$$n = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)} \qquad \text{½}$$



1

For refraction to take place on opposite face

$r_2 < i_c$

$$n = \frac{1}{\sin i_c} \qquad \text{½}$$

$$i_c = \sin^{-1}\left(\frac{1}{n}\right)$$

$$r_2 < \sin^{-1}\left(\frac{1}{n}\right) \qquad \text{½}$$

8. The depth of a swimming pool (refractive index of water = $\frac{4}{3}$) is 1 m. If one looks straight vertically down, the depth of the pool will appear to be :

(A) 0·25 m (B) 0·75 m

(C) 0·95 m (D) 1·0 m

CBSE 2025 C

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(A) 0·25 m (B) 0·75 m

(C) 0·95 m (D) 1·0 m

CBSE 2025 C

(B) 0.75 m

33. (a) (i) An equilateral triangular prism is made of material of refractive index $\sqrt{2}$. Find the value of the angle of minimum deviation for the prism. How will this value be affected when the prism is immersed in water ?

(ii) A ray of light is incident normally on one face of this prism. Find the minimum angle at which the ray must be incident on the adjacent face inside the prism so that the ray experiences total internal reflection at this face. *5*

CBSE 2025 C

(i) $n=\sqrt{2}\,,\ A=60^0$	
$n=\dfrac{\sin\left(\dfrac{D_m+A}{2}\right)}{\sin\left(A/2\right)}$	½
$\sqrt{2}=\dfrac{\sin\left(\dfrac{D_m+60^0}{2}\right)}{\sin\left(60^0/2\right)}$	½
$\sin\left(\dfrac{D_m+60^0}{2}\right)=\dfrac{1}{\sqrt{2}}$	
$\sin\left(\dfrac{D_m}{2}+30^0\right)=45^0$	½
$D_m=30^0$	½
As $n<\dfrac{n}{n_w}$, value of D_m decreases in water	1
(ii) $\sin i_c=\dfrac{1}{n}$	½
$i_c=\sin^{-1}\left(\dfrac{1}{n}\right)=\sin\left(\dfrac{1}{\sqrt{2}}\right)$	½
$i_c=45^0$	½
For total internal reflection, angle of incidence > 45º	½

16. *Assertion (A) :* A ray of light is incident normally on the face of a prism. The emergent ray will graze along the opposite face of the prism when the critical angle at glass-air interface is equal to the angle of the prism.

Reason (R) : The refractive index of a prism depends on angle of the prism.

CBSE 2025

16. *Assertion (A) :* A ray of light is incident normally on the face of a prism. The emergent ray will graze along the opposite face of the prism when the critical angle at glass-air interface is equal to the angle of the prism.

Reason (R) : The refractive index of a prism depends on angle of the prism.

CBSE 2025

(C) Assertion(A) is true and Reason(R) is false.

21. A tank is filled with a liquid of refractive index $\sqrt{2}$, up to a height of 30 cm. A tiny bulb is glowing at the bottom of the tank. Calculate the diameter of an opaque disc floating symmetrically on the liquid surface that can cut off completely the light from the bulb that comes out of the liquid surface.

CBSE 2025 *2*

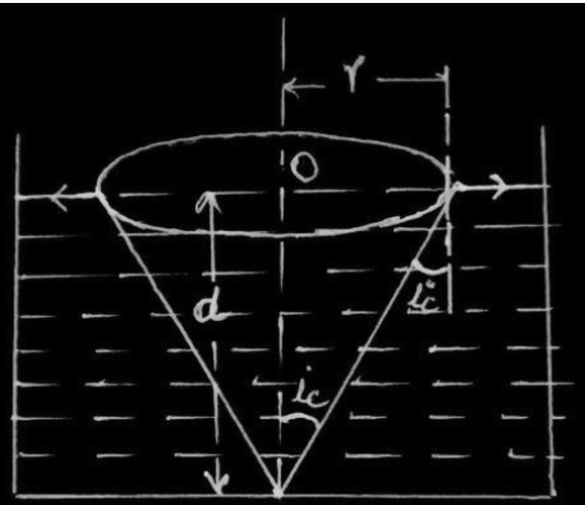
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CBSE 2025 | *2*

$\sin i_c = \frac{1}{\sqrt{2}}$ ½

$i_c = 45^0$

$\tan i_c = \frac{r}{d}$



$r = d \tan 45^0$ ½

$r = 30 \times 1$ cm

$D = 2r = 60$ cm 1

Alternatively:

$D = 2r = \frac{2d}{\sqrt{n^2 - 1}}$ 1

$= \frac{2 \times 30}{\sqrt{(\sqrt{2})^2 - 1}}$ ½

$D = 60$ cm ½

20. A ray of light is incident on face AB of a prism ABC with angle of prism A and emerges out from face AC. The prism is set in the position of minimum deviation with angle of deviation δ. Find (i) the angle of incidence and (ii) the angle of refraction on face AB. CBSE 2025 *2*

20. A ray of light is incident on face AB of a prism ABC with angle of prism A and emerges out from face AC. The prism is set in the position of minimum deviation with angle of deviation δ. Find (i) the angle of incidence and (ii) the angle of refraction on face AB. CBSE 2025 *2*

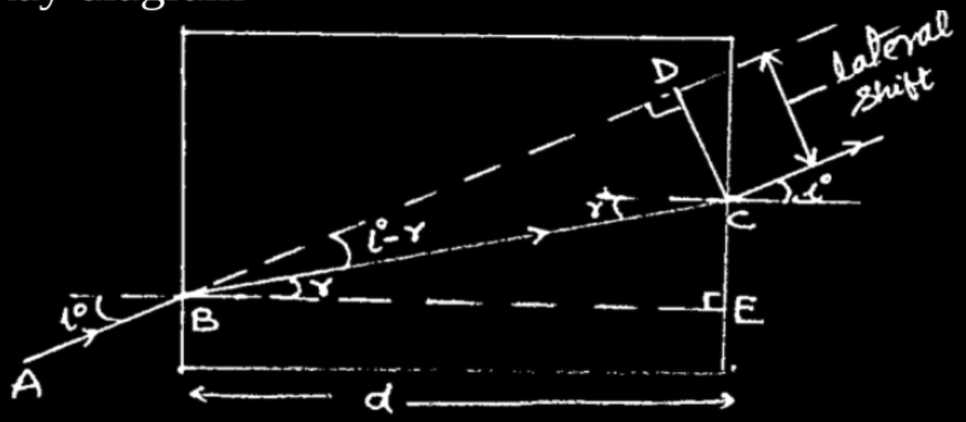
In a prism we know $A+\delta=i+e \quad and\ A=r_1+r_2$	½
At minimum deviation $i=e \ \ and \ \ r_1=r_2=r$	½
$A+D_m=2i$ $i=\frac{A+D_m}{2}$	½
$A=2r$ $r=A/2$	½

24. A ray of light is incident at an angle i on a parallel sided glass slab of thickness 'd' and gets refracted into the slab at angle r. Draw a ray diagram to show its path as its emerges out of the slab. Hence, obtain an expression for the lateral shift of the ray. Under what condition will the shift be minimum ?

CBSE 2025

3

Ray diagram



1

$$\sin(i-r)=\frac{CD}{BC} \qquad \text{---------(1)}$$ ½

$$\cos r=\frac{BE}{BC}=\frac{d}{BC}$$ ½

Putting in equation (1)

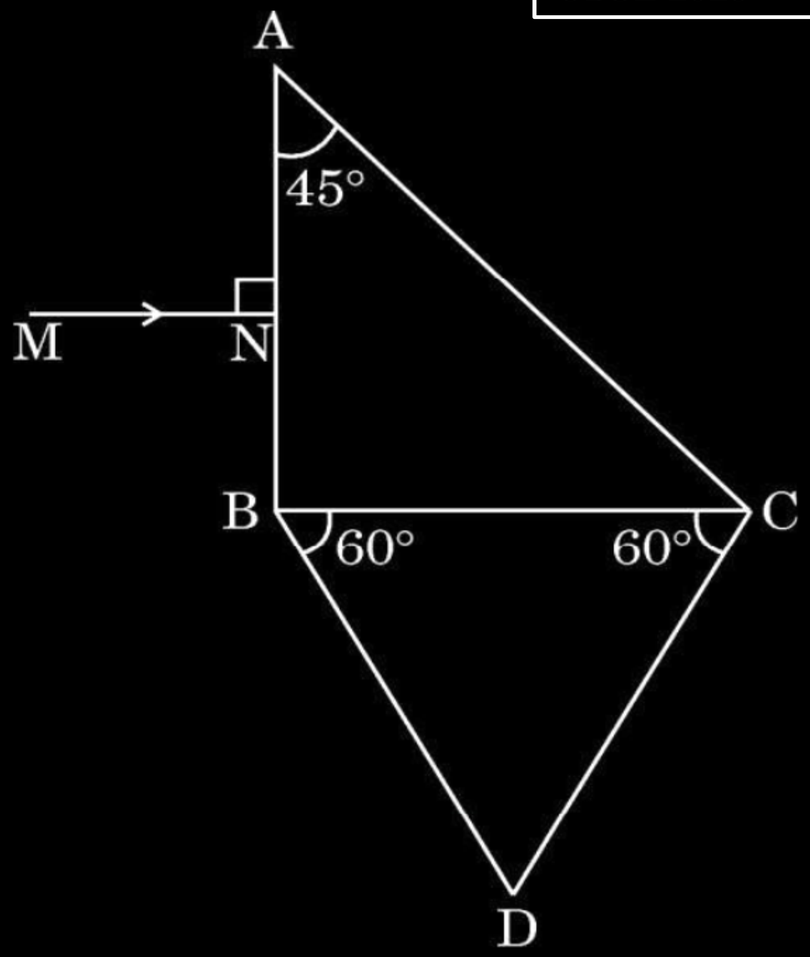
$$CD=\frac{d\sin(i-r)}{\cos r}$$ ½

Shift will be minimum for minimum angle of incidence. ½

20. A right angled isosceles glass prism ABC is kept in contact with an equilateral triangular prism DBC as shown in the figure. Both prisms are made of the same glass of refractive index 1·6. Trace the path of the ray MN incident normally on face AB as it passes through the combination.

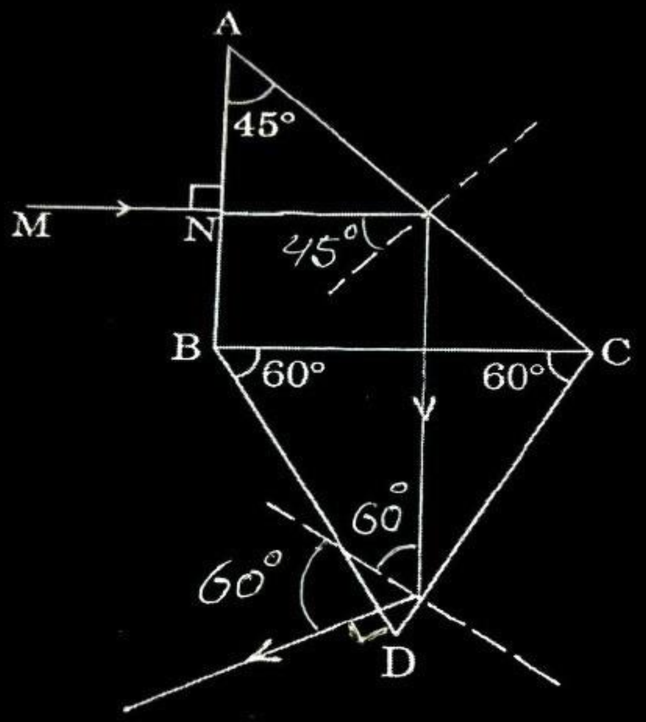
CBSE 2025

2



20. A right angled isosceles glass prism ABC is kept in contact with an equilateral triangular prism DBC as shown in the figure. Both prisms are made of the same glass of refractive index 1·6. Trace the path of the ray MN incident normally on face AB as it passes through the combination.

CBSE 2025 *2*



1

1

Note: Please deduct ½ mark for not showing arrows with the rays.

24. (a) State the conditions under which total internal reflection occurs. **3**

(b) A glass slab 3.0 cm thick, is placed over a dark ink dot. Find the height through which the image of the dot is raised. The refractive index of the glass is 1.5. CBSE 2025

24. (a) State the conditions under which total internal reflection occurs. **3**

(b) A glass slab 3.0 cm thick, is placed over a dark ink dot. Find the height through which the image of the dot is raised. The refractive index of the glass is 1.5. CBSE 2025

a) (i) The ray of light should pass from a denser to a rarer medium	½
b) (ii)The angle of incidence is greater than the critical angle for the pair of media.	½
c) $n = \frac{\text{Real depth}}{\text{Apparant depth}}$	½
$\text{Apparant depth} = \frac{\text{Real depth}}{n}$	
$= \frac{3\text{ cm}}{1.5} = 2\text{cm}$	½
The height through which the image of the dot raised (i.e. Normal shift)	½
= Real depth - Apparent depth	
= 3 - 2 = 1 cm	½

(b) (A) A ray of light is incident at an angle of incidence, $\frac{A}{2}$ on face AB of a prism ABC of angle of prism A. The ray emerges normally from the opposite face AC. If the refractive index of the material of the prism be n, find the relation between A and n.

CBSE 2025

(A) As refracted ray emerges normally from opposite face AC of the prism,
So $r_2 = 0$

$$r_1 + r_2 = A$$ ½

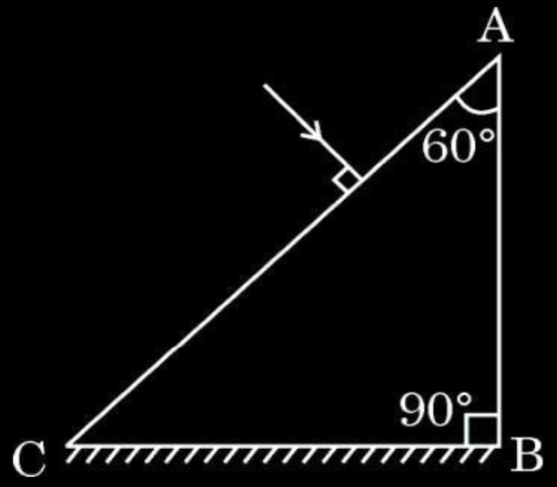
$$r_1 = A$$ ½

$$n = \frac{\text{Sin } i_1}{\text{Sin } r_1} = \frac{\text{Sin } A/2}{\text{Sin } A}$$ ½

$$n = \frac{\text{Sin } A/2}{2\,\text{Sin } A/2\, \text{Cos } A/2}$$ ½

$$n = \frac{1}{2\cos A/2}$$ ½

25. A right-angled prism ABC (refractive index $\sqrt{2}$) is kept on a plane mirror as shown in the figure. A ray of light is incident normally on the face AC.

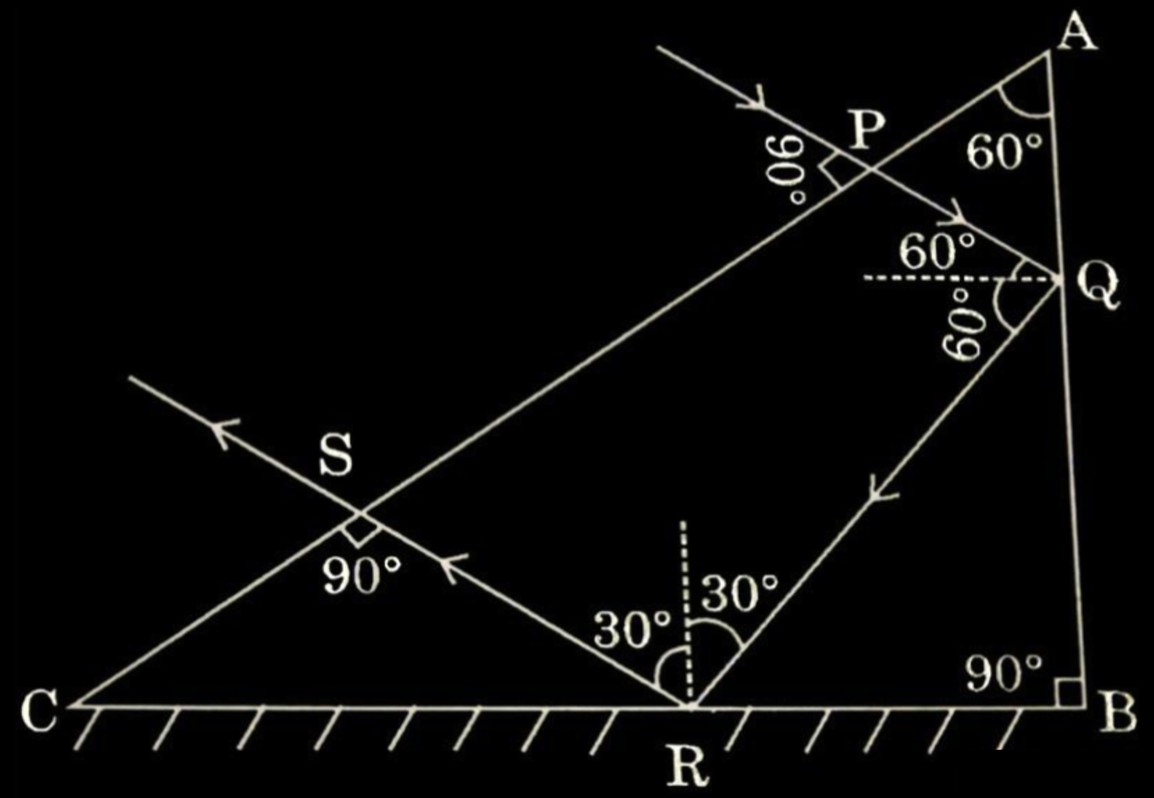


(a) Trace the path of the ray as it passes through the prism.

(b) Find the angle of deviation produced by the prism. CBSE 2025 *3*

(a) $\sin i_c = \frac{1}{\sqrt{2}}$ ½

$i_c = 45^0$

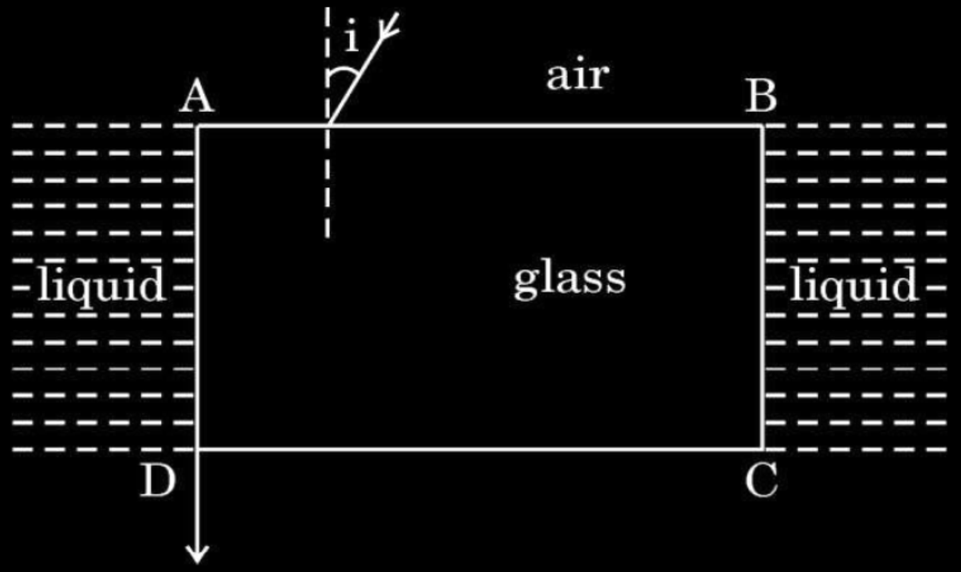


2

(Deduct ½ mark for not showing arrow for the ray)

(b) The angle of deviation is 180° from diagram ½

27. A rectangular glass slab ABCD (refractive index 1·5) is surrounded by a transparent liquid (refractive index 1·25) as shown in the figure. A ray of light is incident on face AB at an angle i such that it is refracted out grazing the face AD. Find the value of angle i. *3*



CBSE 2025

For glass- liquid interface

$$\sin i_c = \frac{1}{n_{21}} \quad \text{½}$$

$$= \frac{1.25}{1.5} \quad \text{½}$$

$$= \frac{5}{6} \quad \text{½}$$

$$i_c + r = 90^0 \quad \text{½}$$

$$\sin r = \sqrt{1-\cos^2 r} = \frac{\sqrt{11}}{6}$$

Since

$$\frac{Sin\ i}{Sin\ r} = n \quad \text{½}$$

Therefore, $Sin\ i = \frac{\sqrt{11}}{4}$ or $i = Sin^{-1}\frac{\sqrt{11}}{4}$ ½

31. (a)

(ii) Calculate the angle of minimum deviation of an equilateral prism. The refractive index of the prism is $\sqrt{3}$. Calculate the angle of incidence for this case of minimum deviation also.

CBSE 2025

(ii)

Angle of minimum deviation

$$\mu = \frac{\text{Sin}\,\frac{(A+\delta_m)}{2}}{\text{Sin}\,\frac{A}{2}}$$

½

$$\sqrt{3} = \frac{\text{Sin}\,\frac{(60^\circ+\delta_m)}{2}}{\text{Sin}\,30}$$

$$\frac{\sqrt{3}}{2} = Sin\,\frac{(A+\delta_m)}{2}$$

$$60^\circ = \frac{(A+\delta_m)}{2}$$

½

$$\delta_m = 60^o$$

Angle of incidence

½

$$i + e = A + \delta$$

$$2i = A + \delta_m$$

$$i = \frac{A+\delta_m}{2}$$

½

$$i = 60^\circ$$

28. (a) When a parallel beam of light enters water surface obliquely at some angle, what is the effect on the width of the beam ? 3

(b) With the help of a ray diagram, show that a straw appears bent when it is partly dipped in water and explain it.

(c) Explain the transmission of optical signal through an optical fibre by a diagram.

CBSE 2025

28. (a) When a parallel beam of light enters water surface obliquely at some angle, what is the effect on the width of the beam ? 3

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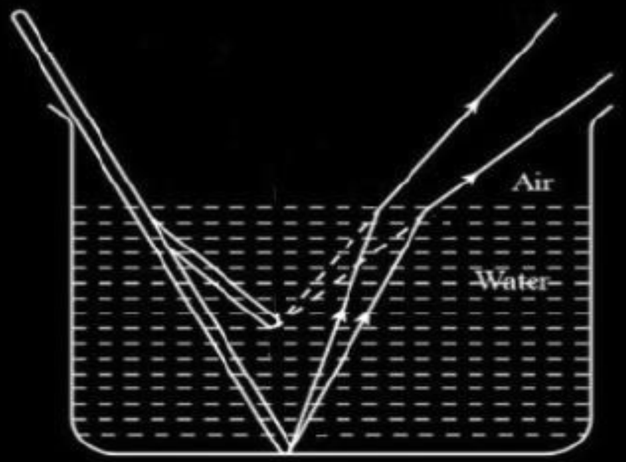
CBSE 2025

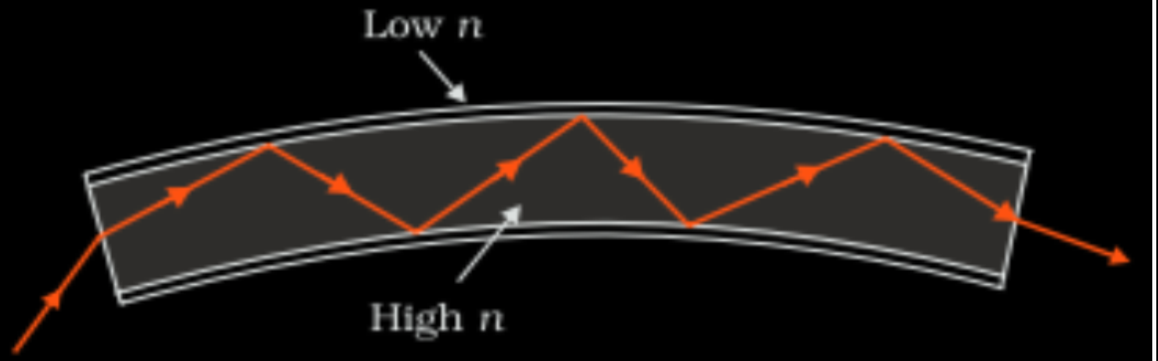
(a) Width of the parallel beam of light increases in water. 1

Alternatively: -
If a student explains using diagram, full credit to be given.

(b) Due to refraction of light, the image of the portion immersed in water appears to be raised.

(c)





32. (a)

(ii) A ray of light is incident on a refracting face AB of a prism ABC at an angle of 45°. The ray emerges from face AC and the angle of deviation is 15°. The angle of prism is 30°. Show that the emergent ray is normal to the face AC from which it emerges out. Find the refraction index of the material of the prism.

CBSE 2025

(ii) For prism; ½

$i + e = A + \delta$ ½

$45^o + e = 30^o + 15^o$

$\therefore e = 0^o$ ½

Hence, $r_2 = 0^o$

$\therefore$ Emergent ray is perpendicular to face AC.

Alternatively: - If the same is shown using diagram full credit to be given.

$r_1 + r_2 = A$ ½

As $r_2 = 0$, hence $r_1 = 30^o$

Refractive index(n) = $\frac{\sin i}{\sin r}$

$= \frac{\sin 45^o}{\sin 30^o}$

½

$n = \sqrt{2}$

9. A tub is filled with a transparent liquid to a height of 30.0 cm. The apparent depth of a coin lying at the bottom of the tub is found to be 16.0 cm. The speed of light in the liquid will be **1**

(A) 1.6×10^8 m s^{-1} (B) 2.0×10^8 m s^{-1}

(C) 3.0×10^8 m s^{-1} (D) 2.5×10^8 m s^{-1}

CBSE 2025

9. A tub is filled with a transparent liquid to a height of 30.0 cm. The apparent depth of a coin lying at the bottom of the tub is found to be 16.0 cm. The speed of light in the liquid will be **1**

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(C) 3.0×10^8 m s^{-1} (D) 2.5×10^8 m s^{-1}

CBSE 2025

(A) 1.6 x 10^8 ms^{-1}

8. A point source is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H. The area of the surface of the liquid through which light from the source can emerge out is **1**

(A) $\frac{\pi H^2}{(n-1)}$

(B) $\frac{\pi H^2}{(n^2-1)}$

(C) $\frac{\pi H^2}{\sqrt{n^2-1}}$

(D) $\frac{\pi H^2}{(n^2+1)}$

CBSE 2025

8. A point source is placed at the bottom of a tank containing a transparent liquid (refractive index n) to a depth H. The area of the surface of the liquid through which light from the source can emerge out is **1**

(A) $\frac{\pi H^2}{(n-1)}$

(B) $\frac{\pi H^2}{(n^2-1)}$

(C) $\frac{\pi H^2}{\sqrt{n^2-1}}$

(D) $\frac{\pi H^2}{(n^2+1)}$

CBSE 2025

(B) $\frac{\pi H^2}{(n^2-1)}$

8. The speed of light in two media '1' and '2' are v_1 and v_2 ($> v_1$) respectively. For a ray of light to undergo total internal reflection at the interface of these two media, it must be incident from **1**

(A) medium '1' and at an angle greater than $\sin^{-1}\left(\frac{v_1}{v_2}\right)$

(B) medium '1' and at an angle greater than $\cos^{-1}\left(\frac{v_1}{v_2}\right)$

(C) medium '2' and at an angle greater than $\sin^{-1}\left(\frac{v_1}{v_2}\right)$

(D) medium '2' and at an angle greater than $\cos^{-1}\left(\frac{v_1}{v_2}\right)$

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(D) medium '2' and at an angle greater than $\cos^{-1}\left(\frac{v_1}{v_2}\right)$

CBSE 2025

(A) medium '1' and at an angle greater that $\sin^{-1}\left(\frac{v_1}{v_2}\right)$

8. A beaker is filled with water (refractive index $\frac{4}{3}$) upto a height H. A coin is placed at its bottom. The depth of the coin, when viewed along the near normal direction, will be **1**

(A) $\frac{H}{4}$

(B) $\frac{3H}{4}$

(C) H

(D) $\frac{4H}{3}$

CBSE 2025

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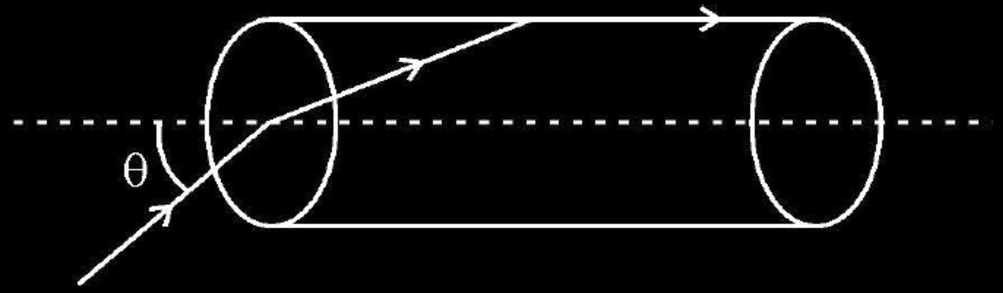
(C) H

(D) $\frac{4H}{3}$

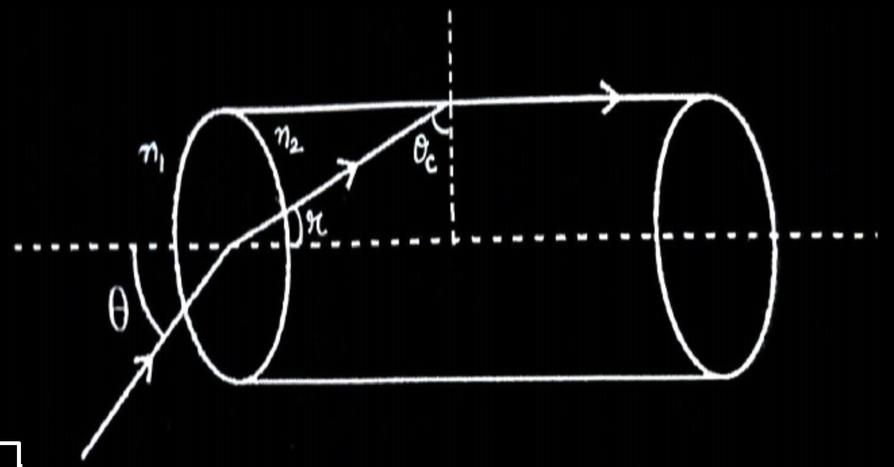
CBSE 2025

(B) $\frac{3H}{4}$

19. A transparent solid cylindrical rod (refractive index $\frac{2}{\sqrt{3}}$) is kept in air. A ray of light incident on its face travels along the surface of the rod, as shown in figure. Calculate the angle θ. **2**



CBSE 2025



For critical Angle

$$\frac{n_2}{n_1} = \frac{1}{\sin\theta_c} \qquad ½$$

$n_1=1$ $n_2=\frac{2}{\sqrt{3}}$ (given)

$$\frac{2}{\sqrt{3}} = \frac{1}{\sin\theta_c}$$

$$\sin\theta_c = \frac{\sqrt{3}}{2}$$

$$\theta_c = 60°$$

$$r = 90 - \theta_c$$

$$= 30° \qquad ½$$

From Snell's law at air rod interface

$$n_1 \sin i = n_2 \sin r \qquad ½$$

$$n_2 = \frac{\sin\theta}{\sin r}$$

$$\frac{2}{\sqrt{3}} = \frac{\sin\theta}{\sin 30^0}$$

$$\frac{2}{\sqrt{3}} \times \frac{1}{2} = \sin\theta$$

$$\frac{1}{\sqrt{3}} = \sin\theta$$

$$\theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right) \qquad ½$$

20. A point light source rests on the bottom of a bucket filled with a liquid of refractive index $\mu = 1.25$ up to height of 10 cm. Calculate :

(a) the critical angle for liquid-air interface

(b) radius of circular light patch formed on the surface by light emerging from the source.

CBSE 2024 C

2

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(a) the critical angle for liquid-air interface

(b) radius of circular light patch formed on the surface by light emerging from the source.

CBSE 2024 C

2

(a) $\sin i_c = \frac{1}{\mu}$ ½

$\sin i_c = \frac{4}{5} \quad (\because \mu = 1.25 = \frac{5}{4})$

$i_c = \sin^{-1}(\frac{4}{5})$ ½

$or\ i_c = 53^o$

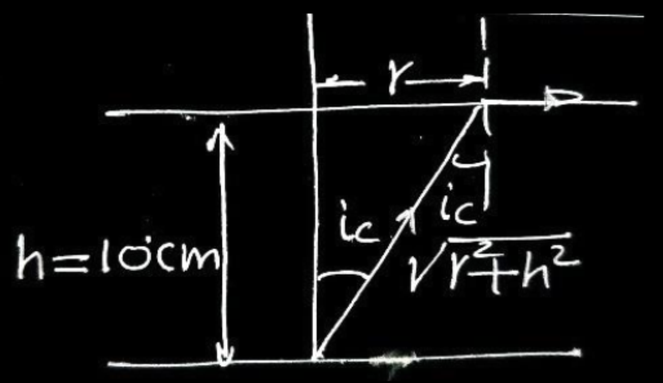
(b) $\sin i_c = \frac{r}{\sqrt{r^2+h^2}}$ ½

$\frac{r^2}{r^2+h^2} = \left(\frac{4}{5}\right)^2$

$25r^2 = 16r^2 + 16h^2$

$9r^2 = 1600$

$r = \frac{40}{3}\text{cm}$ ½



32. (a)

(ii) A ray of light is incident at an angle of 45° at one face of an equilateral triangular prism and passes symmetrically through the prism. Calculate :

(1) the angle of deviation produced by the prism

(2) the refractive index of the material of the prism

CBSE 2024 C

(ii) i+e = D+A

$at \quad D = D_m \quad , \quad i = e$

$2i = D_m + A$

$2 \times 45 = D_m + 60^0$

$D_m = 30^0$

1

$$\mu = \frac{\sin\left(\frac{A + D_m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$

$$\mu = \frac{\sin\left(\frac{60^0 + 30^0}{2}\right)}{\sin\left(\frac{60^0}{2}\right)}$$

$= \sqrt{2}$

1

16. *Assertion (A) :* In optical fibres, the refractive index of cladding must be lesser than that of the core.

Reason (R) : Total internal reflection takes place only when the ray passes from denser to rarer medium.

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(A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

18. (a) Why does a water tank appear shallow ? Explain. Write the relation between real depth and apparent depth of the tank with the refractive index of water. CBSE 2024 C *2*

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It is due to refraction of light. Light coming out of the water surface deviates away from normal and as a result appear to come from a point above the bottom of water tank. Relation between real depth, apparent depth and refractive index of water. $\text{Refractive index} = \frac{\text{Real depth}}{\text{Apparent depth}}$ Alternatively- $\mu_w = \frac{R_d}{A_d}$	1 1

(b) What are optical fibres ? How does a light ray travel through an optical fibre ? Give two applications of optical fibres. CBSE 2024 C *2*

(b) What are optical fibres ? How does a light ray travel through an optical fibre ? Give two applications of optical fibres. CBSE 2024 C *2*

Optical fibres are composite glass/ quartz fibres consisting of a core & a cladding used to transmit audio and video signals through long distances.	½
When a signal in the form of light is directed at one end of the fibre at a suitable angle, it undergoes repeated total internal reflections along the length of the fibre and finally comes out at the other end.	½
Two applications of optical fibres- (1) For communication (2) For decorative purpose (3) For examining internal organs (Any two)	1

29. A prism is an optical medium bounded by three refracting plane surfaces. A ray of light suffers successive refractions on passing through its two surfaces and deviates by a certain angle from its original path. The refractive index of the material of the prism is given by $\mu = \sin\left(\frac{A + \delta m}{2}\right) / \sin\frac{A}{2}$. If the angle of incidence on the second surface is greater than an angle called critical angle, the ray will not be refracted from the second surface and is totally internally reflected.

(i) The critical angle for glass is θ_1 and that for water is θ_2. The critical angle for glass-water surface would be (given ${}_a\mu_g = 1.5$, ${}_a\mu_w = 1.33$)

(A) less than θ_2 (B) between θ_1 and θ_2

(C) greater than θ_2 (D) less than θ_1

CBSE 2024

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(C) greater than θ_2 (D) less than θ_1

CBSE 2024

(C) greater than θ_2

(ii) When a ray of light of wavelength λ and frequency v is refracted into a denser medium

(A) λ and v both increase.

(B) λ increases but v is unchanged.

(C) λ decreases but v is unchanged.

(D) λ and v both decrease.

(ii) When a ray of light of wavelength λ and frequency v is refracted into a denser medium

(A) λ and v both increase.

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(C) λ decreases but v is unchanged.

(D) λ and v both decrease.

(C) λ decreases but ν is unchanged

(iii) (a) The critical angle for a ray of light passing from glass to water is minimum for

(A) red colour

(B) blue colour

(C) yellow colour

(D) violet colour

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(iii) (b) Three beams of red, yellow and violet colours are passed through a prism, one by one under the same condition. When the prism is in the position of minimum deviation, the angles of refraction from the second surface are r_R, r_Y and r_V respectively. Then

(A) $r_V < r_Y < r_R$

(B) $r_Y < r_R < r_V$

(C) $r_R < r_Y < r_V$

(D) $r_R = r_Y = r_V$

(iii) (b) Three beams of red, yellow and violet colours are passed through a prism, one by one under the same condition. When the prism is in the position of minimum deviation, the angles of refraction from the second surface are r_R, r_Y and r_V respectively. Then

(A) $r_V < r_Y < r_R$

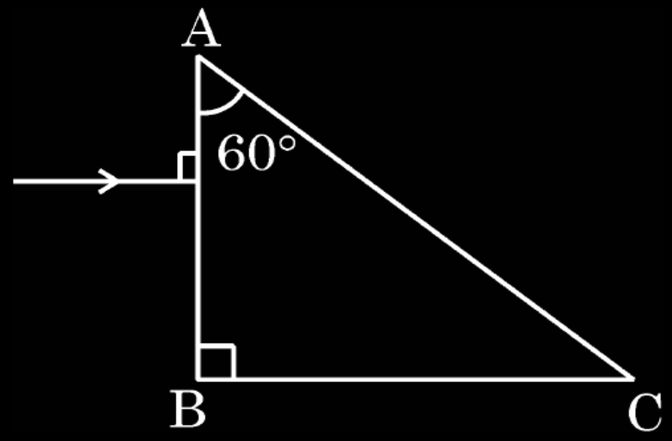
(B) $r_Y < r_R < r_V$

(C) $r_R < r_Y < r_V$

(D) $r_R = r_Y = r_V$

(C) $r_R < r_Y < r_V$

(iv) A ray of light is incident normally on a prism ABC of refractive index $\sqrt{2}$, as shown in figure. After it strikes face AC, it will



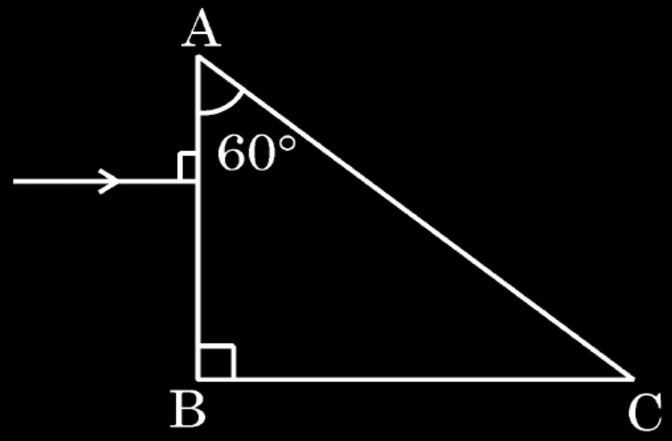
(A) go straight undeviated

(B) just graze along the face AC

(C) refract and go out of the prism

(D) undergo total internal reflection

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(ii) An object is placed between the pole and the focus of a concave mirror. Using mirror formula, prove mathematically that it produces a virtual and an enlarged image.

CBSE 2024

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CBSE 2024

(ii)

$$\frac{1}{f}=\frac{1}{v}+\frac{1}{u}$$

$$v=\frac{uf}{u-f}$$

Following new cartesian sign conversion

$$v=\frac{(-u)(-f)}{-u-(-f)}$$

$$v=\frac{uf}{f-u} \qquad \text{as } f>u$$

1

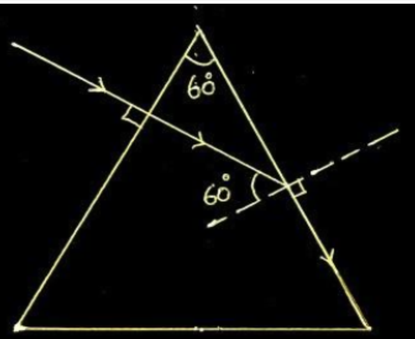
v is +ve, So image is virtual.

$$m=-\frac{v}{u}=\frac{f}{f-u}>1$$ i.e. Enlarged image

1

19. A ray of light is incident normally on one face of an equilateral glass prism of refractive index μ. When the prism is completely immersed in a transparent medium, it is observed that the emergent ray just grazes the adjacent face. Find the refractive index of the medium. CBSE 2024 **2**

19. A ray of light is incident normally on one face of an equilateral glass prism of refractive index μ. When the prism is completely immersed in a transparent medium, it is observed that the emergent ray just grazes the adjacent face. Find the refractive index of the medium. CBSE 2024 **2**



½

From snell's law, $\mu.\sin i = \mu_m.\sin r$ ½

$\mu.\sin 60^0 = \mu_m.\sin 90^0$ ½

$\mu_m = \mu.\frac{\sqrt{3}}{2}$ ½

Alternatively

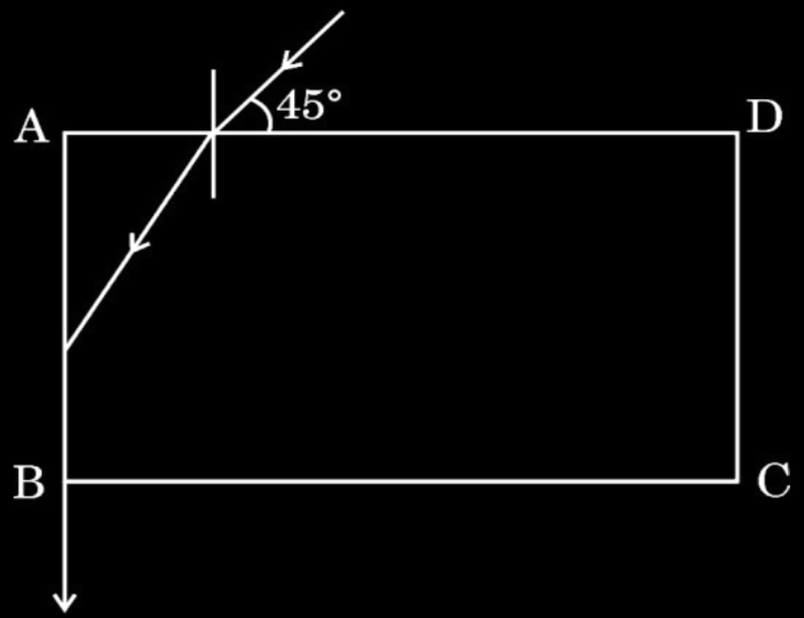
$\mu_{ga} = \frac{1}{\sin C}$ 1

$\frac{\mu}{\mu_m} = \frac{1}{\sin 60^0}$ ½

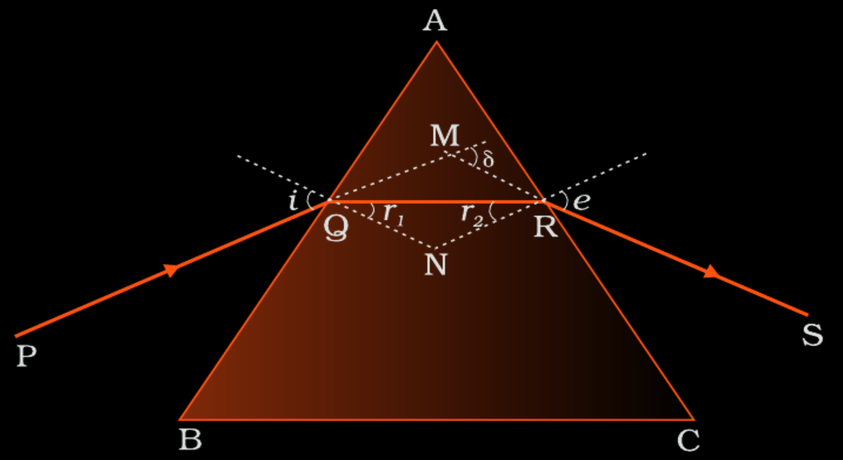
$\mu_m = \frac{\sqrt{3}}{2}\mu$ ½

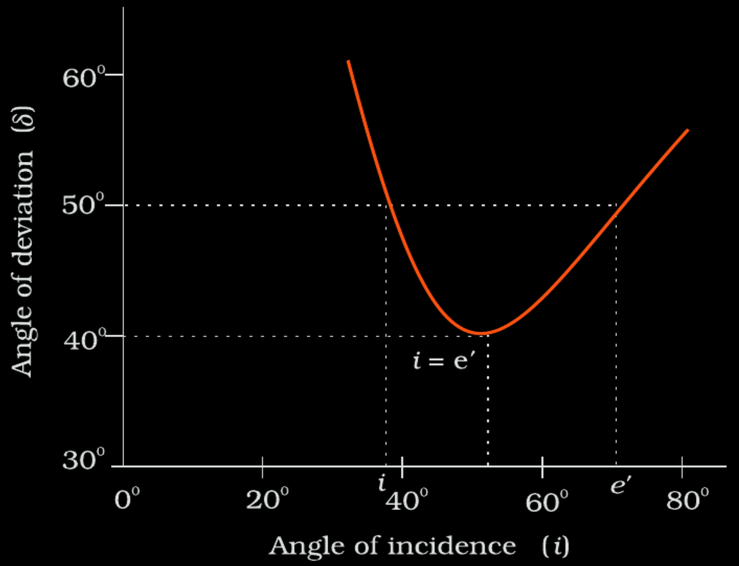
33. (a) (i) Trace the path of a ray of light showing refraction through a triangular prism and hence obtain an expression for angle of deviation (δ) in terms of A, i and e, where symbols have their usual meanings. Draw a graph showing the variation of angle of deviation with the angle of incidence.

(ii) In the figure, a ray of light is incident on a transparent liquid contained in a thin glass box at an angle of 45° with its one face. The emergent ray passes along the face AB. Find the refractive index of the liquid. *5*



CBSE 2024





½

1

For quadrilateral AQNR,

$\angle A + \angle QNR = 180^o$ --- (i)

½

For triangle QNR

$r_1 + r_2 + \angle QNR = 180^o$ ---- (ii)

comparing equation (i) and (ii)

$r_1 + r_2 = A$ ------ (iii)

½

The angle of deviation

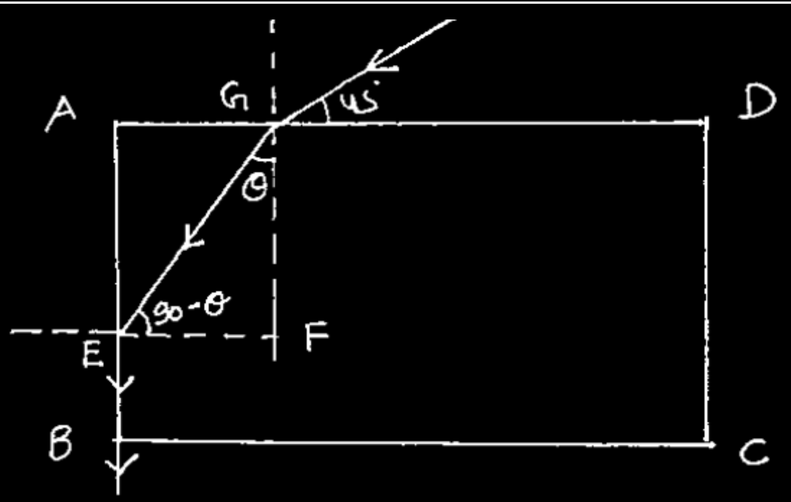
$\delta = (i - r_1) + (e - r_2)$ ------ (iv)

from equation (iii) and (iv)

$\delta = i + e - A$

½

(ii)



$$\frac{\sin 45^0}{\sin \theta} = \mu$$

$$\frac{1}{\sqrt{2}} = \mu \sin \theta$$

½

For second surface,

$$\frac{1}{\sqrt{2}} \frac{\cos \theta}{\sin \theta} = 1$$

½

$$\tan \theta = \frac{1}{\sqrt{2}}$$

½

From the triangle GEF

$$\sin \theta = \frac{1}{\sqrt{3}}$$

$$\mu = \sqrt{\frac{3}{2}}$$

½

29. When a ray of light propagates from a denser medium to a rarer medium, it bends away from the normal. When the incident angle is increased, the refracted ray deviates more from the normal. For a particular angle of incidence in the denser medium, the refracted ray just grazes the interface of the two surfaces. This angle of incidence is called the critical angle for the pair of media involved.

CBSE 2024

(i) For a ray incident at the critical angle, the angle of reflection is : *1*

(A) $0°$ (B) $< 90°$

(C) $> 90°$ (D) $90°$

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CBSE 2024

(i) For a ray incident at the critical angle, the angle of reflection is : *1*

(A) $0°$ (B) $< 90°$

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Since no option is correct, award 1 mark to all students.

(ii) A ray of light of wavelength 600 nm is incident in water $\left(n=\frac{4}{3}\right)$ on the water-air interface at an angle less than the critical angle. The wavelength associated with the refracted ray is : *1*

(A) 400 nm (B) 450 nm

(C) 600 nm (D) 800 nm

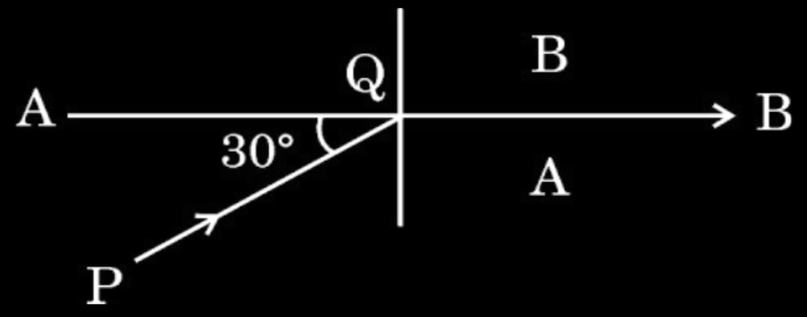
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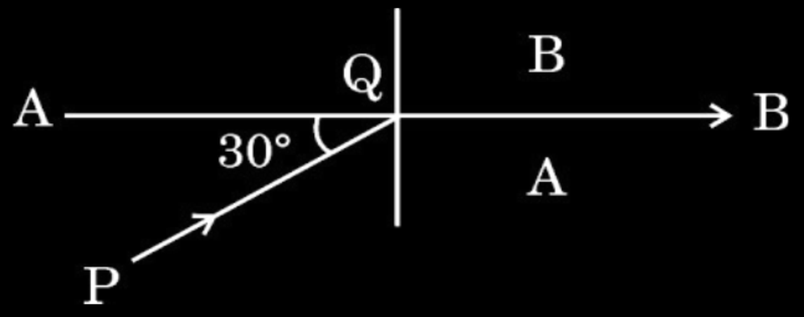
(iii) (a) The interface AB between the two media A and B is shown in the figure. In the denser medium A, the incident ray PQ makes an angle of 30° with the horizontal. The refracted ray is parallel to the interface. The refractive index of medium B w.r.t. medium A is : *1*



(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{\sqrt{5}}{2}$

(C) $\frac{4}{\sqrt{3}}$ (D) $\frac{2}{\sqrt{3}}$

(iii) (a) The interface AB between the two media A and B is shown in the figure. In the denser medium A, the incident ray PQ makes an angle of 30° with the horizontal. The refracted ray is parallel to the interface. The refractive index of medium B w.r.t. medium A is : *1*



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(C) $\frac{4}{\sqrt{3}}$ (D) $\frac{2}{\sqrt{3}}$

(A) $\frac{\sqrt{3}}{2}$

(b) Two media A and B are separated by a plane boundary. The speed of light in medium A and B is 2×10^8 ms^{-1} and $2{\cdot}5 \times 10^8$ ms^{-1} respectively. The critical angle for a ray of light going from medium A to medium B is : *1*

(A) $\sin^{-1} \frac{1}{2}$ (B) $\sin^{-1} \frac{4}{5}$

(C) $\sin^{-1} \frac{3}{5}$ (D) $\sin^{-1} \frac{2}{5}$

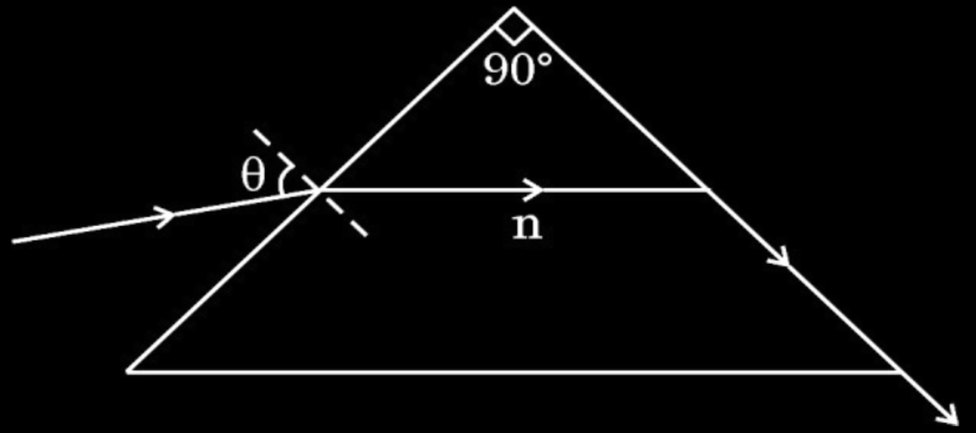
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(A) $\sin^{-1} \frac{1}{2}$ (B) $\sin^{-1} \frac{4}{5}$

(C) $\sin^{-1} \frac{3}{5}$ (D) $\sin^{-1} \frac{2}{5}$

(B) $\sin^{-1}\left(\frac{4}{5}\right)$

(iv) The figure shows the path of a light ray through a triangular prism. In this phenomenon, the angle θ is given by : *1*



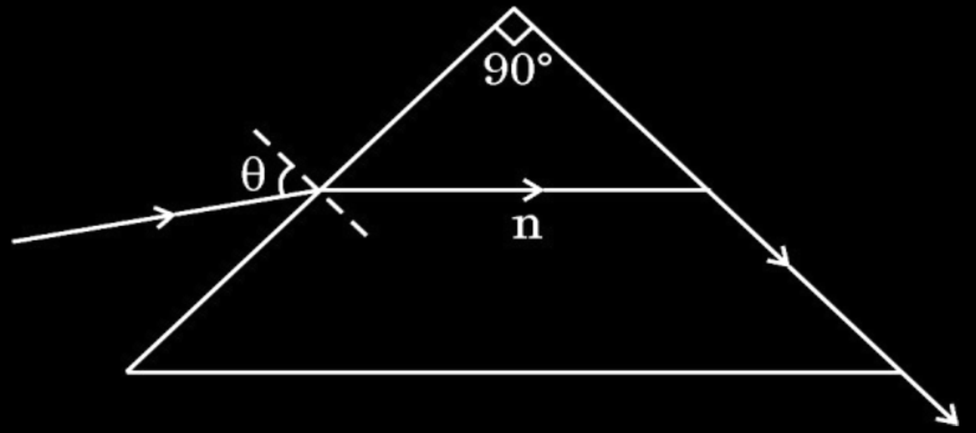
(A) $\sin^{-1}\sqrt{n^2-1}$

(B) $\sin^{-1}(n^2-1)$

(C) $\sin^{-1}\left[\frac{1}{\sqrt{n^2-1}}\right]$

(D) $\sin^{-1}\left[\frac{1}{(n^2-1)}\right]$

(iv) The figure shows the path of a light ray through a triangular prism. In this phenomenon, the angle θ is given by : *1*



(A) $\sin^{-1}\sqrt{n^2-1}$

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(C) $\sin^{-1}\left[\frac{1}{\sqrt{n^2-1}}\right]$

(D) $\sin^{-1}\left[\frac{1}{(n^2-1)}\right]$

(A) $\sin^{-1}\sqrt{n^2-1}$

33. (a) (i) A ray of light passes through a triangular prism. Show graphically, how the angle of deviation varies with the angle of incidence ? Hence define the angle of minimum deviation.

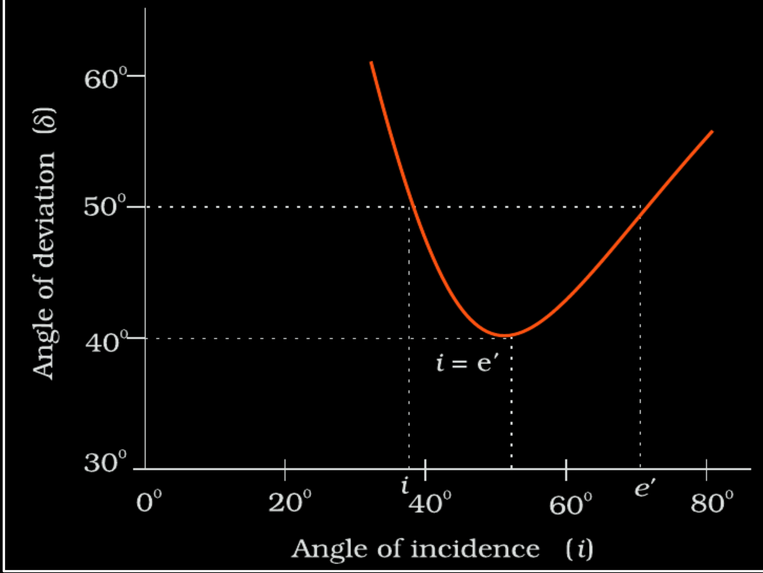
(ii) A ray of light is incident normally on a refracting face of a prism of prism angle A and suffers a deviation of angle δ. Prove that the refractive index n of the material of the prism is given by $n = \frac{\sin (A + \delta)}{\sin A}$.

(iii) The refractive index of the material of a prism is $\sqrt{2}$. If the refracting angle of the prism is 60°, find the

(1) Angle of minimum deviation, and

(2) Angle of incidence.

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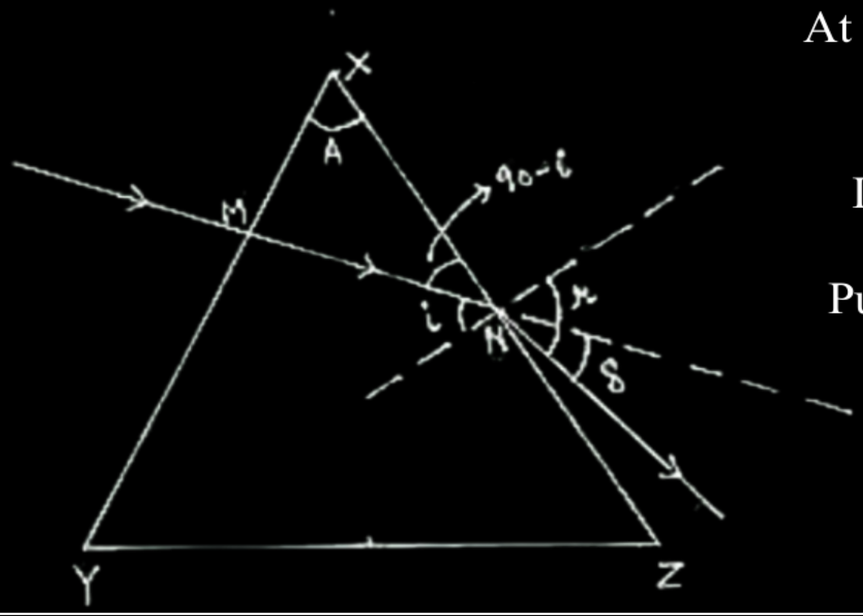
Minimum deviation angle is defined as the angle at which angle of incidence is equal to the angle of emergence.

Alternatively

At minimum deviation refracted ray inside the prism becomes parallel to the base of the prism.

1

1



At the face XZ :-

$\mu \sin i = 1 \times \sin r$ ----- (1)

$r = i + \delta$ [from diagram] ----- (2)

In ΔXMN ; $A + (90 - i) + 90 = 180$

$\Rightarrow A = i$ ----- (3)

Putting eq. (3) & (2) in eq. (1)

$\mu \sin A = \sin (A + \delta)$

$$\mu = \frac{\sin (A + \delta)}{\sin A}$$

½

½

(iii)

(1) $$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin\frac{A}{2}}$$

$$\sqrt{2} = \frac{\sin\left(\frac{60+\delta_m}{2}\right)}{\sin 30^\circ}$$ ½

$$\Rightarrow \sin\left(\frac{60+\delta_m}{2}\right) = \frac{1}{\sqrt{2}} = \sin 45^\circ$$

$$\frac{60+\delta_m}{2} = 45^\circ \Rightarrow \delta_m = 30^\circ$$ ½

(2) $$i = \frac{A+\delta_m}{2}$$ ½

$$\Rightarrow i = \frac{60+30}{2}$$

$$i = 45^\circ$$ ½

16. *Assertion (A) :* Signals through an optical fibre can be transmitted without any appreciable loss of energy.

Reason (R) : An optical fibre works on the principle of total internal reflection.

CBSE 2024

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CBSE 2024

(A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of the Assertion (A).

29. In a homogeneous medium, light travels along a straight line path. But when a ray of light travels from one medium to another, it deviates from its initial path. This bending of light is called refraction of light. The extent to which a ray deviates in medium 2 from its initial path in medium 1, is determined by the refractive index of medium 2 with respect to medium 1. The refractive index is also related to the ratio of speeds of light in the two media.

CBSE 2024

The phenomenon of total internal reflection occurs when light passes from a denser medium into a rarer medium and the angle of incidence exceeds the critical angle for the given pair of media.

(i) A ray of monochromatic light of wavelength λ and speed v travels from air into a glass slab. Inside the glass slab, *1*

(A) both λ and v decrease

(B) both λ and v increase

(C) λ decreases and v increases

(D) λ increases and v decreases

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CBSE 2024

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(i) A ray of monochromatic light of wavelength λ and speed v travels from air into a glass slab. Inside the glass slab, *1*

(A) both λ and v decrease

(B) both λ and v increase

(C) λ decreases and v increases

(D) λ increases and v decreases

(A) both λ & v decrease

(ii) (a) A ray of light travelling from air into a medium, is incident on the interface at an angle of 45°. If the ray deviates by 15° from its initial path, the refractive index of the medium is : *1*

(A) 2 (B) $\sqrt{2}$

(C) $2\sqrt{2}$ (D) $\sqrt{3}$

(ii) (a) A ray of light travelling from air into a medium, is incident on the interface at an angle of 45°. If the ray deviates by 15° from its initial path, the refractive index of the medium is : *1*

(A) 2 (B) $\sqrt{2}$

(C) $2\sqrt{2}$ (D) $\sqrt{3}$

(B) $\sqrt{2}$

(ii) (b) A point source of light is kept at the bottom of a tank filled with a transparent liquid up to a height H. If the radius of the circular patch on the surface from which light can come out is $\frac{H}{\sqrt{3}}$, the refractive index of the liquid is close to : *1*

(A) 1·3 (B) 1·5

(C) 1·7 (D) 2·0

(ii) (b) A point source of light is kept at the bottom of a tank filled with a transparent liquid up to a height H. If the radius of the circular patch on the surface from which light can come out is $\frac{H}{\sqrt{3}}$, the refractive index of the liquid is close to : *1*

(A) 1·3 (B) 1·5

(C) 1·7 (D) 2·0

(D) 2

(iii) In an optical fiber, the refractive indices of the core and cladding are μ_1 and μ_2 respectively. Then : *1*

(A) $\mu_1 = \mu_2$

(B) $\mu_1 < \mu_2$

(C) μ_1 is slightly greater than μ_2

(D) μ_1 is large compared to μ_2

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(C) μ_1 is slightly greater than μ_2

(iv) Light of wavelength λ travelling with speed c in air, enters a medium of refractive index μ. The frequency of light in the medium is : *1*

(A) $\frac{c}{\mu\lambda}$

(B) $\frac{\mu c}{\lambda}$

(C) $\frac{\mu\lambda}{c}$

(D) $\frac{c}{\lambda}$

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(A) $\frac{c}{\mu\lambda}$

(B) $\frac{\mu c}{\lambda}$

(C) $\frac{\mu\lambda}{c}$

(D) $\frac{c}{\lambda}$

(D) $\frac{c}{\lambda}$

35. When a ray of light passes from a denser to a rarer medium, it bends away from the normal. If we go on increasing the angle of incidence in denser medium, then for a particular value of angle of incidence, the incident ray is refracted along the interface of the two media. This angle is called critical angle (i_c). If incident angle (i) is greater than critical angle (i_c), then the ray of light is reflected back in the same denser medium. This phenomenon is called total internal reflection of light.
Based on the above facts, answer the following questions :

(i) If the refractive index of the denser medium increases, how will the corresponding value of critical angle for the pair of mediums change ? *1*

(ii) Explain how this phenomenon can be used in a prism to deviate a ray of light by 90°. *1*

CBSE 2023 C

(iii) What is an optical fibre ? Write its two uses. *2*

OR

(iii) A ray of light travels from an optically denser medium to a rarer medium and is incident at critical angle (i_c). Find the angle of deviation for this ray. *2*

CBSE 2023 C

(i) $\sin i_c = \frac{1}{\mu}$. If μ of denser medium increases, critical angle (i_c) decreases.	1
(ii) For this we take isosceles right angled prism. We allow incident light to fall normally on one of the two mutually perpendicular faces. It will deviated by 90° after total internal reflection	1
(iii) Optical fibres are very fine fibres made of glass. Through these fibres optical signals can be transmitted without appreciable loss of energy, using the phenomena of total internal reflection. Uses: Endoscopy Communication	1 ½ + ½
OR	
(iii) Angle of deviation = $90^o - i_c$	2

35. A prism is a solid transparent medium bounded by three rectangular faces with triangular base and top. A ray of light incident at angle i on one face of a prism suffers two refractions on passing through a prism. Hence it deviates through a certain angle δ from its original path. The angle of deviation becomes minimum ($\delta = \delta_m$) for a certain value of angle i. In such a condition, the refracted ray inside the prism becomes parallel to its base. An expression for refractive index μ of the material of the prism can be obtained in terms of angle A and angle δ_m.

(a) Show in a figure the variation of angle δ with angle of incidence i. *1*

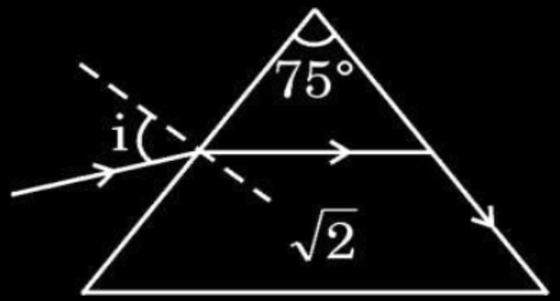
(b) Show that for a prism of small angle A, the refractive index μ of its material can be written as $\mu = 1 + \frac{\delta_m}{A}$. *1*

CBSE 2023 C

(c) A ray of light passes through an equilateral prism such that both the angle of incidence and the angle of emergence are equal to the angle of prism A. Find the refractive index of the material of the prism, in terms of A. *2*

OR

(c) A ray of light passes through a prism of angle 75°, as shown in the figure. The refractive index of the material of the prism, with respect to its surrounding is $\sqrt{2}$. Find the angle of incidence i. *2*



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(b)

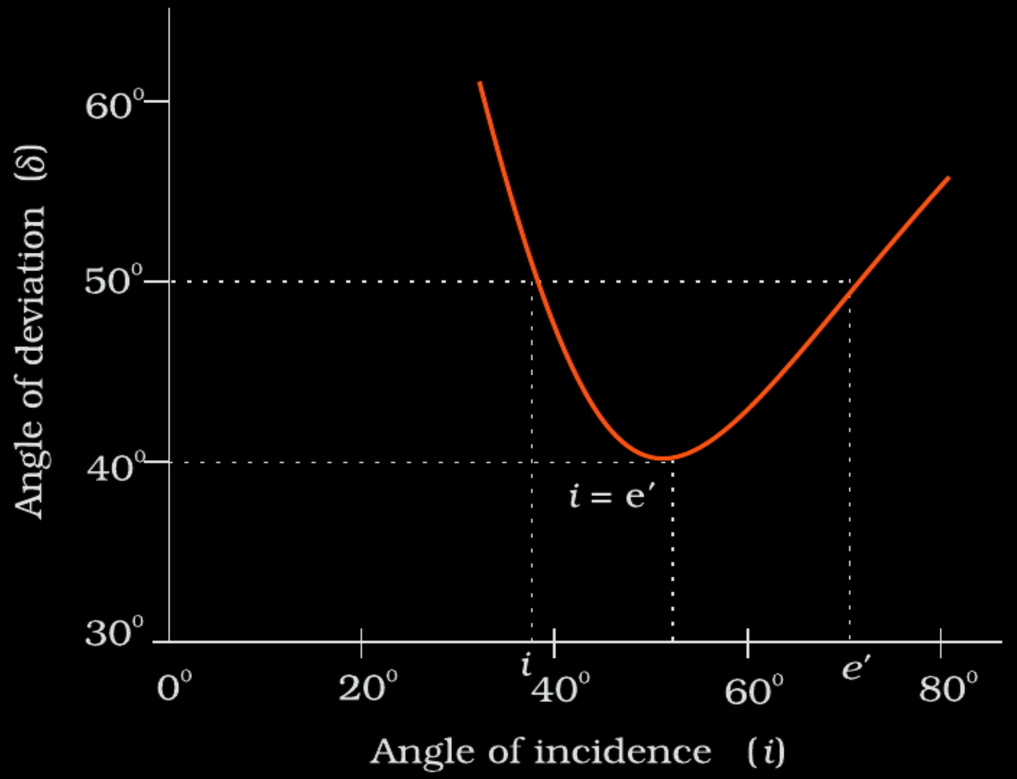
$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin A/2}$$

For small angle

$$\mu = \frac{\frac{A+\delta_m}{2}}{A/2}$$

$$\mu = \frac{A+\delta_m}{A}$$

$$\mu = 1+\frac{\delta_m}{A}$$



½

1

½

(c) $i+e=A$

$$r = \frac{A}{2}$$

$$\mu = \frac{\sin i}{\sin r}$$

$$\mu = \frac{\sin A}{\sin A/2}$$

$$\mu = \frac{2\sin(A/2)\cos(A/2)}{\sin(A/2)} = 2\cos(A/2)$$

½

½

½

½

$$\frac{\sin i}{\sin r} = \sqrt{2}$$

$$\frac{\sin r_2}{\sin 90^0} = \frac{1}{\sqrt{2}}$$ ½

$$\sin r_2 = \frac{1}{\sqrt{2}}$$

$$r_2 = 45^0$$ ½

$$r_1 + r_2 = A$$

$$r_1 + 45^0 = 75^0$$

$$r_1 = 30^0$$ ½

$$\frac{\sin i}{\sin r_1} = \sqrt{2}$$

$$\frac{\sin i}{\sin 30^0} = \sqrt{2}$$

$$\sin i = \sqrt{2} \times \frac{1}{2}$$ ½

$$\sin i = \frac{1}{\sqrt{2}}$$

$$i = 45^0$$

19. A bright point source is placed at the bottom of a tank containing a transparent liquid (refractive index μ) to a height H. Let r be the radius of the circular area of the liquid surface through which light can emerge out. Find the ratio (r/H).

CBSE 2023 C

2

19. A bright point source is placed at the bottom of a tank containing a transparent liquid (refractive index μ) to a height H. Let r be the radius of the circular area of the liquid surface through which light can emerge out. Find the ratio (r/H). CBSE 2023 C *2*

$\text{Sin } i_c = \frac{1}{\mu}$	½
$\cos i_c = \sqrt{1-\sin^2 i} = \frac{\sqrt{\mu^2-1}}{\mu}$	½
$\tan i_c = \frac{1}{\sqrt{\mu^2-1}}$	½
$\frac{r}{H} = \frac{1}{\sqrt{\mu^2-1}}$	½

1. A small object lies at the bottom of a vessel filled with water (refractive index 4/3) up to a height H. When viewed from a point above the surface of water, the object appears raised by n percent of H. The value of n is :

(a) 15 (b) 20

(c) 25 (d) 33

CBSE 2023 C

1. A small object lies at the bottom of a vessel filled with water (refractive index 4/3) up to a height H. When viewed from a point above the surface of water, the object appears raised by n percent of H. The value of n is :

(a) 15 (b) 20

(c) 25 (d) 33

CBSE 2023 C

(c) 25

(b) (i) Draw a ray diagram showing refraction of light through a prism of angle A and obtain the relation between μ, A and the angle of minimum deviation δ_m.

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From the figure

$\angle A+\angle QNR= 180^o$

From $\triangle QNR$

$r_1+ r_2+\angle QNR=180^o$

comparing the above equations

$r_1+ r_2=A$ ------- (1) ½

$\delta = (i-r_1)+(e - r_2)$

$\delta = i + e - A$ ------- (2) ½

At the minimum derivation D_m

$\delta = D_m$, i=e and $r_1 = r_2$

From equation (1) ½

$2r= A$

$r= \frac{A}{2}$

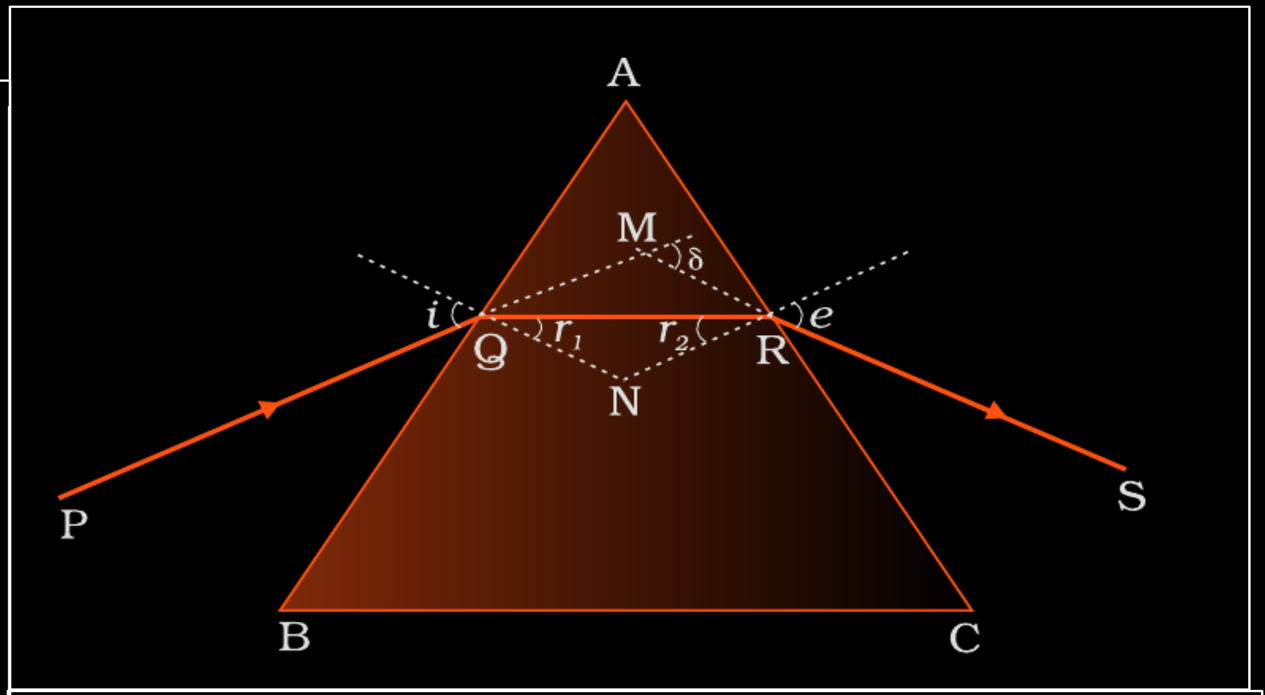
From equation (2)

$D_m =2i-A$

$i= \frac{A+D_m}{2}$

The refractive index of the prism is

$n_{21} = \frac{\sin i}{\sin r} = \frac{\sin[\frac{(A+D_m)}{2}]}{\sin[A/2]}$ ½



16. *Assertion (A) :* The speed of light decreases when it passes from air into a denser medium.

Reason (R) : The speed of light in a denser medium is given by $v = \frac{c}{\mu}$, where c is the speed of light in air and μ is the refractive index of the denser medium.

CBSE 2023

16. *Assertion (A) :* The speed of light decreases when it passes from air into a denser medium.

Reason (R) : The speed of light in a denser medium is given by $v = \frac{c}{\mu}$, where c is the speed of light in air and μ is the refractive index of the denser medium.

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(a) [A]- True [R]- True and correct explanation

21. A glass jug is filled with water to a height of 9·6 cm. The depth of a coin lying on its bottom as measured by a microscope is 7·2 cm. Find the refractive index of the water and the speed of light in water. CBSE 2023 *2*

21. A glass jug is filled with water to a height of 9·6 cm. The depth of a coin lying on its bottom as measured by a microscope is 7·2 cm. Find the refractive index of the water and the speed of light in water. CBSE 2023 *2*

$\mu = \frac{real\ depth}{\text{apparent } depth}$	½
$\mu = \frac{9.6}{7.2} = \frac{4}{3}$	½
$v = \frac{c}{\mu}$	½
$= \frac{3\times10^8}{4}\times3 = 2.25\times10^8\ ms^{-1}$	½

33. (a) (i) What is meant by total internal reflection of light ? Write the two conditions necessary for this phenomenon to occur. Briefly explain one of its technological applications.

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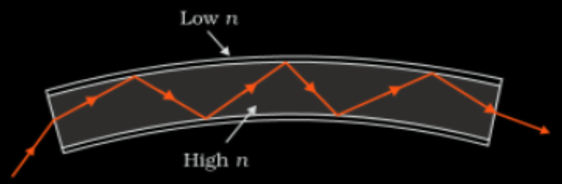
(i) Total internal reflection is the phenomenon in which light waves reflected back in the same medium while travelling from denser to rarer medium without undergoing any refraction.	1
Two conditions:- (i) The incident ray must propagate from denser to rarer medium. (ii) The incident angle must be greater than critical angle.	½+ ½
Application: Telecommunication, medical application, data transmission any one or any other relevant technological application. Through optical fibre a signal goes from its one end to another by multiple total internal reflections without any loss in signal.	1

(b) (i) Explain the working principle of an optical fibre with the help of a diagram. Mention one use of a light pipe.

(ii) A ray of light is incident at an angle of 60° on one face of a prism with the prism angle $A = 60°$. The ray passes symmetrically through the prism. Find the angle of minimum deviation (δ_m) and refractive index of the material of the prism. If the prism is immersed in water, how will δ_m be affected ? Justify your answer. CBSE 2023 *5*

(i) Working Principle:

Optical fibre uses the optical principle of total internal reflection to capture the light transmitted in an optical fibre and confine the light to the core of the fibre. **1**



½

Uses : Transmission of audio and video signal / Examination of internal organs / Endoscopy ½

(ii) $\delta_m = i + e - A$ ½

$\delta_m = 2i - A$

$\delta_m = 60^\circ$ ½

Refractive Index

$$\mu = \frac{\sin\left(\frac{A+\delta_m}{2}\right)}{\sin A/2}$$ ½

$$\mu = \frac{\sin\frac{120^o}{2}}{\sin\frac{60^0}{2}}$$

$$\mu = \frac{\sin 60^0}{\sin 30^0} = \frac{\sqrt{3}/2}{1/2}$$

$\mu = \sqrt{3}$ ½

If the prism is immersed in water μ decreases and consequently angle of minimum deviation decreases. Since δ_m depends on μ through equation given above. 1

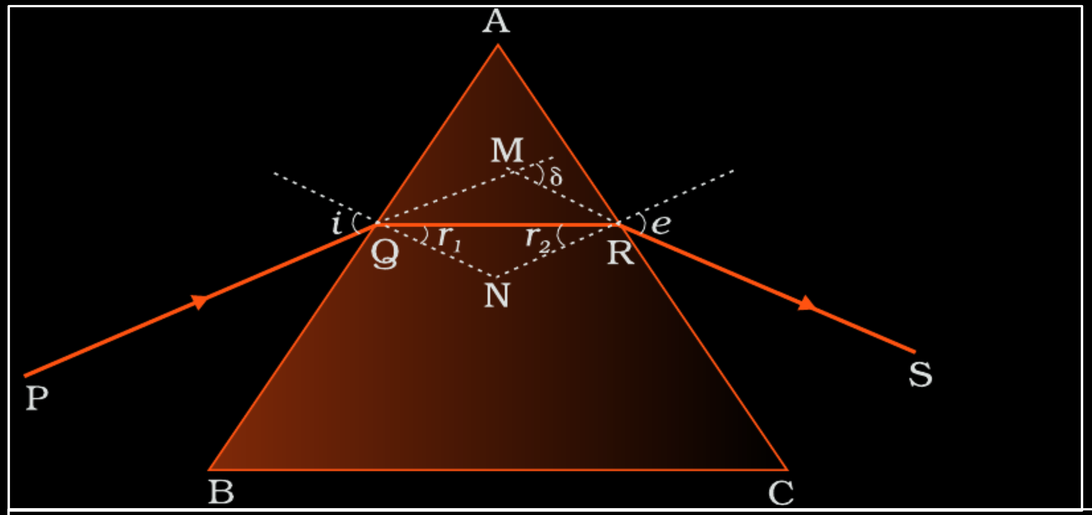
24. The refractive indices of two media A and B are 2 and $\sqrt{2}$ respectively. What is the critical angle for their interface ? CBSE 2023 *2*

24. The refractive indices of two media A and B are 2 and $\sqrt{2}$ respectively. What is the critical angle for their interface ? CBSE 2023 *2*

From Snell 's law:-	
$\mu_A \sin i_c = \mu_B \text{Sin}\, 90°$	½
$2 \times \sin i_c = \sqrt{2} \times 1$	½
$\sin i_c = \frac{1}{\sqrt{2}}$	½
$i_c = 45°$	½

(b) (i) Draw a ray diagram showing refraction of a ray of light through a triangular glass prism. Hence, obtain the relation for the refractive index (μ) in terms of angle of prism (A) and angle of minimum deviation (δ_m).

CBSE 2023



In the quadrilateral AQNR, two of the angles (at the vertices Q and R) are right angles. Therefore, the sum of the other angles of the quadrilateral is 180º.

$\angle A + \angle QNR = 180^\circ$

From the triangle QNR, $r_1 + r_2 + \angle QNR = 180^\circ$

Comparing these two equations, we get

$r_1 + r_2 = A$ -------(i)

The total deviation δ is the sum of deviations at the two faces,

$\delta = (i - r_1) + (e - r_2)$ that is, $\delta = i + e - A$ -------(ii)

When $\delta = \delta_m$; $i = e$ & $r_1 = r_2$

From (i); $2r = A$ or $r = A/2$

From (ii); $\delta_m = 2i - A$ or $i = \frac{A + \delta_m}{2}$

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin\left(\frac{A + \delta_m}{2}\right)}{\sin A/2}$$

½

½

½

½

13. Which of the following pairs of media has the least value of critical angle ?

(a) Glass to air

(b) Glass to water

(c) Diamond to water

(d) Diamond to air

CBSE 2023

13. Which of the following pairs of media has the least value of critical angle ?

(a) Glass to air

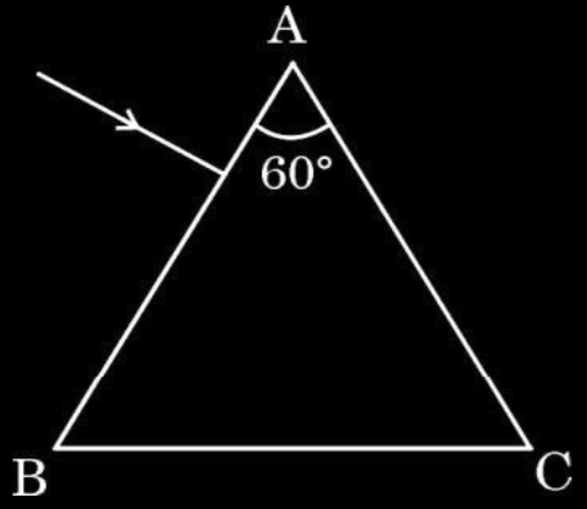
(b) Glass to water

(c) Diamond to water

(d) Diamond to air

CBSE 2023

(d) Diamond to air



35. Strontium titanate is a rare oxide — a natural mineral found in Siberia. It is used as a substitute for diamond because its refractive index and critical angle are 2·41 and 24·5°, respectively, which are approximately equal to the refractive index and critical angle of diamond. It has all the properties of diamond. Even an expert jeweller is unable to differentiate between diamond and strontium titanate. A ray of light is incident normally on one face of an equilateral triangular prism ABC made of strontium titanate.

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Answer the following questions based on the above :

(a) Trace the path of the ray showing its passage through the prism. *1*

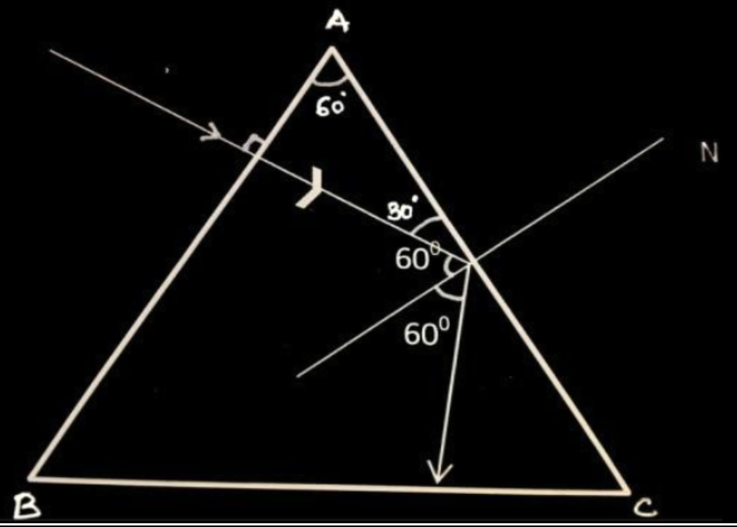
(b) Find the velocity of light through the prism. *1*

(c) Briefly explain two applications of total internal reflection. *2*

OR

(c) Define total internal reflection of light. Give two conditions for it. *2*

a)



From fig. angle of incidence on second face $\angle i = 60^0$
critical angle $\angle i_c = 24.5^o$ — 1

$(\angle i) > (\angle i_c)$

$\therefore$ TIR takes place

b) $n = \frac{c}{v}$

$v = \frac{c}{n} = \frac{3\times10^8}{2.41} = 1.24 \times 10^8$ m/s — 1

c) Optical Fibre / Brilliance of diamond / mirage (any two) — 1+1
Note: Give full credit if students mention the names of applications only.

OR

c) When light travels from optically denser medium to rarer medium at an interface and gets reflected back into the same medium the phenomenon is called as total internal reflection. — 1

Conditions for TIR
1. Light must travel from optically denser medium to rarer medium.
2. Angle of incidence at the interface must be greater than the critical angle for the pair of media. — ½ + ½

12. A ray of light travels a distance of 12.0 m in a transparent sheet in 60 ns. The refractive index of the sheet is **1**

(a) 1.33 (b) 1.50

(c) 1.65 (d) 1.75

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12. A ray of light travels a distance of 12.0 m in a transparent sheet in 60 ns. The refractive index of the sheet is **1**

(a) 1.33 (b) 1.50

(c) 1.65 (d) 1.75

CBSE 2023

(b) 1.50

14. A ray of monochromatic light propagating in air, is incident on the surface of water. Which of the following will be the same for the reflected and refracted rays ? **1**

(a) Energy carried
(b) Speed
(c) Frequency
(d) Wavelength

CBSE 2023

14. A ray of monochromatic light propagating in air, is incident on the surface of water. Which of the following will be the same for the reflected and refracted rays ? **1**

(a) Energy carried
(b) Speed
(c) Frequency
(d) Wavelength

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(c) Frequency

26. A ray of light is refracted by a glass prism. Obtain an expression for the refractive index of the glass in terms of the angle of prism A and the angle of minimum deviation δm.

CBSE 2023 **3**

26. A ray of light is refracted by a glass prism. Obtain an expression for the refractive index of the glass in terms of the angle of prism A and the angle of minimum deviation δm.

CBSE 2023 **3**

At minimum deviation , a ray of light passes through a prism symmetrically

i.e $\angle i = \angle e$, $\angle r_1 = \angle r_2 = \angle r$ ½

We know that $i + e = A + \delta$ and $r_1 + r_2 = A$ ½

$\therefore \quad i + i = A + \delta m$

$$i = \frac{A+\delta m}{2}$$

And also

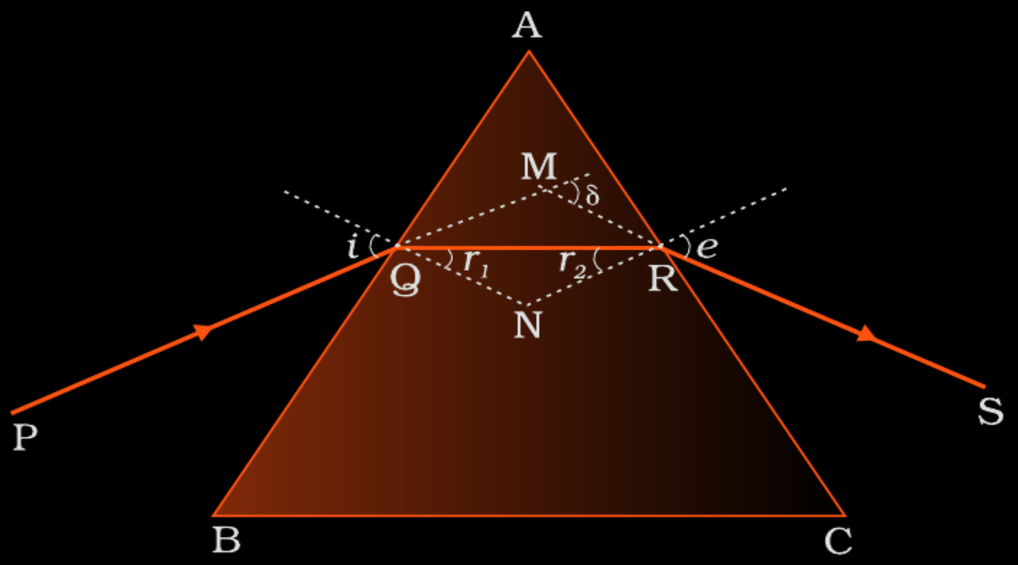
$r + r = A$

$$\Rightarrow r = \frac{A}{2}$$ ½

Using Snell's law

$$\mu = \frac{\sin i}{\sin r}$$ ½

$$\mu = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\left(\frac{A}{2}\right)}$$ ½



14. A ray of light of wavelength 600 nm propagates from air into a medium. If its wavelength in the medium becomes 400 nm, the refractive index of the medium is **1**

(a) 1.4 (b) 1.5

(c) 1.6 (d) 1.8

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14. A ray of light of wavelength 600 nm propagates from air into a medium. If its wavelength in the medium becomes 400 nm, the refractive index of the medium is **1**

(a) 1.4 (b) 1.5

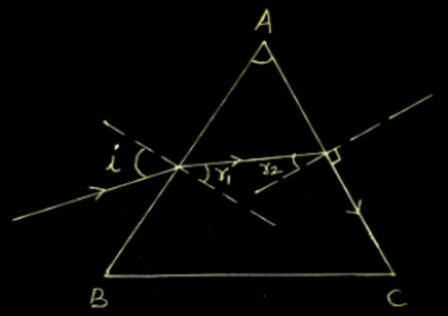
(c) 1.6 (d) 1.8

CBSE 2023

(b) 1.5

27. A ray of light is incident on a glass prism of refractive index μ and refracting angle A. If it just suffers total internal reflection at the other face, obtain a relation between the angle of incidence, angle of prism and critical angle. CBSE 2023 **3**

27. A ray of light is incident on a glass prism of refractive index μ and refracting angle A. If it just suffers total internal reflection at the other face, obtain a relation between the angle of incidence, angle of prism and critical angle. CBSE 2023 **3**



1

∵ the light ray just suffers total internal reflection

∴ $\angle r_2 = \angle i_c$ ½

$$A = r_1 + r_2$$

$$A = r_1 + i_c$$ ½

From Snell's law at surface AB

$$\sin i = \mu . \sin r_1$$ ½

$$\sin i = \mu . \sin(A - i_c)$$ ½

9. A beam of light travels from air into a medium. Its speed and wavelength in the medium are 1.5×10^8 ms^{-1} and 230 nm respectively. The wavelength of light in air will be **1**

(a) 230 nm (b) 345 nm

(c) 460 nm (d) 690 nm

CBSE 2023

9. A beam of light travels from air into a medium. Its speed and wavelength in the medium are 1.5×10^8 ms^{-1} and 230 nm respectively. The wavelength of light in air will be **1**

(a) 230 nm (b) 345 nm

(c) 460 nm (d) 690 nm

CBSE 2023

(c)
460 nm

29. Define critical angle for a given pair of media and total internal reflection. Obtain the relation between the critical angle and refractive index of the medium. **3**

CBSE 2023

29. Define critical angle for a given pair of media and total internal reflection. Obtain the relation between the critical angle and refractive index of the medium. **3**

CBSE 2023

Critical angle - When a ray of light passes from a denser to a rarer medium, the value of angle of incidence for which the angle of refraction becomes 90^0 is called critical angle for that pair of media. 1

Total internal Reflection – When a ray of light passes from a denser to rarer medium and the angle of incidence exceeds the critical angle for pair of media, the ray under goes reflection. This is called total internal reflection. 1

From Snell's law $\dfrac{\sin i}{\sin r} = \mu_{rd}$ ½

When angle of incidence is equal to critical angle ($\angle i = \angle i_c$) , $\angle r = 90^0$

$\therefore \quad \dfrac{\sin i_c}{\sin 90^0} = \dfrac{1}{\mu_{dr}}$

$\therefore \quad \mu_{dr} = \dfrac{1}{\sin i_c}$ ½

10. (a) For an equilateral triangular prism, the angle of minimum deviation is 30°. Find the refractive index of the material of the prism.

CBSE 2022 C

(b) The angle of minimum deviation for a prism is found to be equal to the angle of the prism (A). Show that the refractive index of the material of the prism, $\mu = 2 \cos\left(\frac{A}{2}\right)$. *3*

10. (a) For an equilateral triangular prism, the angle of minimum deviation is 30°. Find the refractive index of the material of the prism.

CBSE 2022 C

(b) The angle of minimum deviation for a prism is found to be equal to the angle of the prism (A). Show that the refractive index of the material of the prism, $\mu = 2\cos\left(\frac{A}{2}\right)$. *3*

(a) $\mu = \dfrac{\sin\left(\dfrac{A+D_m}{2}\right)}{\sin A/2}$	½
$= \dfrac{\sin\left(\dfrac{60^o+30^o}{2}\right)}{\sin 60/2}$	½
$= \dfrac{\sin 45^o}{\sin 30^o} = \sqrt{2}$	½
(b) $\mu = \dfrac{\sin\left(\dfrac{A+D_m}{2}\right)}{\sin A/2}$	½
$= \dfrac{\sin\left(\dfrac{A+A}{2}\right)}{\sin A/2} = \dfrac{\sin A}{\sin A/2}$	½
$\mu = \dfrac{2\sin A/2\cos A/2}{\sin A/2} = 2\cos A/2$	½

(b) Light travels from a rarer to a denser medium. How will

(i) the energy carried, and

(ii) the wavelength of the light wave,

be affected ?

CBSE 2022 C

1

(b) Light travels from a rarer to a denser medium. How will

(i) the energy carried, and

(ii) the wavelength of the light wave,

be affected ?

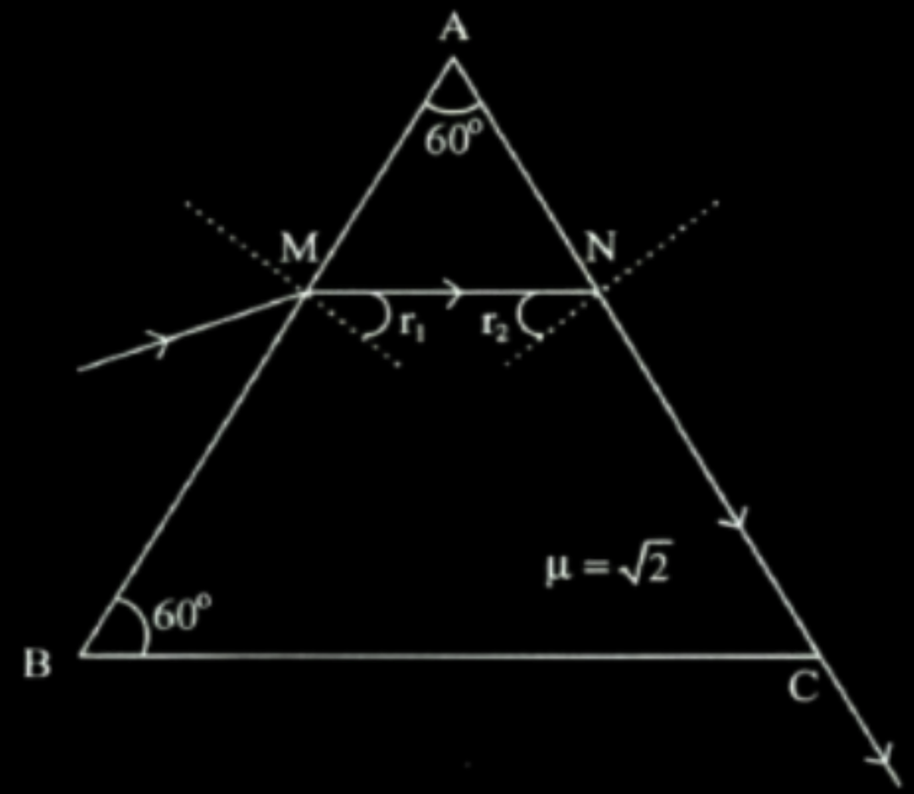
CBSE 2022 C

1

(b) (i) No effect
(c) (ii) Wavelength decreases

10. A ray of light passes through a prism of refractive index $\sqrt{2}$ as shown in the figure. Find 3

CBSE 2022



(i) The angle of incidence ($\angle r_2$) at face AC.

(ii) The angle of minimum deviation for this prism.

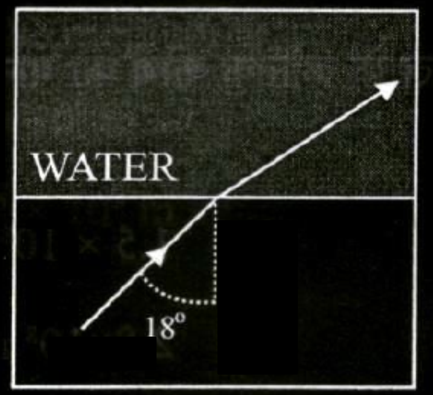
(i) As the emergent ray grazes along the side AC, therefore	
$\frac{1}{\sqrt{2}} = \frac{\sin r_2}{\sin 90^0}$	1
$\therefore \ r_2 = 45°$	½
(ii) $\mu = \frac{\sin(\frac{A+\delta m}{2})}{\sin\frac{A}{2}}$	½
$\sqrt{2} = \frac{\sin(\frac{60^0+\delta m}{2})}{\sin 30^0}$	½
$\therefore \delta_m = 30^0$	½

12. A ray of light travels from a denser to a rarer medium. After refraction, it bends away from the normal. When we keep increasing the angle of incidence, the angle of refraction also increases till the refracted ray grazes along the interface of two media. The angle of incidence for which it happens is called critical angle. If the angle of incidence is increased further the ray will not emerge and it will be reflected back in the denser medium. This phenomenon is called total internal reflection of light. 5

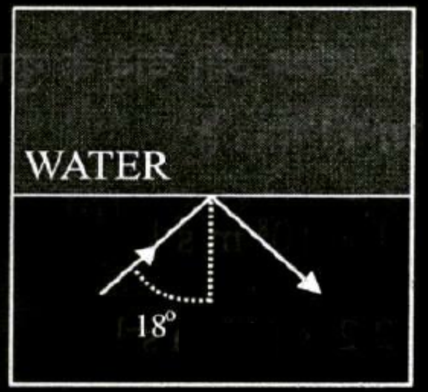
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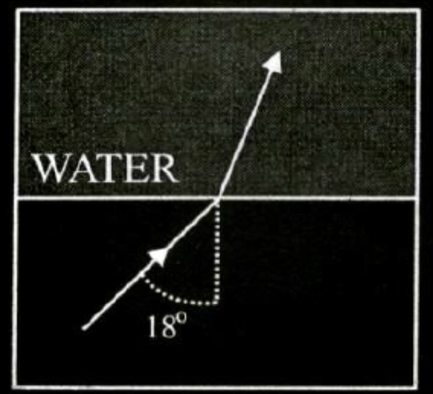
(i) A ray of light travels from a medium into water at an angle of incidence of 18°. The refractive index of the medium is more than that of water and the critical angle for the interface between the two media is 20°. Which one of the following figures best represents the correct path of the ray of light ?

(A)

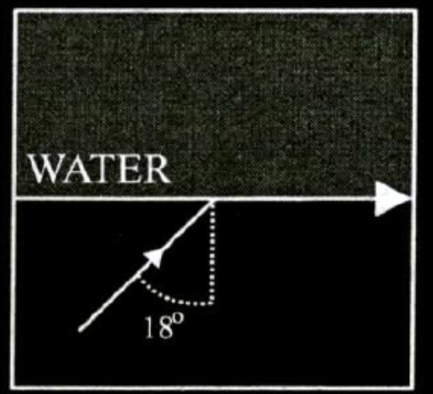


(B)





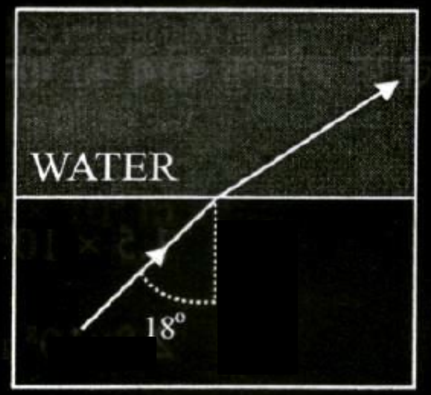
(C)



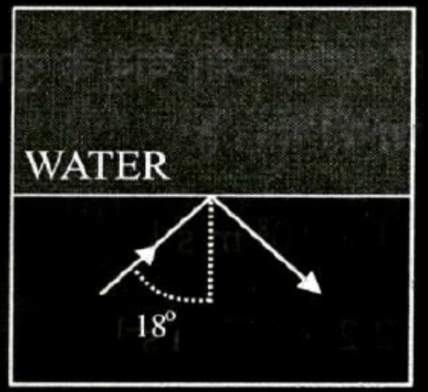
(D)

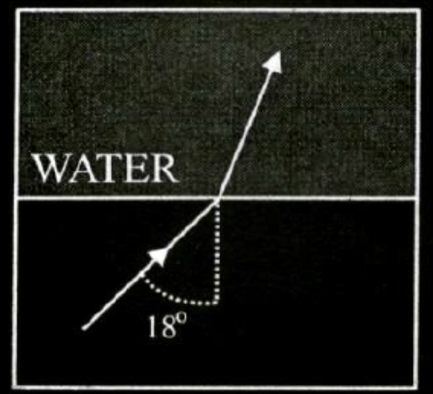
(i) A ray of light travels from a medium into water at an angle of incidence of 18°. The refractive index of the medium is more than that of water and the critical angle for the interface between the two media is 20°. Which one of the following figures best represents the correct path of the ray of light ?

(A)

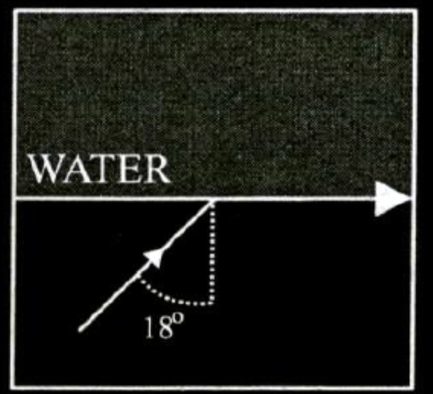


(B)





(C)



(D)

(i)—a

(ii) A point source of light is placed at the bottom of a tank filled with water, of refractive index μ, to a depth d. The area of the surface of water through which light from the source can emerge, is :

(a) $\frac{\pi d^2}{2(\mu^2-1)}$ (b) $\frac{\pi d^2}{(\mu^2-1)}$ (c) $\frac{\pi d^2}{\sqrt{2}\sqrt{\mu^2-1}}$ (d) $\frac{2\pi d^2}{(\mu^2-1)}$

(ii) A point source of light is placed at the bottom of a tank filled with water, of refractive index μ, to a depth d. The area of the surface of water through which light from the source can emerge, is :

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(ii)—b

(iii) For which of the following media, with respect to air, the value of critical angle is maximum ?

(a) Crown glass (b) Flint glass (c) Water (d) Diamond

(iii) For which of the following media, with respect to air, the value of critical angle is maximum ?

(a) Crown glass (b) Flint glass (c) Water (d) Diamond

(iii)—c

(iv) The critical angle for a pair of two media A and B of refractive indices 2.0 and 1.0 respectively is :

(a) 0° (b) 30° (c) 45° (d) 60°

(iv) The critical angle for a pair of two media A and B of refractive indices 2.0 and 1.0 respectively is :

(a) 0° (b) 30° (c) 45° (d) 60°

(iv)—b

(v) The critical angle of pair of a medium and air is 30°. The speed of light in the medium is :

(a) $1 \times 10^{8}\ m\ s^{-1}$

(b) $1.5 \times 10^{8}\ m\ s^{-1}$

(c) $2.2 \times 10^{8}\ m\ s^{-1}$

(d) $2.8 \times 10^{8}\ m\ s^{-1}$

(v) The critical angle of pair of a medium and air is 30°. The speed of light in the medium is :

(a) 1×10^8 m s^{-1}

(b) 1.5×10^8 m s^{-1}

(c) 2.2×10^8 m s^{-1}

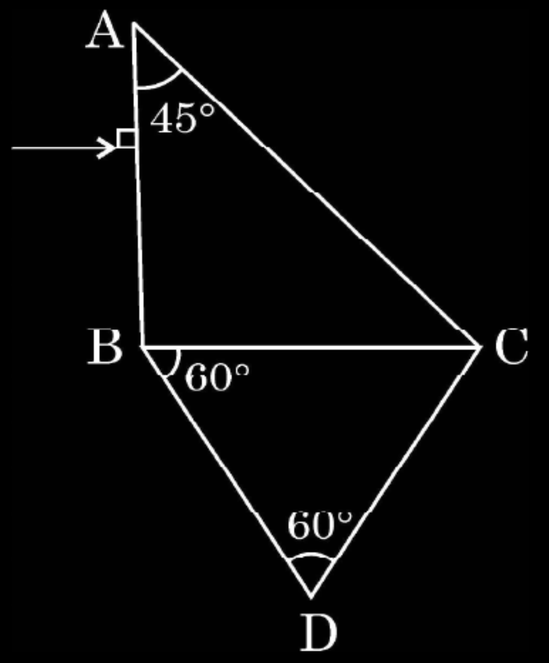
(d) 2.8×10^8 m s^{-1}

(v) —b

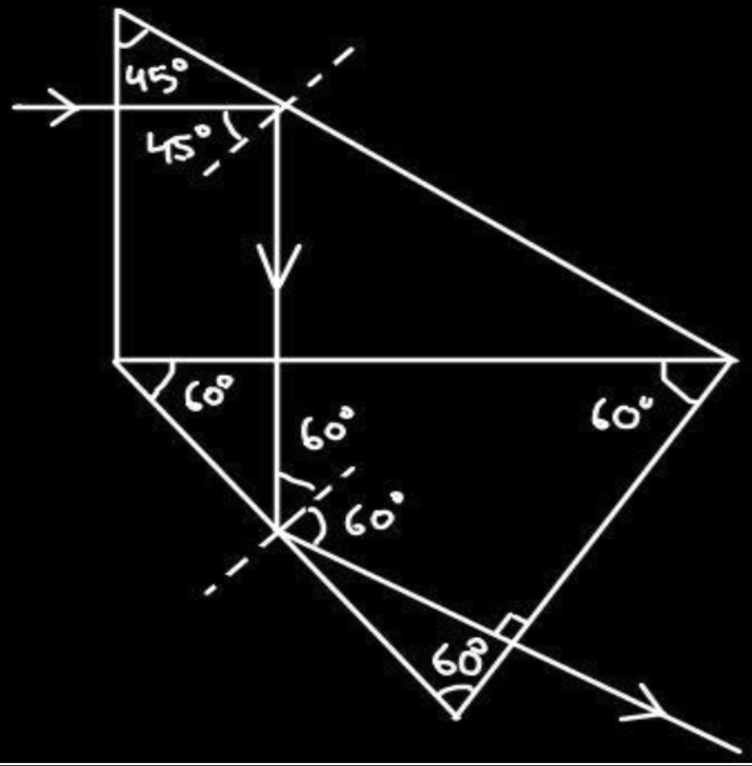
CBSE 2022

4. (a) Write two necessary conditions for total internal reflection.

(b) Two prisms ABC and DBC are arranged as shown in figure.



The critical angles for the two prisms with respect to air are 41.1° and 45° respectively. Trace the path of the ray through the combination. **3**

a) Two conditions	
(i) The light must travel from an optically denser medium to a rarer medium.	1
(ii) Angle of incidence should be greater than the critical angle.	1
(b) 	1

5. Monochromatic light of wavelength 600 nm is incident from air on a water surface. The refractive index of water is 1.33. Find the (i) wavelength, (ii) frequency and (iii) speed, of reflected and refracted light. CBSE 2022 **3**

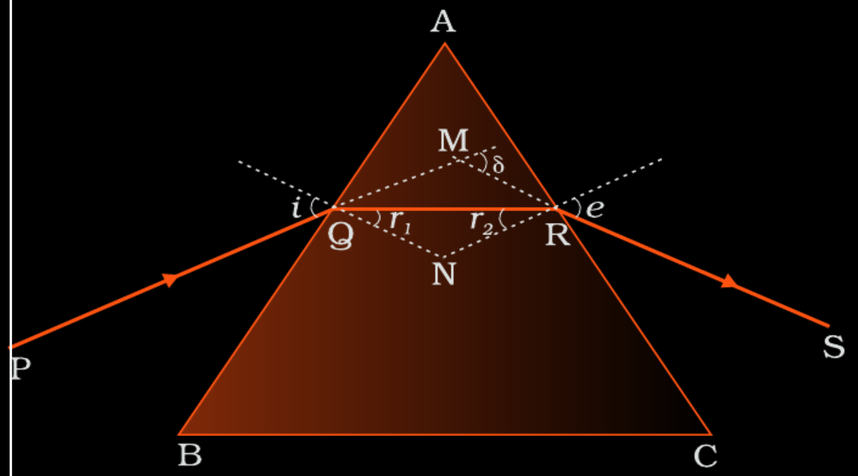
5. Monochromatic light of wavelength 600 nm is incident from air on a water surface. The refractive index of water is 1.33. Find the (i) wavelength, (ii) frequency and (iii) speed, of reflected and refracted light. CBSE 2022 **3**

(a) Reflected high	
$\lambda' = \lambda = 600$ nm.	
No change in wavelength	½
c = 3 x 10^8 m/s in air	
	½
$\upsilon = \frac{c}{\lambda} = \frac{3 \times 10^8}{5 \times 10^{14}}$ Hz	½
Refracted light	
$\lambda' = \frac{\lambda}{\mu} = \frac{600}{1.33} = 451.1$ nm	½
$\simeq 450$nm	
$\upsilon = \frac{c}{\mu} = \frac{3\ X10^8}{1.33}$	½
$= 2.26 \times 10^8$ m/s	
$= 5 \times 10^{14}$ Hz	½

(b) (i) Trace the path of a ray of light PQ which is incident at an angle i on one face of a glass prism of angle A. It then emerges out from the other face at an angle e. Use the ray diagram to prove that the angle through which the ray is deviated is given by $\angle\delta = \angle i + \angle e - \angle A$.

(ii) What will be the minimum value of δ if the ray passes symmetrically through the prism ?

CBSE 2022



$\delta = (i - r_1) + (e - r_2)$ ½

$\delta = (i + e) - (r_1 + r_2)$

In Quadrilateral AQOR

$\angle Q = \angle R = 90^\circ$ $\quad \therefore \angle A + \angle O = 180^\circ$ ---------(1)

In ΔQOR

$O + r_1 + r_2 = 180^\circ$ ---------(2) ½

Comparing (1) and (2)

$\therefore A = r_1 + r_2$ ½

$\therefore \delta = (i + e) - A$

i) If a ray passes symmetrically through a prism (parallel to base of prism), the value of angle of deviation is minimum. ½
At this angle $\angle i = \angle e$ and $\angle r_1 = \angle r_2$

11. For a glass prism of refractive index $\sqrt{3}$, the value of angle of minimum deviation δ is equal to the angle of prism A. Find the angle of prism A. How will the angle δ be affected if the prism were immersed in water ? **3**

CBSE 2022

11. For a glass prism of refractive index $\sqrt{3}$, the value of angle of minimum deviation δ is equal to the angle of prism A. Find the angle of prism A. How will the angle δ be affected if the prism were immersed in water ? **3**

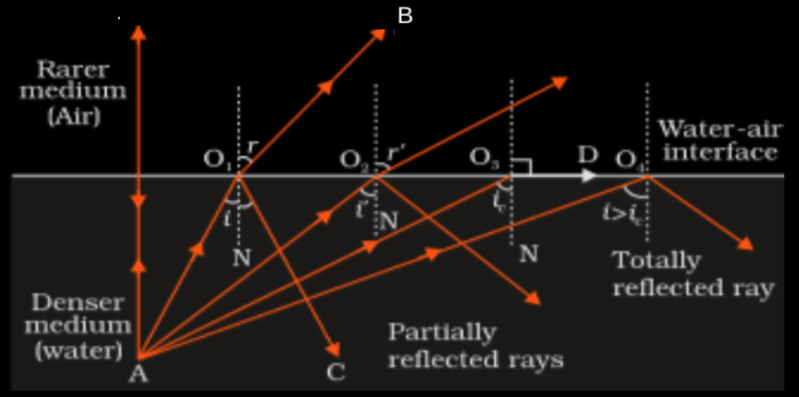
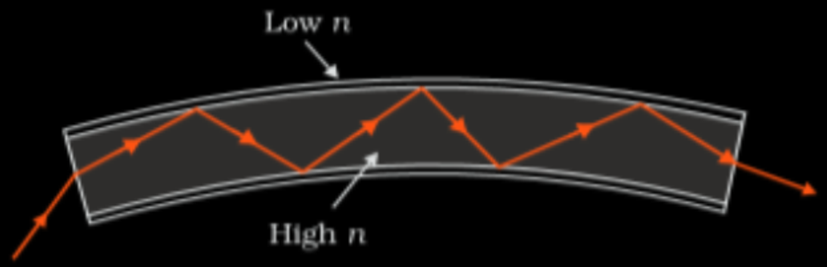
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$\mu = \sqrt{3}$, $\delta = A$	
$\mu = \dfrac{\sin\dfrac{(A+\delta_m)}{2}}{\sin(A/2)}$	½
$\sqrt{3} = \dfrac{\sin\dfrac{(A+A)}{2}}{\sin(A/2)} = \dfrac{\sin A}{\sin\dfrac{A}{2}}$	½
$\sqrt{3} = \dfrac{2\sin\dfrac{A}{2}\cos\dfrac{A}{2}}{\sin\dfrac{A}{2}} = 2\cos\dfrac{A}{2}$	½
$\dfrac{\sqrt{3}}{2} = \cos\dfrac{A}{2}$	
$\dfrac{A}{2} = 30$	
$\therefore A = 60^\circ$	½
With increase in refractive index of surrounding medium, angle of deviation (δ_m) will decrease	1

4. Explain with the help of a suitable diagram, the phenomenon on which an optical fibre works. Mention any two uses of optical fibres. CBSE 2022 **3**

4. Explain with the help of a suitable diagram, the phenomenon on which an optical fibre works. Mention any two uses of optical fibres. **3**

CBSE 2022

An optical fibre works on the principle of Total internal reflection.	1
	1
Alternatively 	
Uses of optical fibres(any two) i) Medical and optical examination (endoscopy). ii) Transmission and reception of signals iii) Photometric sensors.	½+½

7. A ray of light passes through an equilateral triangular prism and suffers minimum deviation of 30°. Calculate :

(a) the refractive index of the material of the prism.

(b) the angle at which the ray is incident on the prism. *3*

CBSE 2022

7. A ray of light passes through an equilateral triangular prism and suffers minimum deviation of 30°. Calculate :

(a) the refractive index of the material of the prism.

(b) the angle at which the ray is incident on the prism. *3*

CBSE 2022

(b) $\angle i = \dfrac{\angle A + \angle \delta m}{2} = \dfrac{60^\circ + 30^\circ}{2} = 45^\circ$ — 1½

(a) $\mu = \dfrac{\sin\dfrac{(A+\delta m)}{2}}{\sin(\dfrac{A}{2})} = \dfrac{\sin(\dfrac{60^\circ+30^\circ}{2})}{\sin\dfrac{60^\circ}{2}} = \dfrac{\sin 45^\circ}{\sin 30^\circ}$ — 1

$= \dfrac{\dfrac{1}{\sqrt{2}}}{\dfrac{1}{2}} = \dfrac{2}{\sqrt{2}} = \sqrt{2}$ — ½

(b) (i) What is meant by total internal reflection of light ? Write the two necessary conditions for total internal reflection to occur.

(ii) Calculate the speed of light in a medium for which the critical angle with respect to air is 45°. CBSE 2022 *3*

(b) (i) What is meant by total internal reflection of light ? Write the two necessary conditions for total internal reflection to occur.

(ii) Calculate the speed of light in a medium for which the critical angle with respect to air is 45°. CBSE 2022 *3*

I. (i) The phenomenon due to which the light entering from a optically denser to a rarer medium gets reflected to the denser medium.	½
(ii) Two conditions:	
• Light should travel from a denser medium to rarer medium	½
• Angle of incidence should be more than the critical angle.	½
II. $n = \frac{1}{\sin C} = \frac{c_o}{c}$	½
$c = c_o \sin C$ where c_o- speed of light in vacuum (3×10^8 m/s)	
$= 3\times10^8 \text{ m/s} \times \frac{1}{\sqrt{2}} = \frac{3}{\sqrt{2}}\times10^8 \text{ m/s}$	½ + ½

11. A diver looking up through water ($\mu = \frac{4}{3}$) sees the outside world contained in a circular area on the surface of water. If the diver's eyes are $\sqrt{7}$ m below the surface of water, then calculate the area of the circle. *3*

CBSE 2022

11. A diver looking up through water ($\mu = \frac{4}{3}$) sees the outside world contained in a circular area on the surface of water. If the diver's eyes are $\sqrt{7}$ m below the surface of water, then calculate the area of the circle. *3*

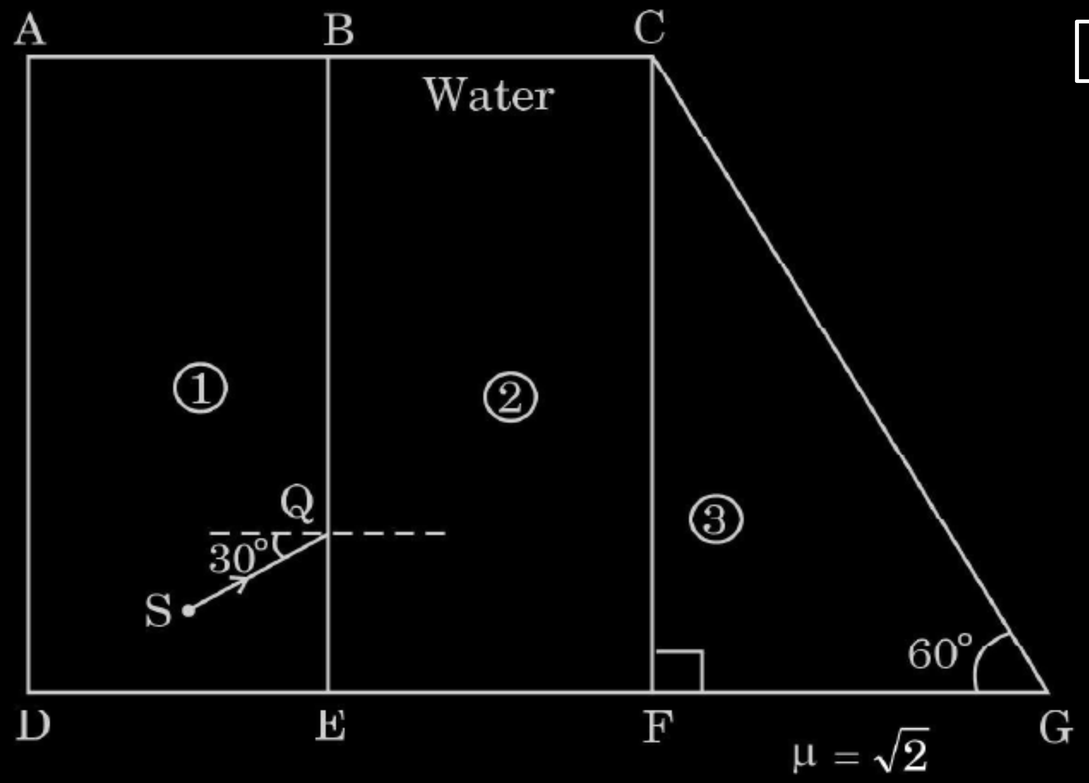
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$\text{Sin C} = \frac{1}{\mu}$	½
$\text{Sin C} = \frac{3}{4}$	½
$\text{Sin C} = \frac{R}{\sqrt{R^2+7}}$	½
$\frac{3}{4} = \frac{R}{\sqrt{R^2+7}}$	½
$R = 3\text{m}$	½
$\text{Area} = \pi R^2$	
$= 3.14 \times (3)^2$	
$= 28.26\ \text{m}^2$	½

11. The figure shows a water column BCFE surrounded by two media ① and ③ of the same refractive index $\sqrt{2}$. A ray of light from a point source S is incident on surface BE at an angle of 30°.

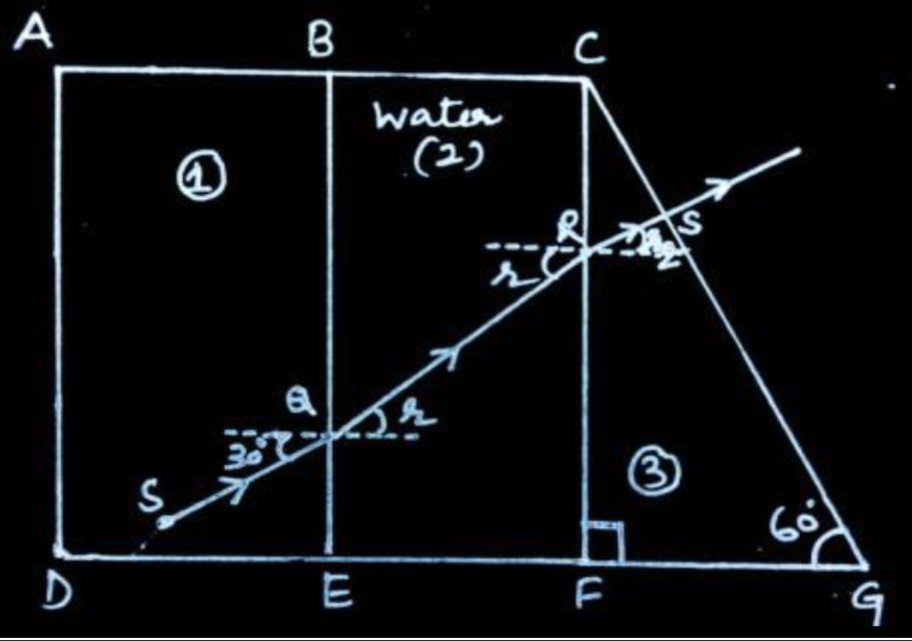
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(a) Trace the path of ray through media ② and ③ as it emerges out of face CG.

(b) Find the angle of emergence at face CG. *3*

a)



b) Refraction at Q , $\sqrt{2}\ \text{Sin}\ 30^{o} = \frac{4}{3}\ \text{Sin r}$

$\text{Sin r} = \frac{3}{4\sqrt{2}}$ ½

Refraction from face CF

$\frac{4}{3}\ \text{Sin r} = \sqrt{2}\ \text{Sin r}_2$ ½

$\frac{4}{3} \times \frac{3}{4\sqrt{2}} = \sqrt{2}\ \text{Sin r}_2$ ½

$r_2 = 30^{0}$

$\angle C + \angle CRS + \angle CSR = 180^{0}$

$30^{o} + 60^{o} + \angle CSR = 180^{0}$ ½

$\angle CSR = 90^{0}$

Ray goes perpendicularly out from face CG

$e = 0$

∴ Ray RS incidents perpendicular to CS

∴ angle of emergence of the ray is zero.

Alternatively-

As medium is same on both side of the water column.

Angle of emergence

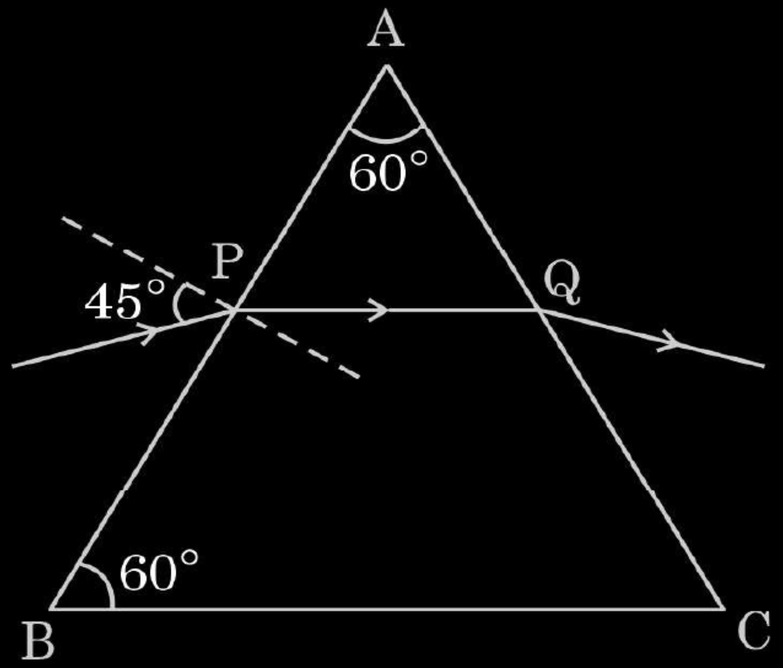
Now $r_1 + r_2 = A$, ½

$r_2 = 30^{0}$ ½

$r_1 + 30 = 30$ ½

$r_1 = 0$, Hence $e = 0$ ½

10. A ray of light is incident on a prism at an angle of 45° and passes symmetrically as shown in the figure. Calculate : CBSE 2022 *3*



(a) the angle of minimum deviation,

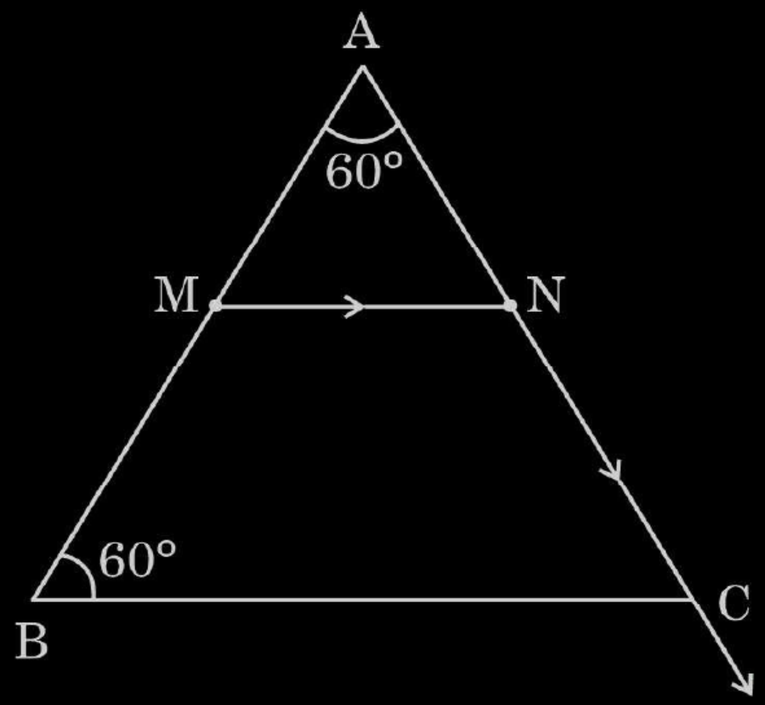
(b) the refractive index of the material of the prism, and

(c) the angle of refraction at the point P.

(a)	As the ray passes symmetrically	
	$\angle i = \angle e$ and $\delta = \delta_m$	
	$\therefore \quad i + e = A + \delta$	½
	$2i = A + \delta_m$	
	$\delta_m = 2i - A = 90° - 60°$	
	$= 30°$	½
(b)		
	$\mu = \dfrac{\sin \dfrac{(A + \delta_m)}{2}}{\sin A/2}$	½
	$= \dfrac{\sin 45°}{\sin 30°} = \sqrt{2}$	½
(c)		
	As $\delta = \delta_m$	
	$r_1 = r_2$ (Alternatively)	
	Hence, r_1 (Angle of refraction at P)	
	$1 \times \sin 45 = \mu \sin r_1$	
	$1 \times \dfrac{1}{\sqrt{2}} = \sqrt{2} \sin r_1$	½
	$\therefore \angle r_1 = 30°$	½

7. A ray is incident on a prism of material of refractive index $\sqrt{2}$ at point M such that it grazes along NC after emerging from the prism, as shown in the figure.

CBSE 2022



Find : *3*

(a) the critical angle for the prism.

(b) the angle of refraction at face AB.

a)	
$\mu = \frac{1}{\sin i_c} \Rightarrow \sqrt{2} = \frac{1}{\sin i_c}$	½
$i_c = 45°$	½
b)	
$r_2 = 45° = i_c$	½
$A = r_1 + r_2$	½
$60 = r_1 + 45°$	½
$r_1 = 15°$	½

10. (a) Is the speed of light in glass independent of the colour of light ? Give reason.

(b) A small bulb is placed at the bottom of a tank containing water to a depth of 70 cm. Find the area of the surface of water through which light from the bulb can emerge out. Given refractive index of water is $\frac{4}{3}$. *3*

CBSE 2022

a) No

$$\mu \alpha \frac{1}{v_m} \alpha \frac{1}{\lambda}$$

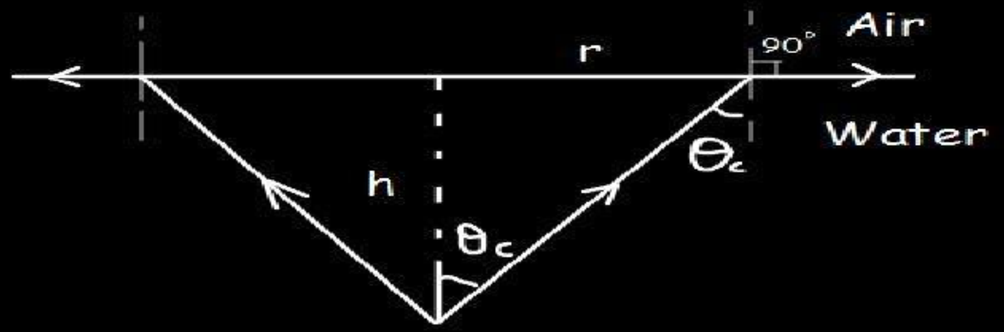
½

Where v_m velocity of light in medium

$\therefore$ velocity depends on wavelength

½

b)



½

$$\mu \sin\theta_c = 1 \Rightarrow \sin\theta_c = \frac{1}{\mu} = \frac{3}{4}$$

½

$$\tan\theta_c = \frac{r}{h} = \frac{1}{\mu} \times \frac{1}{\sqrt{1-\frac{1}{\mu^2}}} = \frac{1}{\sqrt{\mu^2-1}}$$

Area $A = \pi r^2 = \pi(h\tan\theta_c)^2$

½

$$A = \frac{\pi h^2}{\mu^2 - 1}$$

$$A = \frac{\pi \times 70 \times 70}{\left(\frac{4}{3}\right)^2 - 1} = 19800 \text{ cm}^2$$

½

11. The refractive index of an equilateral triangular prism kept in air is $\sqrt{2}$.

Calculate : *3*

(a) the angle of minimum deviation.

(b) the critical angle for the prism.

CBSE 2022

a)

$$\mu = \frac{\sin(\frac{\delta_m + A)}{2}}{\sin(A/2)}$$

½

$$\sqrt{2} = \frac{\sin\left(\frac{\delta_m + 60°}{2}\right)}{\sin 30°} \Rightarrow \sin\left(\frac{\delta_m + 60°}{2}\right) = \sqrt{2} \times \frac{1}{2}$$

½

$$\sin\left(\frac{\delta_m + 60°}{2}\right) = \frac{1}{\sqrt{2}} \Rightarrow \left(\frac{\delta_m + 60°}{2}\right) = 45°$$

½

$$\delta_m = 90° - 60° \Rightarrow \delta_m = 30°$$

½

b) $\sin\theta_c = \frac{1}{\mu} = \frac{1}{\sqrt{2}}$

½

$\theta_c = 45°$

½

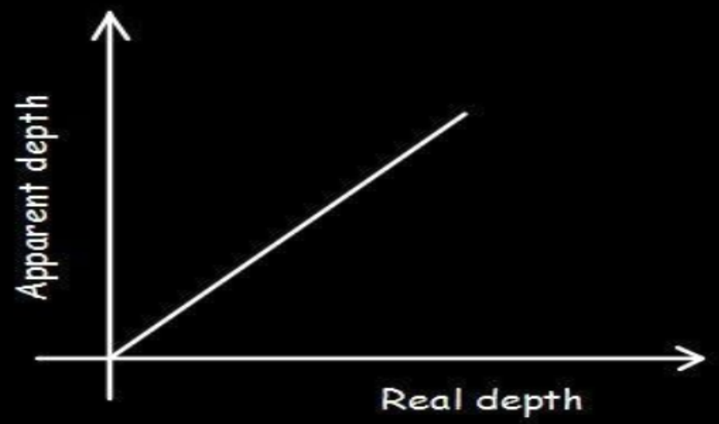
5. (a) State the conditions for total internal reflection to take place.

(b) A tank is filled with a transparent liquid to height 'H'. A coin suspended by a thread in the liquid is gradually lowered till it touches the bottom. The apparent depth is determined corresponding to different positions of the coin.

(i) Plot a graph showing variation of the apparent depth with the real depth of the coin.

(ii) What is the physical significance of the slope of the graph ? *3*

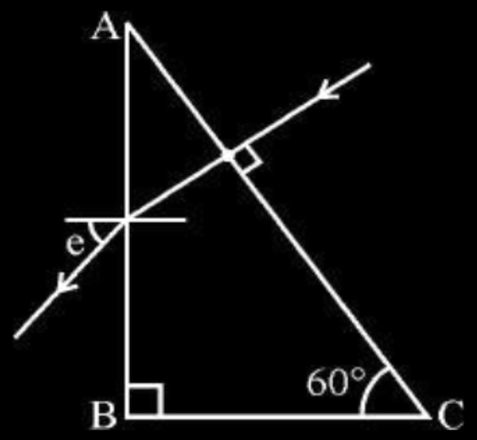
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(a) The conditions for total internal reflection to take place are:	
(i) The light should travel from denser to rarer medium.	½
(ii) The angle of incidence should be greater than the critical angle for a given pair of media.	½
(b) (i) 	1
(ii)The slope will give 1/μ where μ is refractive index as per the graph above.	1

35.

(b) Calculate the angle of emergence (e) of the ray of light incident normally on the face AC of a glass prism ABC of refractive index $\sqrt{3}$. How will the angle of emergence change qualitatively, if the ray of light emerges from the prism into a liquid of refractive index 1.3 instead of air ?

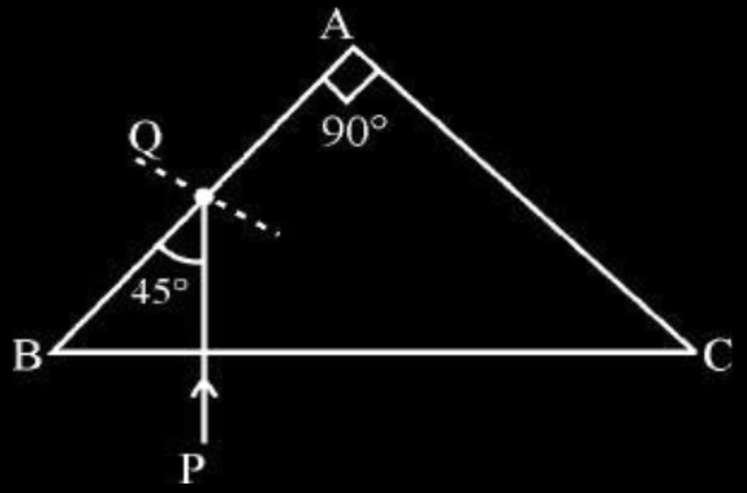
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(b) Applying Snell's law at face AB, we get	
$\sqrt{3}\sin 30 = 1 \cdot \sin e$	½
$\sqrt{3} \times \frac{1}{2} = \sin e$	½
$\frac{\sqrt{3}}{2} = \sin e$	
$\sin 60 = \sin e$	
$e = 60^0$	½
When the medium (the air) in which the prism is kept is replaced with a liquid of refractive index 1.3 the angle of emergence would decrease. It is because bending in the ray of light will be lesser.	½ ½

36. (a) Draw the ray diagram showing refraction of ray of light through a glass prism. Derive the expression for the refractive index μ of the material of prism in terms of the angle of prism A and angle of minimum deviation δ_m.

(b) A ray of light PQ enters an isosceles right angled prism ABC of refractive index 1.5 as shown in figure.

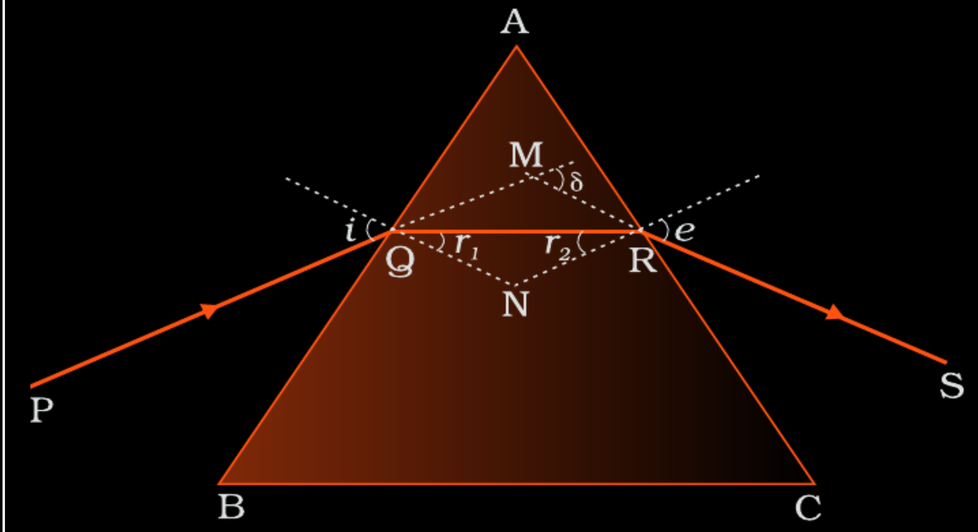


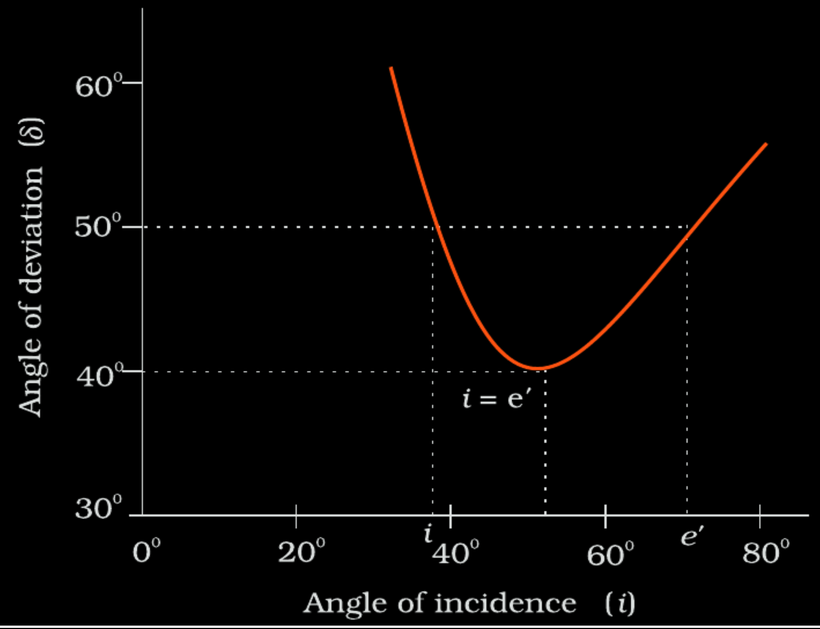
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(i) Trace the path of the ray through the prism.

(ii) What will be the effect on the path of the ray if the refractive index of the prism is 1·4 ?

5





(ii) Derivation

From the figure

$\angle A + \angle QNR = 180^0$ (i) ½

In $\Delta QNR \quad r_1 + r_2 + \angle QNR = 180^0$ (ii)

Comparing equation (i) and (ii) we get

$r_1 + r_2 = A$ (iii)

Total deviation produced $\delta = (i - r_1) + (e - r_2)$

$\delta = i + e - (r_1 + r_2) = i + e - A$ (iv) ½

From the graph δ *vs* i we find that when δ becomes minimum i.e. δ_m

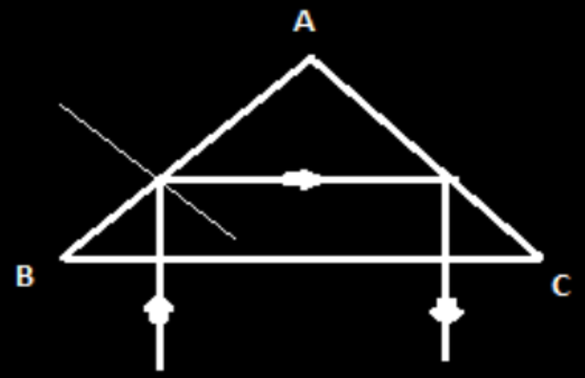
i = e and $r_1 = r_2$ ½

From (iv) $i = \frac{(A + \delta_m)}{2}$

and from (iii) $r = \frac{A}{2}$

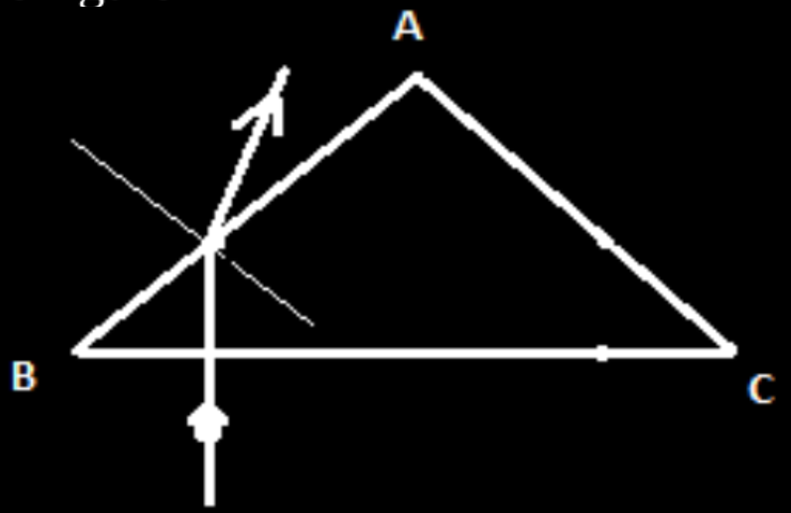
$\mu = \frac{\sin i}{\sin r} = \frac{\sin {}^{(A+\delta_m)}/_2}{\sin \frac{A}{2}}$ ½

(a) (i)



1

(ii) If $\mu = 1.4$ Total Internal Reflection will not occur as shown in the figure



1

(Note: Award this last one mark if student does not draw the diagram and conclude correctly.)

8. For a glass prism, the angle of minimum deviation will be smallest for the light of *1*

(A) red colour.

(B) blue colour.

(C) yellow colour.

(D) green colour.

CBSE 2020

8. For a glass prism, the angle of minimum deviation will be smallest for the light of *1*

(A) red colour.

(B) blue colour.

(C) yellow colour.

(D) green colour.

CBSE 2020

(A)

red colour

15. A ray of light on passing through an equilateral glass prism, suffers a minimum deviation equal to the angle of the prism. The value of refractive index of the material of the prism is _____ . CBSE 2020 *1*

15. A ray of light on passing through an equilateral glass prism, suffers a minimum deviation equal to the angle of the prism. The value of refractive index of the material of the prism is $\sqrt{3}$. CBSE 2020 *1*

(a) Plot a graph for angle of deviation as a function of angle of incidence for a triangular prism.

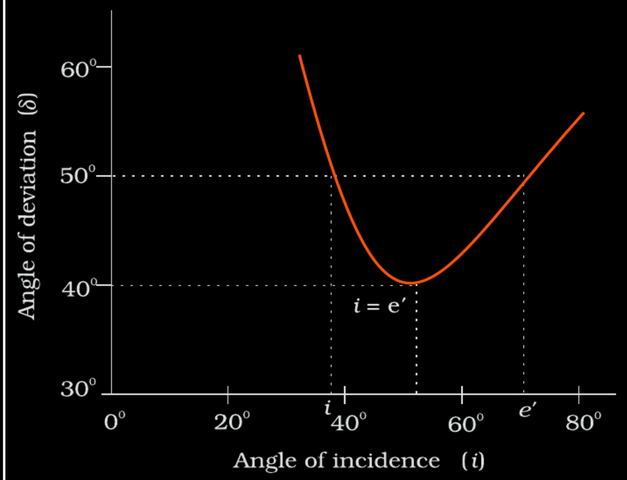
(b) Derive the relation for the refractive index of the prism in terms of the angle of minimum deviation and angle of prism.

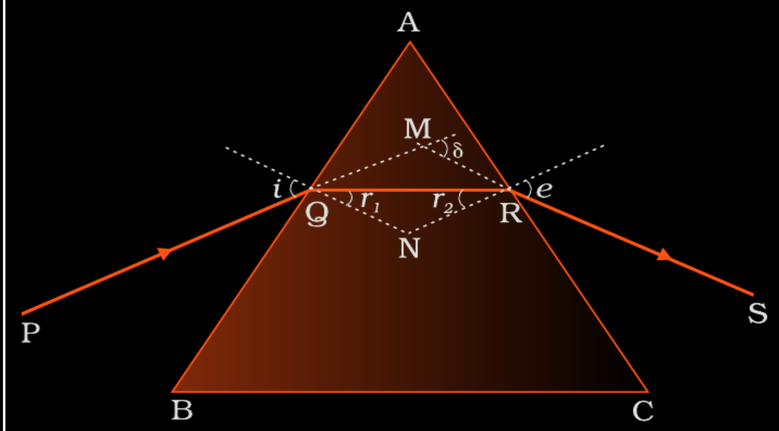
CBSE 2019

(a) Plot a graph for angle of deviation as a function of angle of incidence for a triangular prism.

(b) Derive the relation for the refractive index of the prism in terms of the angle of minimum deviation and angle of prism.

CBSE 2019





Since $n_{21} = \dfrac{\sin i}{\sin r}$ ½

From the figure and calculations

$r_1 + r_2 = A$

At minimum deviation i.e. $\delta = \delta m$, $i = e$ and $r_1 = r_2 = r$ 1

$\therefore \quad r = A/2$ ..... (eqⁿ 1)

From the figure

$\delta = (i - r_1) + (e - r_2)$

$\therefore \quad \delta_m = (i + e) + (r_1 + r_2)$ ½

$i = \dfrac{A + \delta_m}{2}$ ..... (eqⁿ 2)

$\therefore \quad n_{21} = \dfrac{\sin i}{\sin r} = \dfrac{\text{Sin}\,\dfrac{A + \delta_m}{2}}{\sin A/2}$ ½

½

How does the angle of minimum deviation of a glass prism vary if the incident violet light is replaced by red light ?

CBSE 2019

State the condition under which a large magnification can be achieved in an astronomical telescope.

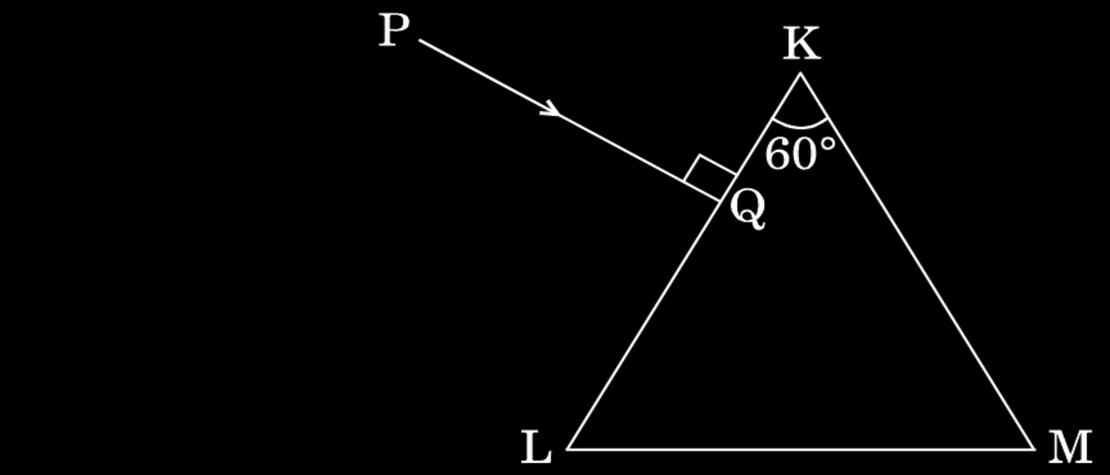
OR

How does the angle of minimum deviation of a glass prism vary if the incident violet light is replaced by red light ?

CBSE 2019

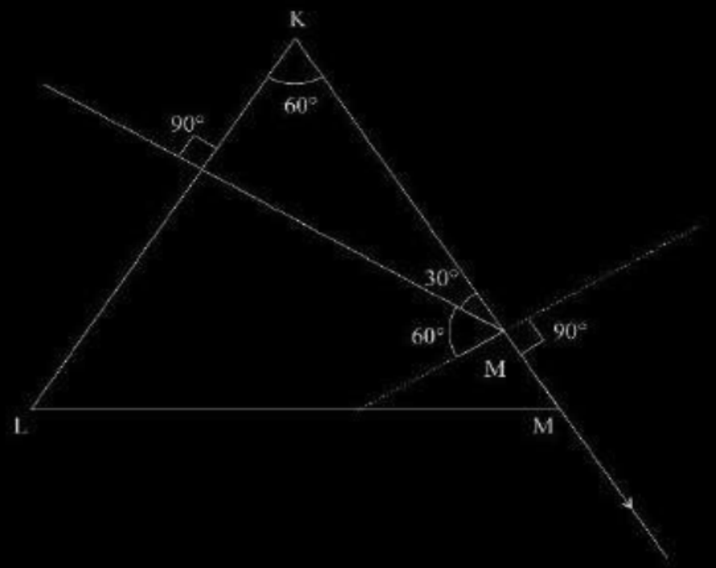
Angle of minimum deviation decreases.	1

A triangular prism of refracting angle $60°$ is made of a transparent material of refractive index $2/\sqrt{3}$. A ray of light is incident normally on the face KL as shown in the figure. Trace the path of the ray as it passes through the prism and calculate the angle of emergence and angle of deviation.



CBSE 2019

Ray diagram:



1

- A=60°
- $\frac{2}{\sqrt{3}}\sin 60^o = \sin r$

1½

- $\sin r = \frac{2\sqrt{3}}{2\sqrt{3}} = 1$
- $r = 90^o$

½

Angle of deviation is equal to 30°

An equilateral glass prism has a refractive index 1.6 in air. Calculate the angle of minimum deviation of the prism, when kept in a medium of refractive index $4\sqrt{2}/5$.

CBSE 2019

An equilateral glass prism has a refractive index 1.6 in air. Calculate the angle of minimum deviation of the prism, when kept in a medium of refractive index $4\sqrt{2}/5$.

CBSE 2019

$$\mu = \frac{\sin\frac{(A + D_m)}{2}}{\sin A/2}$$ ½

$$\mu = \frac{\mu_2}{\mu_1} = \frac{1.6}{\frac{4}{5}\sqrt{2}} = \frac{8}{4\sqrt{2}} = \sqrt{2}$$ ½

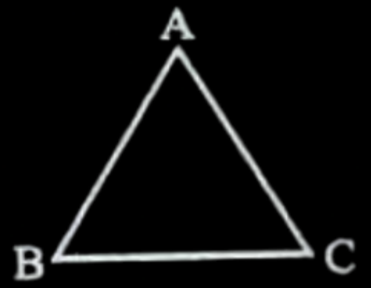
$$\sqrt{2} = \frac{\sin(\frac{60+D_m}{2})}{\sin 60/2} = \frac{\sin\frac{(60+D_m)}{2}}{\sin 30}$$

$$\therefore \quad \sin(\frac{60+D_m}{2}) = \sqrt{2} \cdot \frac{1}{2} = \frac{1}{\sqrt{2}} = \text{Sin } 45^0$$ ½

$$\therefore \frac{60+D_m}{2} = 45^o$$ ½

$$\therefore \quad D_m = 30^0$$

(i) A ray of light incident on face AB of an equilateral glass prism, shows minimum deviation of 30°. Calculate the speed of light through the prism.



(ii) Find the angle of incidence at face AB so that the emergent ray grazes along the face AC.

CBSE 2017

(b) A ray of light passing from air through an equilateral glass prism undergoes minimum deviation when the angle of incidence is $\frac{3}{4}^{\text{th}}$ of the angle of prism. Calculate the speed of light in the prism.

CBSE 2017

b) At minimum deviation

$$r = {}^{A}/_{2} = 30^{0}$$

We are given that

$$i = \frac{3}{4}A = 45^{0}$$

$$\therefore \mu = \frac{\sin 45^{0}}{\sin 30^{0}} = \sqrt{2}$$

$\therefore$ Speed of light in the prism $= \frac{c}{\sqrt{2}}$

$$(\cong 2.1 \times 10^{8}\ \mathrm{ms}^{-1})$$

[Award ½ mark if the student writes the formula:

$$\mu = \frac{\sin(A + D_m)/2}{\sin({}^{A}/_{2})}$$

but does not do any calculations.]

½

½

½

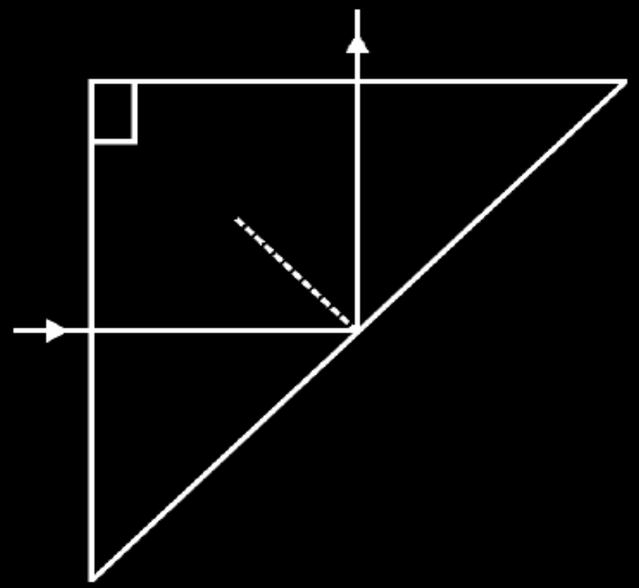
½

(i) Plot a graph to show variation of the angle of deviation as a function of angle of incidence for light passing through a prism. Derive an expression for refractive index of the prism in terms of angle of minimum deviation and angle of prism.

(ii) What is dispersion of light ? What is its cause ?

(iii) A ray of light incident normally on one face of a right isosceles prism is totally reflected as shown in fig. What must be the minimum value of refractive index of glass ? Give relevant calculations.

CBSE 2016



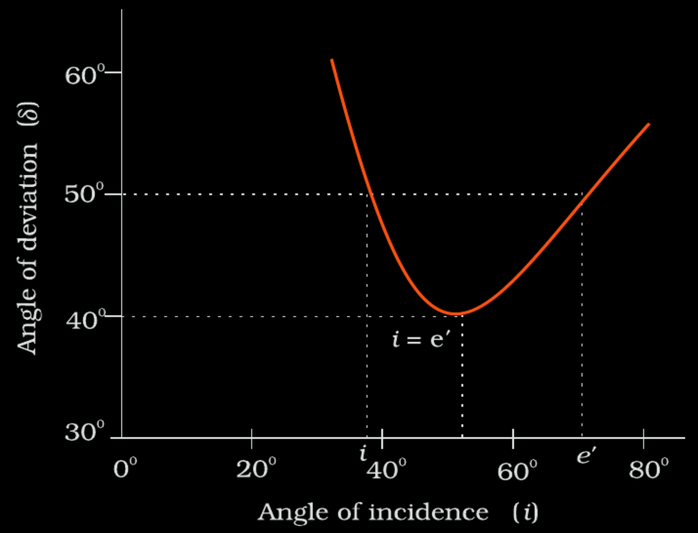
From figure $\delta = D_m, i = e$ which implies $r_1 = r_2$

$2r$ = A, or $r = {}^{A}/_{2}$ ½

Using $\delta = i + e - A$
$D_m = 2i - A$

$$i = \frac{A + D_m}{2}$$ ½

$$\mu = \frac{\sin i}{\sin r} = \frac{\sin(\frac{A + D_m}{2})}{\sin {}^{A}/_{2}}$$ ½



(ii) The phenomenon of splitting of white light into its constituent colours. ½

Cause: Refractive index of the material is different for different colours
According to the equation, $\delta \cong (\mu - 1)A$, where A is the angle of prism, different colours will deviate through different amount. ½

For total internal reflection,
$\angle i \geq \angle i_c$ (critical angle)

$\Rightarrow 45^0 \geq \angle i_c$, i.e. , $\angle i_c \leq 45^0$
$\sin i_c \leq \sin 45^0$

$$\leq \frac{1}{\sqrt{2}}$$

$$\frac{1}{\sin i_c} \geq \sqrt{2}$$

$\Rightarrow \mu \geq \sqrt{2}$

Hence, the minimum value of refractive index must be $\sqrt{2}$

(i) For a glass prism $\left(\mu = \sqrt{3}\right)$ the angle of minimum deviation is equal to the angle of the prism. Calculate the angle of the prism.

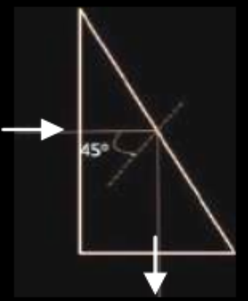
(ii) Draw ray diagram when incident ray falls normally on one of the two equal sides of a right angled isosceles prism having refractive index $\mu = \sqrt{3}$.

CBSE 2016

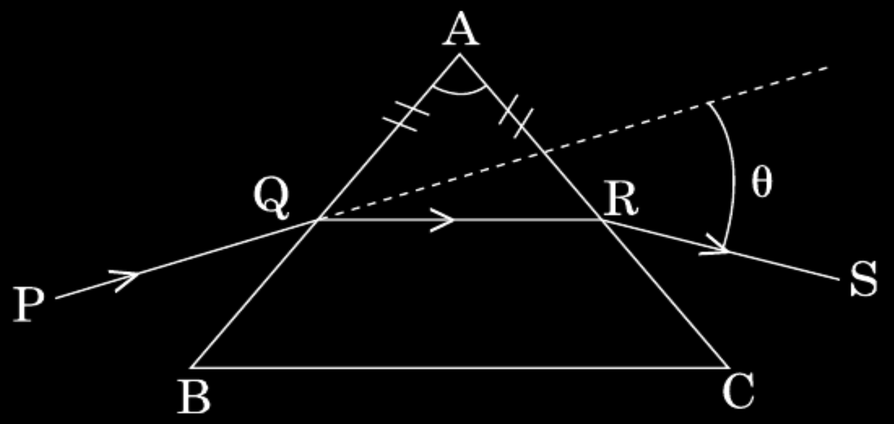
(i) For a glass prism $\left(\mu = \sqrt{3}\right)$ the angle of minimum deviation is equal to the angle of the prism. Calculate the angle of the prism.

(ii) Draw ray diagram when incident ray falls normally on one of the two equal sides of a right angled isosceles prism having refractive index $\mu = \sqrt{3}$.

CBSE 2016

(i) $\mu = \frac{sin\left(\frac{A+D}{2}\right)}{sin\frac{A}{2}}$	½
$= \frac{sin\left(\frac{2A}{2}\right)}{Sin\frac{A}{2}} = 2\cos A/2 = \sqrt{3}$	½
$\therefore A = 60°$	½
(ii) $\mu = \sqrt{3} = \frac{1}{Sini_c}$ 	
$\therefore\ Sini_c = \frac{1}{\sqrt{3}} \cong 0.58$	½
Lies between 30^0 and 45^0 Hence, TIR takes place. **Alternatively,** $sinc = \frac{1}{\sqrt{3}}$ which is less than $\frac{1}{\sqrt{2}}$ $\therefore$ angle of incidence > i_c $\therefore$TIR	1

A ray PQ incident on the refracting face BA is refracted in the prism BAC as shown in the figure and emerges from the other refracting face AC as RS such that AQ = AR. If the angle of prism A = 60° and refractive index of material of prism is $\sqrt{3}$, calculate angle θ.



CBSE 2016

Since AQ = AR, we have
QR|| BC
$\therefore$ θ is the angle of minimum deviation.

(Alternatively: Since AQ=AR, we get
$\angle r_1 = \angle r_2$
$\therefore$ θ is the angle of minimum deviation.) ½

$$\mu = \frac{\sin\left(\frac{A+\delta m}{2}\right)}{\sin\left({}^{A}/_{2}\right)}$$ ½

$$\therefore \sqrt{3} = \frac{\sin\left(\frac{60+\delta m}{2}\right)}{\sin 30^0}$$ ½

$$\therefore \frac{\sqrt{3}}{2} = \sin\left(\frac{60+\delta m}{2}\right)$$

$$\therefore \frac{60+\delta m}{2} = 60$$

or $\delta m = 60^0$ ½

(i) What is total internal reflection ? Under what conditions does it occur ?

(ii) Find a relation between critical angle and refractive index.

CBSE 2016

(iii) Name one phenomenon which is based on total internal reflection.

(i) What is total internal reflection ? Under what conditions does it occur ?

(ii) Find a relation between critical angle and refractive index.

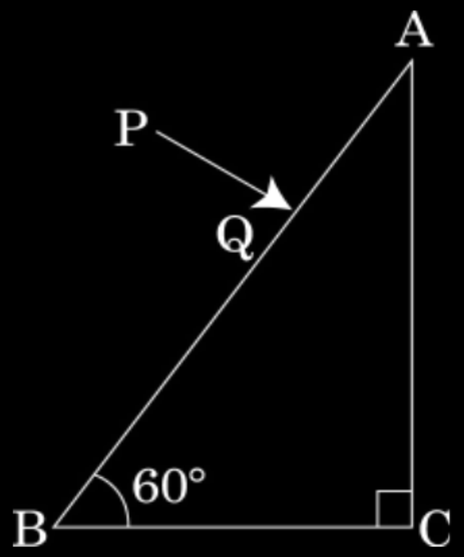
CBSE 2016

(iii) Name one phenomenon which is based on total internal reflection.

(i) When a ray of light travels from a denser medium into a rarer medium at an angle greater than the critical angle, it reflects back into the denser medium. This phenomenon is called total internal reflection. Conditions for total internal reflection	½
(a) Light should travel from denser medium to rarer medium.	½
(b) Angle of incidence should be greater than critical angle.	½
(ii) $\frac{1}{\mu} = \frac{\sin i}{\sin r}$, for total internal reflection to occur $i \geq i_c$ at critical angle, angle of refraction $r = 90^\circ$, hence $\frac{1}{\mu} = \frac{\sin i_c}{\sin 90^\circ}$	½
$\Rightarrow \mu = \frac{1}{\sin i_c}$	½
(iii) Mirage/ sparkling of diamond/ optical fiber/ totally reflecting Prism/ shinning of air bubbles in water.(any one)	½

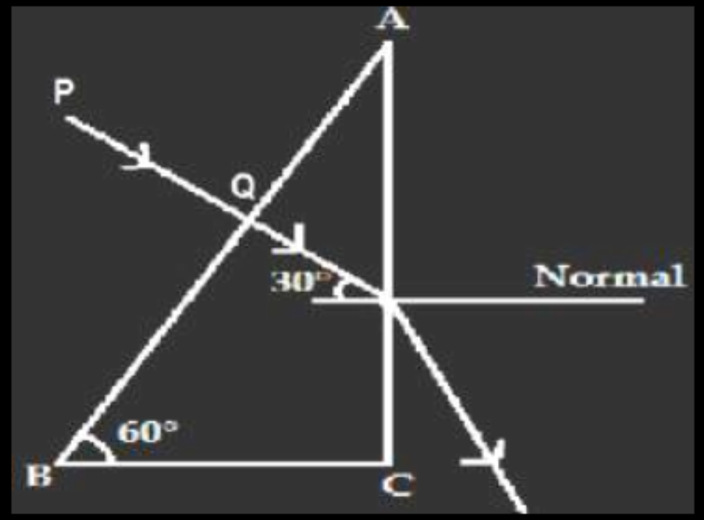
A ray PQ incident normally on the refracting face BA is refracted in the prism BAC made of material of refractive index 1.5. Complete the path of ray through the prism. From which face will the ray emerge ? Justify your answer.

CBSE 2016



A ray PQ incident normally on the refracting face BA is refracted in the prism BAC made of material of refractive index 1.5. Complete the path of ray through the prism. From which face will the ray emerge ? Justify your answer.

CBSE 2016



1

Face-AC

Here $i_c = \sin^{-1}(\frac{2}{3})$ ½

$= \sin^{-1}(0.6)$

$\angle i$ on face AC is 30° which is less than $\angle i_c$.Hence the ray get replaced here. ½

9.3 A tank is filled with water to a height of 12.5 cm. The apparent depth of a needle lying at the bottom of the tank is measured by a microscope to be 9.4 cm. What is the refractive index of water? If water is replaced by a liquid of refractive index 1.63 up to the same height, by what distance would the microscope have to be moved to focus on the needle again? NCERT

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1.33; 1.7 cm

9.4 Figures 9.31(a) and (b) show refraction of a ray in air incident at 60° with the normal to a glass-air and water-air interface, respectively. Predict the angle of refraction in glass when the angle of incidence in water is 45° with the normal to a water-glass interface [Fig. 9.31(c)].

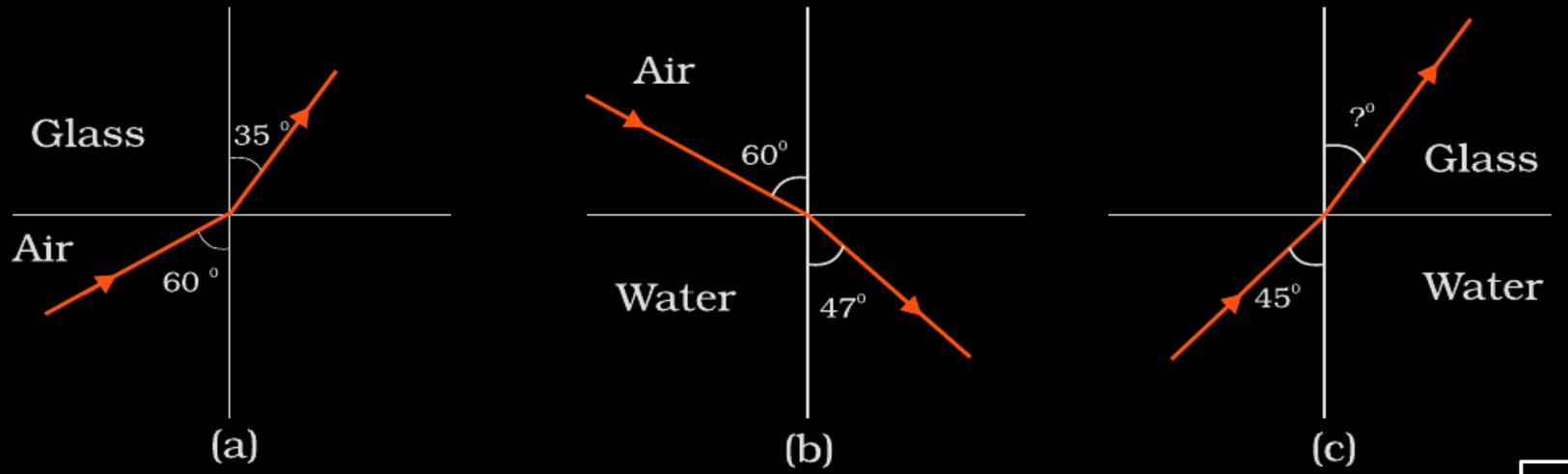


FIGURE 9.31

NCERT

9.4 Figures 9.31(a) and (b) show refraction of a ray in air incident at 60° with the normal to a glass-air and water-air interface, respectively. Predict the angle of refraction in glass when the angle of incidence in water is 45° with the normal to a water-glass interface [Fig. 9.31(c)].

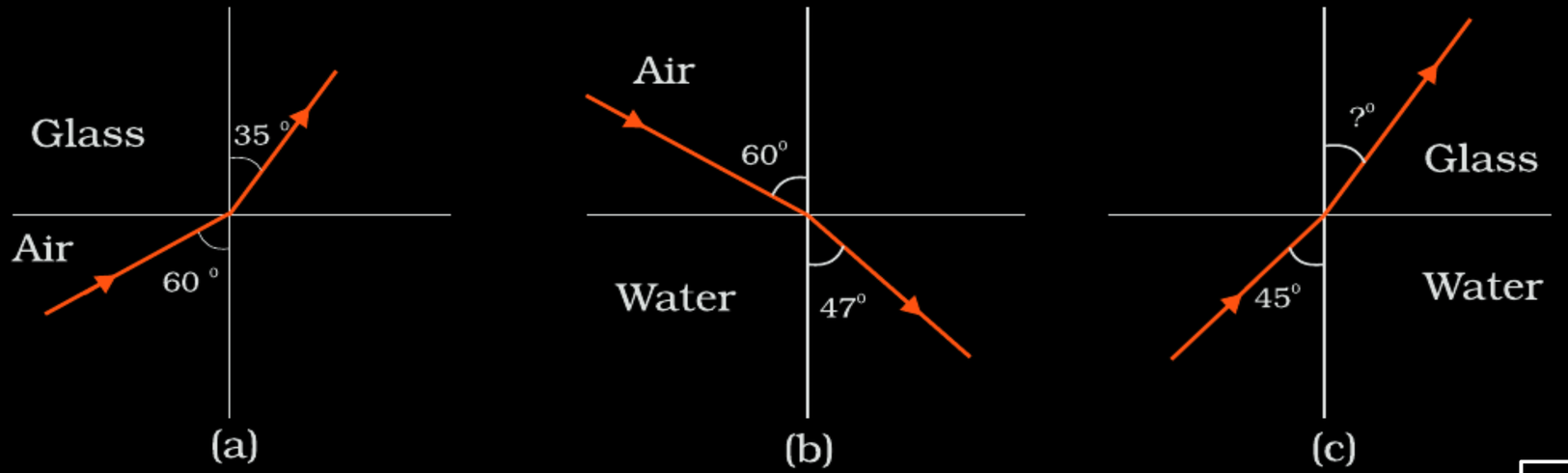


FIGURE 9.31

NCERT

$n_{ga} = 1.51$; $n_{wa} = 1.32$; $n_{gw} = 1.144$; which gives sin $r = 0.6181$ i.e., $r \simeq 38°$.

9.5 A small bulb is placed at the bottom of a tank containing water to a depth of 80cm. What is the area of the surface of water through which light from the bulb can emerge out? Refractive index of water is 1.33. (Consider the bulb to be a point source.) NCERT

9.5 A small bulb is placed at the bottom of a tank containing water to a depth of 80cm. What is the area of the surface of water through which light from the bulb can emerge out? Refractive index of water is 1.33. (Consider the bulb to be a point source.) NCERT

$r = 0.8 \times \tan i_c$ and $\sin i_c = 1/1.33 \cong 0.75$, where r is the radius (in m) of the largest circle from which light comes out and i_c is the critical angle for water-air interface, Area = $2.6\,\text{m}^2$

9.6 A prism is made of glass of unknown refractive index. A parallel beam of light is incident on a face of the prism. The angle of minimum deviation is measured to be 40°. What is the refractive index of the material of the prism? The refracting angle of the prism is 60°. If the prism is placed in water (refractive index 1.33), predict the new angle of minimum deviation of a parallel beam of light. NCERT

9.6 A prism is made of glass of unknown refractive index. A parallel beam of light is incident on a face of the prism. The angle of minimum deviation is measured to be 40°. What is the refractive index of the material of the prism? The refracting angle of the prism is 60°. If the prism is placed in water (refractive index 1.33), predict the new angle of minimum deviation of a parallel beam of light. NCERT

$n \cong 1.53$ and D_m for prism in water $\cong 10°$

9.16 A small pin fixed on a table top is viewed from above from a distance of 50 cm. By what distance would the pin appear to be raised if it is viewed from the same point through a 15 cm thick glass slab held parallel to the table? Refractive index of glass = 1.5. Does the answer depend on the location of the slab?

NCERT

9.16 A small pin fixed on a table top is viewed from above from a distance of 50 cm. By what distance would the pin appear to be raised if it is viewed from the same point through a 15 cm thick glass slab held parallel to the table? Refractive index of glass = 1.5. Does the answer depend on the location of the slab?

NCERT

The pin appears raised by 5.0 cm. It can be seen with an explicit ray diagram that the answer is independent of the location of the slab (for small angles of incidence).

9.17 (a) Figure 9.32 shows a cross-section of a 'light pipe' made of a glass fibre of refractive index 1.68. The outer covering of the pipe is made of a material of refractive index 1.44. What is the range of the angles of the incident rays with the axis of the pipe for which total reflections inside the pipe take place, as shown in the figure.

NCERT

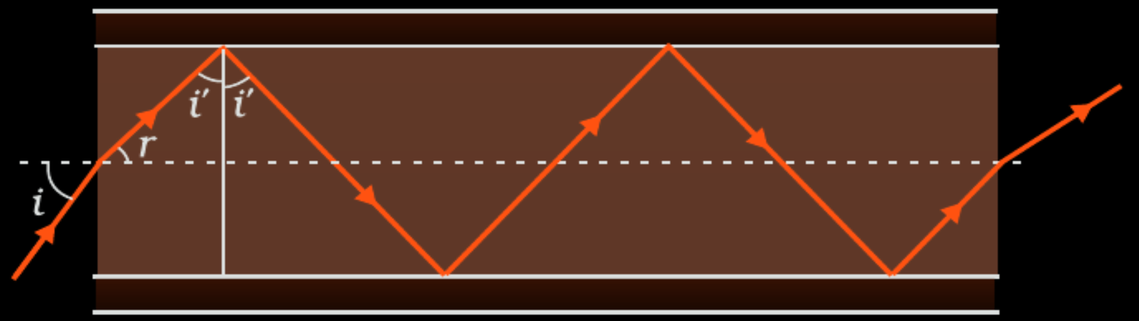


FIGURE 9.32

(b) What is the answer if there is no outer covering of the pipe?

(a) $\sin i_c' = 1.44/1.68$ which gives $i_c' = 59°$. Total internal reflection takes place when $i > 59°$ or when $r < r_{max} = 31°$. Now, $(\sin i_{max}/\sin r_{max}) = 1.68$, which gives $i_{max} \simeq 60°$. Thus, all incident rays of angles in the range $0 < i < 60°$ will suffer total internal reflections in the pipe. (If the length of the pipe is finite, which it is in practice, there will be a lower limit on i determined by the ratio of the diameter to the length of the pipe.)

(b) If there is no outer coating, $i_c' = \sin^{-1}(1/1.68) = 36.5°$. Now, $i = 90°$ will have $r = 36.5°$ and $i' = 53.5°$ which is greater than i_c'. Thus, *all* incident rays (in the range $53.5° < i < 90°$) will suffer total internal reflections.

9.22 At what angle should a ray of light be incident on the face of a prism of refracting angle 60° so that it just suffers total internal reflection at the other face? The refractive index of the material of the prism is 1.524.

NCERT

9.22 At what angle should a ray of light be incident on the face of a prism of refracting angle 60° so that it just suffers total internal reflection at the other face? The refractive index of the material of the prism is 1.524.

NCERT

If the refracted ray in the prism is incident on the second face at the critical angle i_c, the angle of refraction r at the first face is $(60° - i_c)$.

Now, $i_c = \sin^{-1}(1/1.524) \simeq 41°$

Therefore, $r = 19°$

$\sin i = 0.4962$; $i \simeq 30°$